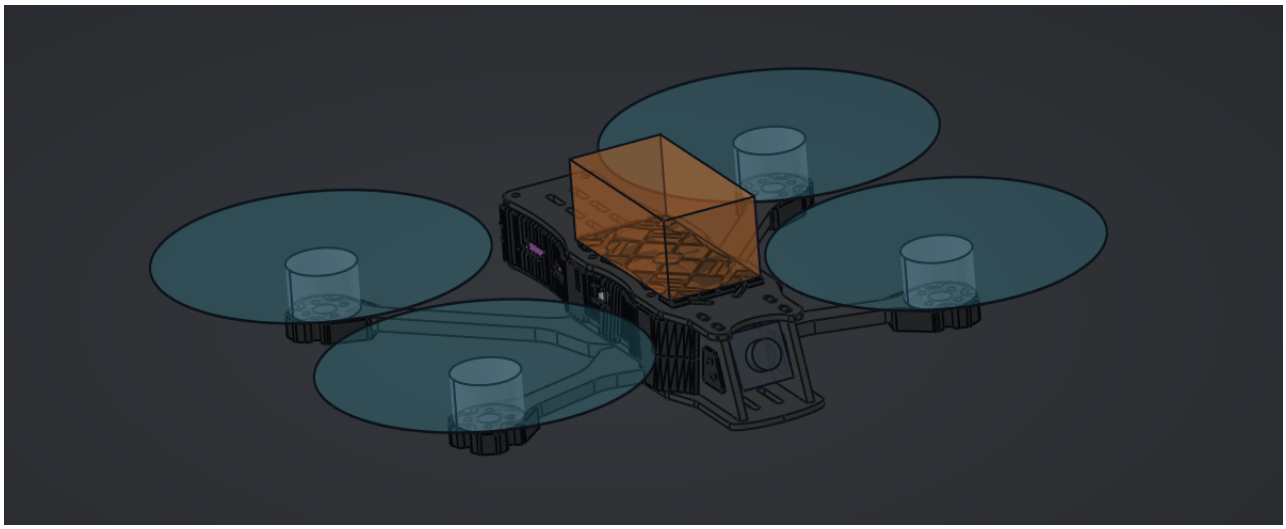
RotorForge

Drone Frame Editor
Complete user guide

Version 0.90.10 • Michael Struggl



RotorForge frame designed by Michael Struggl.

Design the motor layout, carbon plates, arm joints, electronics mounts and cutting profiles in one editable FreeCAD project. This guide follows the editor workflow and describes every visible parameter and command in its relevant chapter.

Dimensions are in millimetres unless stated otherwise. Frame +Y is forward, +X is right and +Z is up. Open the separate Installation PDF for Windows, macOS and Linux installation paths.

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Use the linked chapter names to navigate. Tables use the names shown in the editor. Defaults describe a fresh frame; loading saved settings keeps that design's values. Disabled controls retain their settings for when the relevant mode is enabled.

01 Start a frame and use the editor

Install the complete RotorForge folder in your FreeCAD user Mod directory. Confirm the file freecad/rotorforge/init_gui.py inside that folder. Close FreeCAD and replace the old folder completely when upgrading; merging folders can leave obsolete startup files behind. The Installation PDF gives the steps for each operating system.

Activate the RotorForge workbench, choose New frame, set the plate design and motor layout, then work through the main section icons and subsection buttons. The top view updates as dimensions change. The FreeCAD model is created only when you press Create frame; editing an existing model uses Update frame.

Start with the main proportions, then fit the arm roots, stack, standoffs and FPV hardware. Set plate reaches and centre widths before placing airflow openings or edge notches. The Checks area reports geometric errors and layout warnings. Correct blocking errors before building or exporting.

Control	What it does
New frame	Opens an editor for another parametric frame.
Edit frame	Edits the selected frame, parameter object or part. With one frame in the document it can be found automatically.
Create frame / Update frame	Creates the assembly or rebuilds affected components and their dependent parts. Detailed part and clearance checks can still reject a configuration that passed the fast editor checks.
Cancel	Closes the editor without committing its frame changes.
Maximize window / Restore window	Toggles the editor between a maximized window and its previous size. The normal window controls remain available.
Front isometric	Looks toward the +Y nose of the assembled frame.
Toggle references	Shows or hides approximate propeller, motor, stack and FPV electronics envelopes. These are references, not manufacturing parts.
Help manual (PDF)	Opens this guide in the system PDF viewer.
Load settings / Save settings	Loads or saves a frame JSON, including saved reference-hardware choices and visibility when available. After loading settings, Create / Update restores those references. Save the FreeCAD project separately to retain the generated model and cutting layouts.
Export frame	Writes manufacturing files and refreshes the native cutting-profile layout. The export chapter explains the thickness sheets and coordinate systems.

Where to find controls

The main categories contain focused pages underneath. Choose an icon in the vertical strip on the left, then a text button for the subsection. Hover over an icon to see its name. Subsection buttons wrap into extra rows when needed. Maximize and restore remain available. Group headings separate related dimensions within each page. Disabled fields keep their saved values until the relevant design mode is enabled.

Control	What it does
Frame	Layout: plate design, FPV, wheelbase and propellers. Bottom / Mid / Top plate: outline, reaches, end shapes, fillets and thickness; top-front camera recess. Centre widths: widening and shoulders.
Arms	Roots: beams, fitted joints and root dimensions. Bolts: spacing, coordinates and stack reliefs. Clamps & key: arm sandwich and keystone. Motors: pad outline, rotation and fixing pattern.
Hardware	Standoffs: counts, positions and heights. Stack: FC/ESC pattern, offsets and screw access. Straps: one/two straps, slots or outline cutouts. Nut retention: hex pockets and press nuts. Weight estimates: equipment masses.
Battery	Placement: LiPo size, mass, mounting face and pose. LiPo pad: outline, reaches, grip and density.
FPV	Camera: body dimensions and side screws. Tilt & position: fixed/variable tilt and fitting. Side plates: contacts, captured tabs and reliefs. VTX: mounting holes and envelope.
TPU	Appearance: shared printed-part colour. Camera isolation, Front bumper and Motor protectors. Side guards: fit mode, VTX clearance, USB and wire access. Side guard vents: separate VTX/FC sections. Side guard straps: top/bottom strap passages. Rear antenna: VTX pigtail tabs, tubes or side-entry clips. ELRS antenna: T-antenna hooks, extensions and centre pass-through.
Finishing	Detailed references: motors, props, stack, camera, VTX and LiPo. Airflow / outlines: cuts, notches and patterns. Checks: minimum material, density and visibility.

Find a control

Use Find a control at the bottom beside Create / Update frame, Maximize and Cancel when you know a control name or topic. Type a term such as keystone, strap or camera tilt, then select a result. The editor opens the matching section and subsection and scrolls to the field. Results include fit buttons and airflow pattern controls. Searching does not change parameter values or enable a disabled option. Clear the search field with its small clear button to start another search.

Inspect one part or all flat profiles

Scroll the mouse wheel over the main preview to zoom around the pointer. Drag with the left or middle mouse button to pan. Double-click or press Fit view to reset. Changing the preview mode resets zoom; changing dimensions keeps it. These controls affect the editor preview, separate from the FreeCAD 3D view. Zoom and pan reuse a vector drawing; changed dimensions, preview mode and window size refresh it. This preserves sharp outlines while avoiding repeated geometry work during movement.

Use the preview selector above the drawing to switch between Overall frame, All parts, All four arms, or one active component. Individual parts automatically fit the available space; arms are aligned motor-end upward for inspection. Plate and arm views include their fixing holes, access slots, tip features, applied airflow openings and edge notches. Camera sides appear as flat profiles rather than their thin assembled top edges. Inactive parts disappear from the selector when you change the plate or locking options. This inspection creates no FreeCAD document objects and does not change the exported CNC layout.

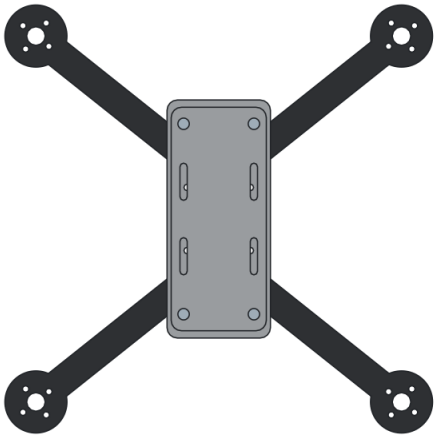
The overall view shows the LiPo footprint in orange when references are visible. After Calculate frame weight / CG, a red cross marks the active estimate CG projected into the top view (equipped plus LiPo when both estimates are enabled). A design or mass-setting change clears the old mass/CG result and marker until you calculate again. Changing only colour or the preview selection keeps the result.

Carbon arms, plates, clamps, keys and camera side plates use satin graphite in the model. TPU > Appearance selects one matte colour for every printed or soft part, including the LiPo pad. Electronics reference colours are independent. The guide uses matching orthographic diagrams: grey carbon, darker-grey soft/printed parts, neutral hardware and amber reference electronics. Lighter grey separates overlapping carbon layers. Shape catalogues follow the same grey palette; teal is reserved for captions and navigation. Colours in the guide identify parts; choose the desired model colour in the editor.

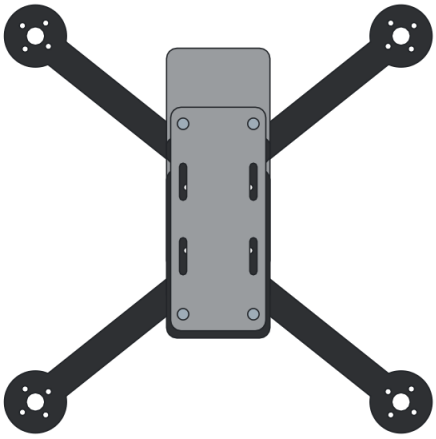
02 Motor layout and frame proportions

Diagonal wheelbase sets the base distance between opposite motor centres. It is not the length of a plate or the overall carbon outline. True X is symmetric. Stretched X increases the left/right motor span on the roll axis. Deadcat widens the front motor pair and automatically compensates rear reach as front spread increases. With zero extra rear extension, both diagonals retain the configured wheelbase.

2 plates



3 plates



The frame uses +Y at the front. Arm names refer to physical positions, not flight-controller motor numbering.

Control	What it does
Plate design	Two plates: choose arms above/below the bottom and one optional clamp on the opposite side of the arms. The upper clamp carries the FC stack; below-bottom arms keep the FC stack on the bottom. Three plates: arms between bottom and mid, with an optional captured keystone. **Choices:** 2 plates; 3 plates.
FPV camera drone	Add camera side plates captured by tabs in the bottom and top, plus rear VTX holes. The nose and rear decks extend to support them.
Motor layout	Select True X for a symmetric motor layout, Stretched X for a wider left/right span, or Deadcat for wider front motor spacing. **Choices:** True X; Stretched X; Deadcat.
Diagonal wheelbase	Sets the base diagonal between opposite motor centres. It is not the carbon outline. With zero rear extension this is the actual diagonal; a Deadcat rear extension increases the actual diagonal shown in the editor summary.
Length / rear width	Stretched X: left–right motor span divided by front–rear spacing. Deadcat: length divided by rear span. Ignored for True X.

Control	What it does
Front / rear motor span	Deadcat only: front motor span / rear span in the base layout. Automatic rear reach compensates for front spread while preserving the base diagonals. Independent rear arm extension is applied afterwards and changes the final rear span and diagonal.
Deadcat rear arm extension	Deadcat only. Extra length, in mm, added to each rear arm along its existing root-to-motor direction. Zero keeps the original layout. Front motor positions and all arm roots stay fixed. Rear span and actual diagonals grow; the base wheelbase is unchanged. The value is retained but ignored in True X and Stretched X.
Propeller diameter	Full propeller diameter: 5 inch = 127 mm; 5.1 inch = 129.54 mm; 7 inch = 177.8 mm.
Desired prop gap	Desired minimum clearance between neighbouring propeller discs. A smaller calculated gap produces a layout warning; changing this value does not move the motors.
Corner Radius	Radius at the main plate corners. Widened-centre transitions use straight tapered shoulders.

Independent Deadcat rear arm length

On Frame > Layout, increase Deadcat rear arm extension to lengthen both rear arms by the entered amount. For example, 20 mm adds 20 mm to each rear root-to-motor length. The motors move outward along the existing arm axes, without moving the front motors or the arm-root references. The rear fixing pattern and motor pad follow the motor. Zero restores the original layout. True X and Stretched X ignore this value but keep it saved.

An extension increases the final rear span and motor diagonal. While it is active, the wheelbase and front/rear span controls are labelled as base values; the summary shows the actual diagonal and front/rear arm lengths. Propeller clearance, preview, generated arms and manufacturing exports follow the extended positions.

Propeller diameter and desired prop gap compare the propeller discs. They do not change the motor locations automatically. Review the resulting minimum gap and the camera view against the actual propellers and camera lens.

03 Plate arrangements, reaches and centre widths

A two-plate frame has a bottom and top. Arms may sit above or below the bottom, with one optional clamp on the opposite side of the arms. The FC stack sits on the upper clamp when arms are above the bottom; below-bottom arms keep the stack on the bottom plate. A three-plate frame sandwiches the arms between bottom and mid. The bottom nose extends toward the camera; the mid body runs from the stack toward the rear.

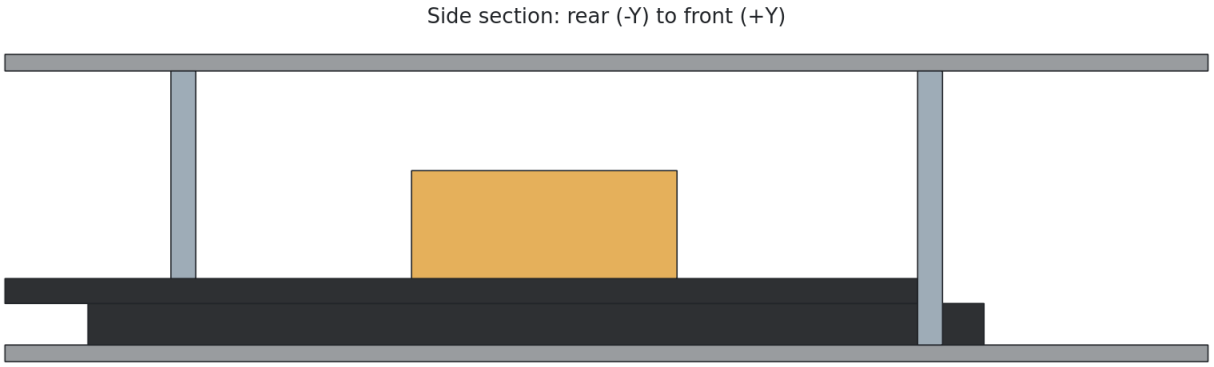


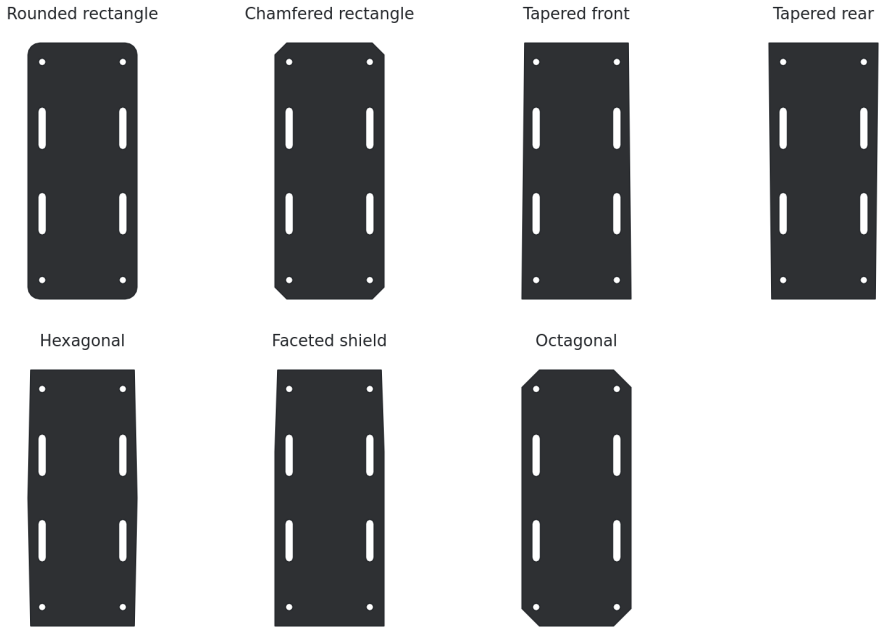
Plate thicknesses and arm position determine the mounting levels used by the stack, standoffs and camera cage.

Choose the base outline of each plate

Bottom, mid and top plates each have their own outline selector. The clamp selector applies to whichever upper or lower clamp is active. Rounded rectangle retains the original shape. Chamfered rectangle clips the corners using the common corner radius; Octagonal uses broader facets. Tapered front and Tapered rear narrow the named end. Hexagonal narrows both ends; Faceted shield keeps broad rear shoulders and tapers the nose. Plate end width controls the narrow ends as a percentage of base width. The stack, bolts and standoffs do not move with the outline.

Control	What it does
Bottom plate outline	Base outline before centre widening, arm-matched tabs and applied cutouts. Reaches and width retain their existing meaning. Check fastener support after narrowing an end. **Choices:** Rounded rectangle; Chamfered rectangle; Tapered front; Tapered rear; Hexagonal; Faceted shield; Octagonal.
Plate end width	Narrow-end width as a percentage of the base width for tapered, hexagonal and shield outlines. Does not alter the independently widened centre. 100 percent removes the taper.
Mid plate outline	Independent mid-plate base outline. The fitted front-arm tabs still follow the actual arms; changing the base does not move mounting holes. **Choices:** Rounded rectangle; Chamfered rectangle; Tapered front; Tapered rear; Hexagonal; Faceted shield; Octagonal.

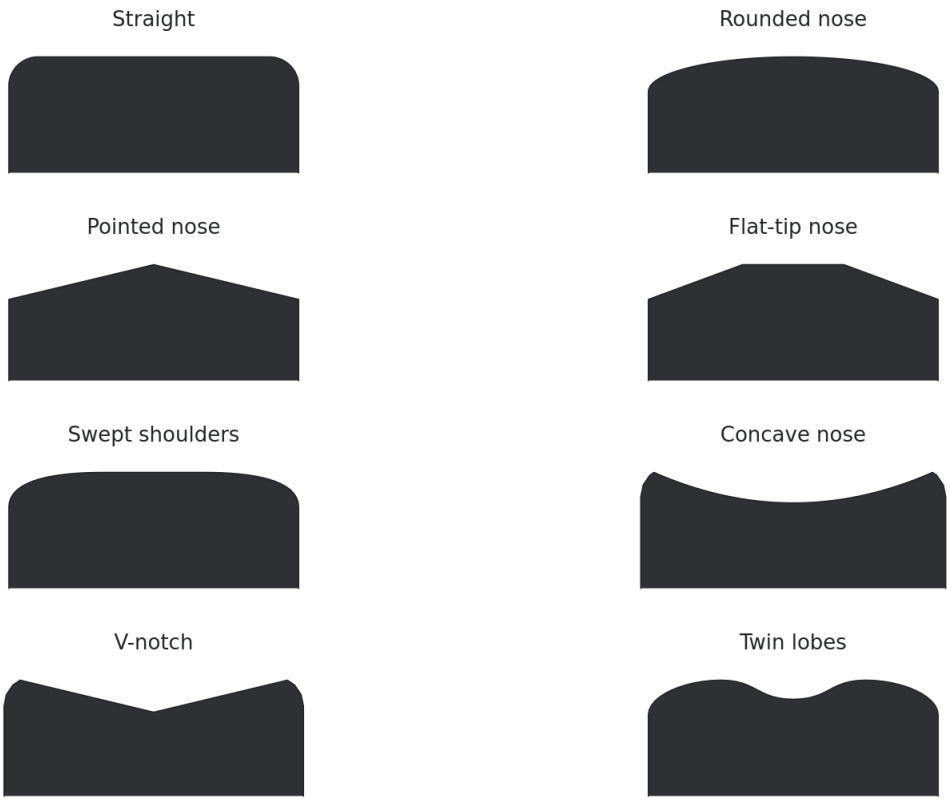
Control	What it does
Top plate outline	Top outline before strap slots, camera sockets, front edge profiles and custom cuts. Tapered ends may require more width for standoffs and camera tabs. **Choices:** Rounded rectangle; Chamfered rectangle; Tapered front; Tapered rear; Hexagonal; Faceted shield; Octagonal.
Clamp plate outline	Base shape for the active upper or lower arm clamp. Existing stack and arm fixings retain their positions. **Choices:** Rounded rectangle; Chamfered rectangle; Tapered front; Tapered rear; Hexagonal; Faceted shield; Octagonal.



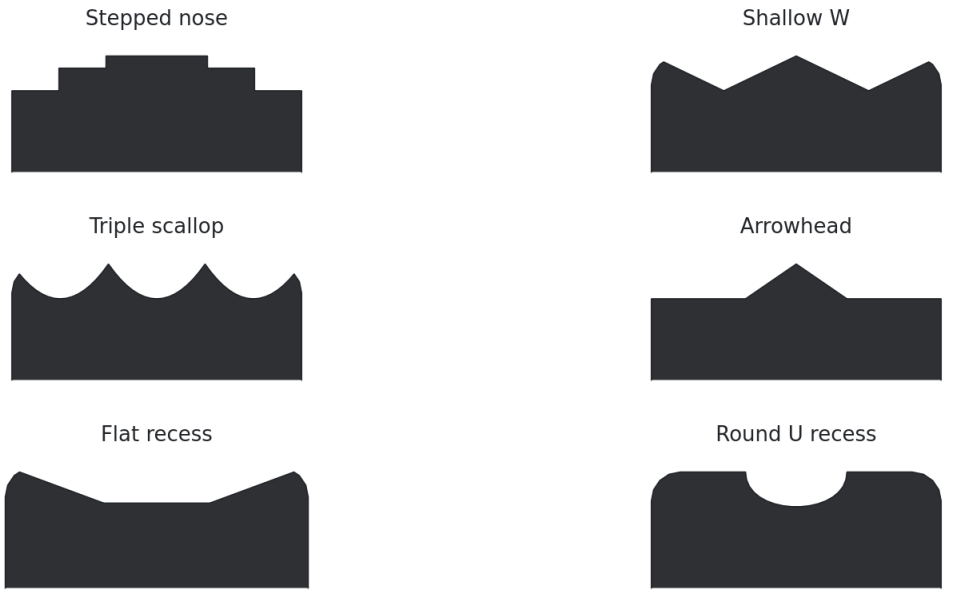
Seven base outlines. Centre widening, arm-matched tabs and applied cutouts are added to the selected base.

Shape the absolute front of the bottom plate

Open Frame / Bottom plate / Front edge. This is separate from the base outline selector and affects only the bottom plate nose. Straight keeps the existing edge. Rounded nose makes a smooth convex cap; Pointed nose meets at a central tip; Flat-tip nose leaves a small central flat; Swept shoulders blend smoothly into a flat tip. Concave nose makes a smooth central recess, V-notch makes an angular recess, and Twin lobes makes two rounded tips with a shallow central dip.



Profile views of the original eight bottom-front options. Front is upward; each example uses 6 mm shaping depth.



Six additional bottom-front profiles, using the same 6 mm shaping zone.

Control	What it does
Front edge shape	Trim only the absolute front of the bottom plate. Straight retains the existing outline. The nose never extends beyond the front reach. Arm and standoff positions stay fixed.
Front edge depth	Fore/aft shaping depth in mm. Convex noses retreat at the shoulders; concave noses retreat at the centre. Keep enough stock around front standoffs and camera sockets.
Front tip width	Width of the flat front tip as a percentage of the nose span. Used by flat-tip, swept, stepped and recessed profiles; other profiles ignore it.

Front edge depth controls the length of the shaping zone. Stepped nose has two shoulder levels; Shallow W and Triple scallop create repeated recesses; Arrowhead leaves a narrow central point; Flat recess and Round U recess leave a central opening. Front tip width controls Flat-tip nose, Swept shoulders, Stepped nose, Flat recess and Round U recess; it is a percentage of the nose span. The span follows the local width at the nose; widening only the middle leaves the nose shape unchanged. The front reach remains the forward limit: these shapes trim material inside it and never extend the plate. Centre widening cannot fill the nose back in. Existing base corners, outline cutouts and the bottom plate fillet still apply. Arm and standoff fixings retain their coordinates and material checks. When Follow plates is enabled, camera contacts and their sockets can move rearward to retain full support on the new nose. Manual camera attachment positions keep their support checks. If a deeper shape removes required support, reduce its depth or increase the bottom front reach. Other plate base outlines and motor arms keep their own settings; camera-following sockets follow the fitted camera cage. Shaped fronts are included in cutting profiles; smooth fronts use exact cubic splines in DXF and cubic paths in SVG.

Top-front shapes for camera uptilt

Open Frame > Top plate > Front edge and camera recess. Straight keeps the current front. The selector also offers the general front profiles shown in the bottom-front catalogue, plus six camera-specific recesses. These shapes open the centre while keeping side rails for the upper camera tabs. They trim inside the existing top front reach and work in either plate design, including non-FPV frames.

The U recess has a flat back with rounded inner corners. Flared recess opens toward the front with a narrower back. Swept recess uses curved shoulders; Crescent makes a smooth bowl. Stepped recess narrows in two stages, and Chamfered recess uses straight sides with clipped rear corners. Every recess is symmetric about the frame centreline.

Control	What it does
Front edge shape	Top front outline inside the existing front reach. Camera recesses clear the centre and can set back the shoulders; Straight preserves the existing profile. This profile is the sole built-in top-front camera opening.
Front edge depth	Fore/aft shaping depth. For camera recesses, measured behind the set-back shoulder line to the deepest point. Mounting holes and camera tabs retain support checks.
Front feature width	Central feature width as a percentage of the front span for flat-tip, swept, stepped and general recessed profiles. Camera-specific recesses use Camera recess width instead.
Camera recess width (0 = auto)	Camera-specific opening width in mm. Zero uses camera width plus twice the top front camera clearance. Flared and swept mouths open up to 20 percent wider; stepped/flared bottoms are narrower. This sets a profile, not a measured guarantee of camera clearance.
Camera recess clearance	Extra clearance on each side of the camera body when Camera recess width is automatic. Used only by top-front camera recesses; custom width is used directly.
Camera shoulder setback	Retract both front shoulders from the current top front reach, in mm. Applied only to camera-specific profiles. Depth is measured from this new shoulder line; the full recess reaches setback plus depth behind the original front.

For camera-specific shapes, the shoulder line is **Top front reach - Camera shoulder setback**. The deepest point is another Front edge depth behind that line. For example, a 90 mm front reach, 6 mm setback and 16 mm depth put the shoulders at Y = 84 mm and the deepest part of the recess at Y = 68 mm. Setback is useful when the plate corners project too far forward. Recess width is in millimetres; zero follows the camera body width plus twice Camera recess clearance. Flared and swept mouths can be 20% wider, while stepped and flared backs are narrower. General front profiles use the percentage Front feature width where applicable and ignore the camera-specific width and setback.

Camera recess catalogue

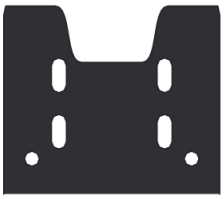
U recess



Flared recess



Swept recess



Crescent



Stepped recess



Chamfered recess

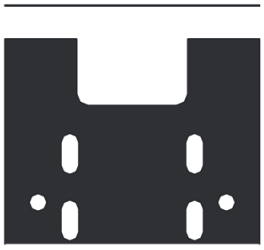


Top-front profiles create the complete camera recess. Front is upward (+Y). Tab sockets follow the supported camera contacts.

Straight front



Recess + 6 mm setback



Teal marks the original front reach. A recess opens the centre; shoulder setback also retracts the outside front edge.

The top-front profile defines the camera opening. Its width, depth, clearance and shoulder setback are all on Frame > Top plate. Straight keeps a solid front. Tilt and camera position remain independent. Follow plates refits the upper contact and sockets; custom attachments retain support checks. The builder checks the configured camera envelope through the tilt range. Adjust depth, width, reach or standoff height if clearance or mounting stock is insufficient.

Rear edge shapes on each deck

Frame > Bottom plate, Mid plate and Top plate each have independent Rear edge controls. Bottom rear shaping is active only with two plates; the mid rear selector is active only with three. Top rear shaping works in both designs. Straight preserves the existing end. The rear library mirrors all 14 front profiles toward -Y and adds Cable channel, Dual cable channels, Split tail and Stepped electronics tail. These profiles trim inside the current rear reach; they do not move holes or extend the plate.

Rear edge depth sets the shaping zone along the frame. Rear feature width changes the flat, step or recess span in profiles that use it. A cable channel has rounded inner corners; Dual cable channels leaves a central bridge. Split tail creates a rectangular central opening, and Stepped electronics tail leaves a narrower central tongue. These names describe outlines; they do not create antenna or GPS holders. Check connector and wire access yourself.

Control	What it does
Bottom rear shape	Independent rear outline for two-plate designs. Straight retains the current end.
Bottom rear depth	Shaping depth in mm, limited to half this plate length.
Bottom rear feature width	Percentage of the rear span used by flat tips, steps and recesses.
Mid rear shape	Independent rear outline for three-plate designs. Straight retains the current end.
Mid rear depth	Shaping depth in mm, limited to half this plate length.
Mid rear feature width	Percentage of the rear span used by flat tips, steps and recesses.
Top rear shape	Independent rear outline for both plate designs. Straight retains the current end.
Top rear depth	Shaping depth in mm, limited to half this plate length.
Top rear feature width	Percentage of the rear span used by flat tips, steps and recesses.

Required stock around standoffs, VTX holes, camera sockets and arm bolts is still checked. If an edge profile removes support, use a shallower depth, a wider tip or a longer rear reach. Base outlines, centre widening, outline cutouts and optional corner fillets combine with the end profile. Side guards and a plate-following LiPo pad follow the resulting deck. Exported CNC profiles use the final shaped boundary.

Rear edge catalogue

Straight



Rounded end



Pointed end



Flat-tip end



Swept shoulders



Concave end



V-notch



Twin lobes



Stepped end



Shallow W



Triple scallop



Arrowhead



Flat recess



Round U recess



Cable channel



Dual cable channels



Split tail



Stepped electronics tail



Rear is downward (-Y). Select the shape independently on each active rear deck.

Independent front and rear body reaches

Automatic reach starts from the configured base size and the support the design requires. Custom reach uses the entered limits and reports missing support instead of silently extending the plate. The plate-bounds readout shows the resulting rear/front Y coordinates.

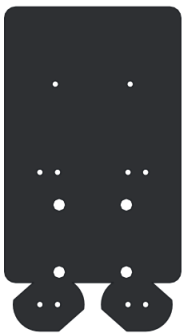
The mid front body is measured from the stack: **front Y = Stack Offset Y + Mid body reach ahead of stack**. The bottom rear body is measured the other way: **rear Y = Stack Offset Y - Bottom body reach behind stack**. Increase the bottom rear body reach to fill more of the centre opening between the rear arm tabs.

The mid front clamping tabs follow the front arm roots; the bottom rear tabs follow the rear arm roots, including their width, taper, bolt positions, fitted joints and key pockets. These small arm-matched tabs can extend beyond the chosen main-body edge. The main body remains controlled by its own width and reach.

Control	What it does
Bottom plate reach	Automatic computes the outer limits. Bottom body reach behind stack independently controls the three-plate body edge. Rear clamping tabs follow the actual rear arm-root outlines. Custom front/rear reaches cap the outline without automatic growth; two-plate bottoms keep their full profile. **Choices:** Automatic; Custom.
Bottom base length	Automatic base length before the front extension. Three-plate bottoms are trimmed behind the rear arm sandwich; use Custom for exact independent reaches.
Bottom front extension (3 plates)	Extra bottom length toward +Y in three-plate mode. Its rear edge does not move. Ignored in two-plate mode.
Bottom reach toward front (+Y)	In Custom bottom reach mode, distance from frame Y=0 to the front body edge. It is independent of the stack-referenced rear body dimension.
Bottom outer rear limit (frame Y=0)	Maximum rearward extent of the bottom plate in Custom mode. In three-plate designs the rear tabs follow the arms, including fitted joints and keystone pockets; extending this limit does not add a full-width rear deck.
Bottom body reach behind stack	Three-plate main-body distance behind the FC stack centre. Rear edge Y = Stack Offset Y minus this reach. Increase it to fill more of the centre between the rear arm tabs; decrease it to shorten the body. Tabs still follow the arms beyond that edge. Mounting support is checked.
Bottom / mid base width	Normal width of the bottom and mid bodies. Centre-width controls can widen their stack and arm-clamping regions independently.
Bottom Thickness	Stock thickness of the bottom plate; changes the arm sandwich, camera lower seats and actual standoff support levels.

Control	What it does
Mid plate reach	Automatic can extend the mid rear deck to support the VTX. Custom uses the exact front/rear reaches measured from the stack centre. **Choices:** Automatic; Custom.
Mid body reach ahead of stack	Front reach of the main stack deck. Its front arm-clamping tabs follow the actual arm-root outlines, including width, fitted joints and key pockets, within the bottom support.
Mid body reach behind stack	Main mid plate extends this far rearward from the stack centre. Independent of bottom plate length.
Mid plate thickness	Stock thickness of the mid plate in three-plate mode. The mid carries the FC stack and rear electronics.
Top plate reach	Automatic covers the camera and rear VTX. Custom uses the exact independent reaches from frame centre; insufficient coverage is reported. **Choices:** Automatic; Custom.
Top plate length	Initial centred top-plate length in Automatic mode. Required camera, VTX and mounting support may extend its ends. Custom mode uses the separate front and rear reaches.
Top reach toward front (+Y)	In Custom top reach mode, distance from frame Y=0 to the front edge. Shortening it changes the sloping camera-wall attachment when plate following is enabled.
Top reach toward rear (−Y)	In Custom top reach mode, distance from frame Y=0 toward the rear edge. Keep the top long enough to cover the VTX and support the rear standoffs.
Top base width	Normal top-plate body width. Top centre width adds a wider middle section when enabled.
Top Thickness	Stock thickness of the top plate. Camera upper tongues and their receiving sockets follow this thickness.

Shorter rear reach

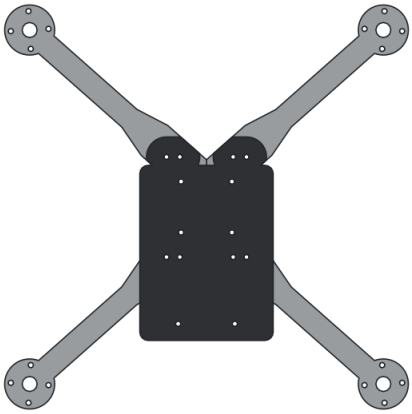


Longer rear reach



Bottom rear body reach is referenced to the FC stack. The rear tabs continue to match the arms beyond that body.

Arm roots / mid-plate tabs



Front mid-plate tabs follow the actual arm-root footprint rather than creating broad independent overhangs.

Smooth plate and cutout corners

Bottom, mid and top plates have independent corner fillet controls on their own Frame pages. The clamp fillet on Arms > Clamps & key applies to the active upper or lower clamp. Zero retains the finished plate. A positive target radius rounds outside corners, inward shoulders and outline notches after all cuts. Closed internal openings stay unchanged by default. Enable the separate Round internal openings checkbox for that plate to include their corners; each plate has its own switch, and the clamp switch follows the active clamp. Circular drills and already tangent slots retain their geometry, as do plate thickness and countersinks. It is separate from the base Corner radius and motor-pad transition blending.

External corners

Include internal openings



Profile previews with a 3 mm target: outside corners only (default), and internal cutout corners included by explicit choice. Mounting drills and tangent slots retain their geometry.

Control	What it does
Bottom plate / cutout fillet	Target plan-view corner radius in mm; zero keeps the finished plate unchanged. Rounds outer edges and outline notches after cutting. Closed cutout corners are included only when Round internal openings is enabled for this plate. Tight corners use smaller local radii to preserve edge lengths, mounting support and minimum web; constrained corners are reported. Circular drills, tangent slots, plate thickness and countersinks retain their geometry.
Round bottom internal openings	Opt in to rounding closed airflow/cutout boundaries using the plate fillet radius. Disabled preserves internal openings; external corners and outline notches still use the plate fillet. Circular drills and tangent slots remain unchanged.
Mid plate / cutout fillet	Target plan-view corner radius in mm; zero keeps the finished plate unchanged. Rounds outer edges and outline notches after cutting. Closed cutout corners are included only when Round internal openings is enabled for this plate. Tight corners use smaller local radii to preserve edge lengths, mounting support and minimum web; constrained corners are reported. Circular drills, tangent slots, plate thickness and countersinks retain their geometry.
Round mid internal openings	Opt in to rounding closed airflow/cutout boundaries using the plate fillet radius. Disabled preserves internal openings; external corners and outline notches still use the plate fillet. Circular drills and tangent slots remain unchanged.
Top plate / cutout fillet	Target plan-view corner radius in mm; zero keeps the finished plate unchanged. Rounds outer edges and outline notches after cutting. Closed cutout corners are included only when Round internal openings is enabled for this plate. Tight corners use smaller local radii to preserve edge lengths, mounting support and minimum web; constrained corners are reported. Circular drills, tangent slots, plate thickness and countersinks retain their geometry.
Round top internal openings	Opt in to rounding closed airflow/cutout boundaries using the plate fillet radius. Disabled preserves internal openings; external corners and outline notches still use the plate fillet. Circular drills and tangent slots remain unchanged.
Clamp plate / cutout fillet	Target plan-view corner radius in mm; zero keeps the finished plate unchanged. Rounds outer edges and outline notches after cutting. Closed cutout corners are included only when Round internal openings is enabled for this plate. Tight corners use smaller local radii to preserve edge lengths, mounting support and minimum web; constrained corners are reported. Circular drills, tangent slots, plate thickness and countersinks retain their geometry.

Control	What it does
Round clamp internal openings	Opt in to rounding closed airflow/cutout boundaries using the plate fillet radius. Disabled preserves internal openings; external corners and outline notches still use the plate fillet. Circular drills and tangent slots remain unchanged.

The entered value is a target maximum, not a requirement that every corner have the same radius. A 3 mm target uses 3 mm where it fits and smaller local radii beside short edges, mounting footprints or narrow webs. This avoids rejecting the complete plate when one corner lacks space. The FreeCAD report lists local radius adjustments and any constrained corners retained to preserve the part. Inspect these messages and the generated shape when exact radii matter. The editor follows the same outside-only or internal-opening selection schematically; created parts and CNC profiles use the exact final arcs. Fillets do not repair a source cutout that already disconnects the plate or crosses protected hardware.

A wider stack and arm-mounting section

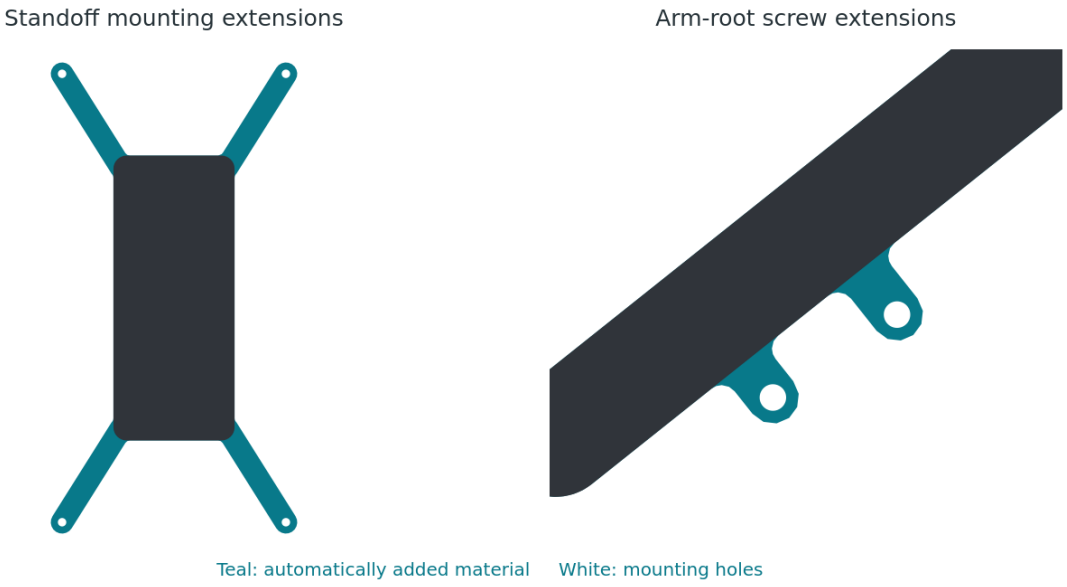
Each plate can widen around the stack independently of its normal base width. A value of zero disables the extra width. Full-width centre length defines the middle plateau; the shoulder length tapers it back to the ordinary width at both ends. The common centre Y shifts this section along the frame.

Control	What it does
Bottom centre width (0 = off)	Zero keeps the base width. A larger value widens only the middle stack/arm section, with tapered shoulders. Front/rear reach stays exact.
Mid centre width (0 = off)	Independent centre widening for the mid body. The integrated front arm tabs still follow the actual roots.
Top centre width (0 = off)	Independent centre widening for the top. Camera and rear sections retain their normal width outside the taper.
Clamp centre width (0 = off)	Independent centre widening for the active upper or lower arm clamp in two-plate designs.
Full-width centre length	Length of the full-width region shared by the plate centre profiles. Each plate is clipped to its own front/rear reach.
Taper length at each end	Length of each sloping transition from the normal plate width to its widened centre.
Centre section position Y	Position of the widened section, measured from frame centre. Positive is front.
Arm roots follow wider bottom centre	When enabled, the arm root inset is measured from the local widened bottom edge. Disable to retain the original arm-root spacing. Deadcat compensation follows the actual roots.

04 Arms, fitted roots, bolts and locking

Automatic mounting extensions: moving an arm bolt beyond its root now grows a local rounded extension with the configured minimum material around the hole. Receiving bottom, mid and clamp plates gain supporting stock where needed. Standoffs moved outside a plate similarly gain rounded bearing pads connected to the original deck. The top plate and the selected lower mounting plate follow the post; screw and post coordinates stay unchanged. This behaviour is automatic and is recalculated from the original outline on each edit, so moving mounts back removes unnecessary growth. Nominal width and reach settings still describe the main body; these local extensions may exceed them. Previews, CNC profiles and exported solids include the same extensions. Fitted arm joints, keystone pockets, other holes and custom cuts retain their clearance checks; extensions cannot make overlapping parts or holes valid.

Mounting extensions automatically receive tangent fillets where they meet the original arm or plate, even with the general edge-fillet controls set to zero. These junction fillets add material; they do not trim the hole support. The target radius is at least 2 mm and follows a larger minimum-material setting, reducing locally where space requires. If no valid rounded join fits, the builder asks for a geometry adjustment instead of exporting a sharp mounting junction. With press nuts, extensions cover the larger of the bore, shank and full flange or flange-counterbore footprint, plus the configured minimum material (at least 1 mm). Larger inserts and recessed-flange clearances therefore enlarge the supporting pad. This also applies to FC-stack press nuts. General edge fillets and final seat checks protect that full bearing margin.



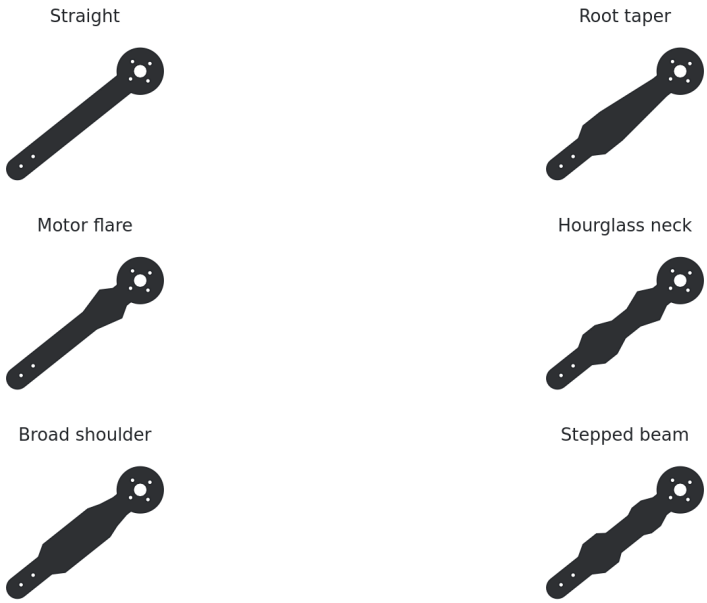
Moving a fixing outside the original outline adds local material around that fixing. The original mounting coordinates are retained.

Arm width sets the ordinary beam. A larger root width adds material only around the inner mounting region and tapers toward the beam. Root reference positions, tail extension and bolt coordinates are separate controls: moving a bolt does not move a motor or root reference.

Straight-axis beam designs

The arm axes, roots, bolts and motors stay fixed. Choose a beam section on Arms > Roots. Root taper adds material near the inner end; Motor flare adds a shoulder near the motor. Hourglass neck has wider ends and a straight narrower middle. Broad shoulder carries a long wide section, and Stepped beam has distinct straight shoulders.

Arm profile widening adds material around the straight centreline while keeping Arm width as the minimum beam width. Zero removes the extra widening. All six beam choices use solid carbon sections.



Six solid beam profiles around fixed straight axes. The roots, motor positions and bolt coordinates remain unchanged.

Control	What it does
Arm beam profile	Changes the beam outline around its straight centreline. Root references, bolt positions and motor centres remain fixed. The shaping begins beyond the arm bolts and widened root shoulder. **Choices:** Straight; Root taper; Motor flare; Hourglass neck; Broad shoulder; Stepped beam.
Arm profile widening	Maximum extra beam width as a percentage of Arm width. Arm width remains the minimum beam stock width. Zero removes extra widening. Choose Straight for an unshaped solid beam. Root fitting and plate-following tabs use the finished profile.

Arm corner fillets

Arms use the same adaptive plan-view fillet system as the plates. Set the target radius on Arms > Roots. Only corners beyond the root mounting area, on the beam and motor-pad outline, are rounded after custom cuts. The complete root outline, fitted joint and mounting stock remain unchanged. The protected length reaches at least Arm root length and extends farther if a custom arm bolt needs more supporting material. Radii reduce locally to preserve minimum webs. Thickness edges remain square. Closed internal openings are left unchanged unless Fillet arm internal openings is enabled. Circular fixing holes remain circular. Zero retains the existing outline; motor-pad transition blending is a separate control.



Exact arm outlines before and after adaptive fillets, with bolt positions unchanged.

Control	What it does
Arm corner fillet	Adaptive beam/motor-pad corner radius from the root outward. The root outline and mounting stock remain unchanged; radii reduce locally to preserve material.
Fillet arm internal openings	Also round corners in arm cutouts. Disabled by default; circular fixing holes remain circular.

Root fit and custom bolts

Independent roots keep their rounded tails. Fitted miters trim each root at the shared joint lines so neighbouring arms can meet without overlapping. Zero joint clearance gives nominal contact where the tails reach; positive clearance creates a deliberate gap. Increase the tail reach or move roots inward if there is still space between them.

Custom bolt coordinates use the **frame axes**, not the arm angle. The front-right root is the reference. X is mirrored for left arms and Y for rear arms. Bolt 1 is the inner shared-mount choice and bolt 2 the outer choice. Use current spacing for custom bolts copies the current along-arm pattern into these frame-aligned offsets.

Control	What it does
Arm Thickness	Carbon stock thickness of each arm and the matching keystone. Changes the stack of material clamped by the root bolts.
Arm Width	Width of the ordinary arm beam. Use Arm root width to enlarge only its inner mounting region.

Control	What it does
Arm root width (0 = follow arm)	Width of the root before joint trimming. Zero follows Arm width. A larger value widens only the root and tapers back into the beam; fitted joints and keystone pockets still trim this material.
Wide root length toward motor	Distance from the arm root reference centre toward the motor before the wider root begins to taper. The tail behind the reference centre uses Arm tail behind root centre.
Root taper length	Length of the transition from the wider root to the ordinary arm beam. The transition must finish before the motor pad.
Arm root distance from centre	The front/rear arm root reference centres sit at +Y and -Y of this distance. Moving custom bolts does not move these references or the motors.
Root inset from local bottom edge	Horizontal distance from the bottom plate side edge to each arm root centre.
Arm tail behind root centre	Untrimmed tail reach behind the midpoint of the two root bolts. Fitted miters trim it at the joint lines and compact key pocket. Increase it until the required seats meet.
Arm root fit	Independent keeps the rounded tails. Fitted miters trim the roots along shared joint lines, allowing them to meet without overlapping. Root centres, mounting holes and motor positions stay where you set them. **Choices:** Independent; Fitted miters.
Arm joint clearance	Complete gap between neighbouring fitted arms. Zero gives nominal contact where the tails reach each other; use a small positive gap for your cutting process. Increase the tail extension if the roots do not yet meet.
Arm bolt placement	Along arm uses the centred pair and Arm root bolt spacing. Custom coordinates uses independent frame-aligned X/Y offsets from each root reference centre. The front-right reference pattern is mirrored across the frame. **Choices:** Along arm; Custom coordinates.
Arm root bolt spacing	Two bolts per arm. Three plates: bottom/arm/mid sandwich. Two plates: arms above or below the bottom, with the optional combined clamp on the same bolts.
Root Hole Diameter	Diameter of the two through-holes in each arm root and their matching plate holes. Shared standoff mounts must use compatible diameters.
Bolt 1 X offset	Frame X offset for bolt 1 on the front-right arm: + right, - left, relative to the root reference centre. Mirrored on the left arms. Shared inner-hole standoffs follow bolt 1.
Bolt 1 Y offset	Frame Y offset for bolt 1 on the front-right arm: + front, - rear, relative to the root reference centre. Mirrored on the rear arms. Changing Y leaves X unchanged.

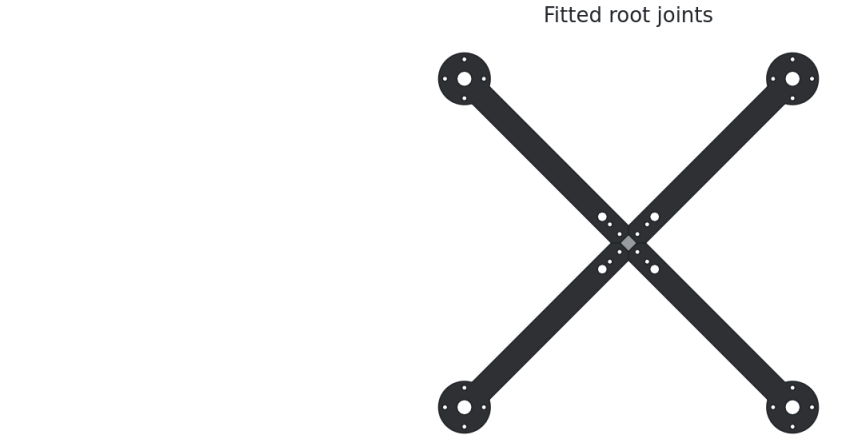
Control	What it does
Bolt 2 X offset	Frame X offset for bolt 2 on the front-right arm: + right, - left, relative to the root reference centre. Mirrored on the left arms. Shared outer-hole standoffs follow bolt 2.
Bolt 2 Y offset	Frame Y offset for bolt 2 on the front-right arm: + front, - rear, relative to the root reference centre. Mirrored on the rear arms. Changing Y leaves X unchanged.
Relieve arm roots for stack access	Enable reliefs where arm roots cross stack screws. Arm stack relief shape selects the original round hole or an inward-opening escape slot. Disable to keep uninterrupted roots when screw access is already clear.
Arm stack relief shape	Round hole keeps the existing circular stack-access relief in each arm. Slot toward stack centre opens an intersecting relief inward, including the configured stack offset, so the arm can slide away from the stack screw. Slot width follows Stack screw access diameter on Hardware > Stack. Slots are cut only for screws intersecting that arm. Arm-bolt support and one-piece geometry remain checked in CAD. **Choices:** Round hole; Slot toward stack centre.

Round reliefs or stack-screw escape slots

On Arms > Bolts, enable Relieve arm roots for stack access, then choose Arm stack relief shape. Round hole keeps the original circular cutout. Slot toward stack centre extends each intersecting relief through the inward arm edge, pointing at the actual FC stack centre rather than the frame origin. A nonzero Stack Offset Y moves that target as well.

The slot width uses Stack screw access diameter on Hardware > Stack, normally 6 mm. Only screws which cross that arm receive a slot. Slots remain present in the arm DXF/SVG and CNC profiles. Disabling reliefs preserves the selected shape for later.

To replace an arm, release its arm bolts and any clamp or keystone capture, then slide the arm away from the stack screw along its open slot. Check the removal path against your actual screw heads and neighbouring parts. With multiple intersecting screws, confirm that their slot paths allow the intended movement. CAD creation still rejects slots that cut into arm-bolt bearing material or split an arm into separate pieces.

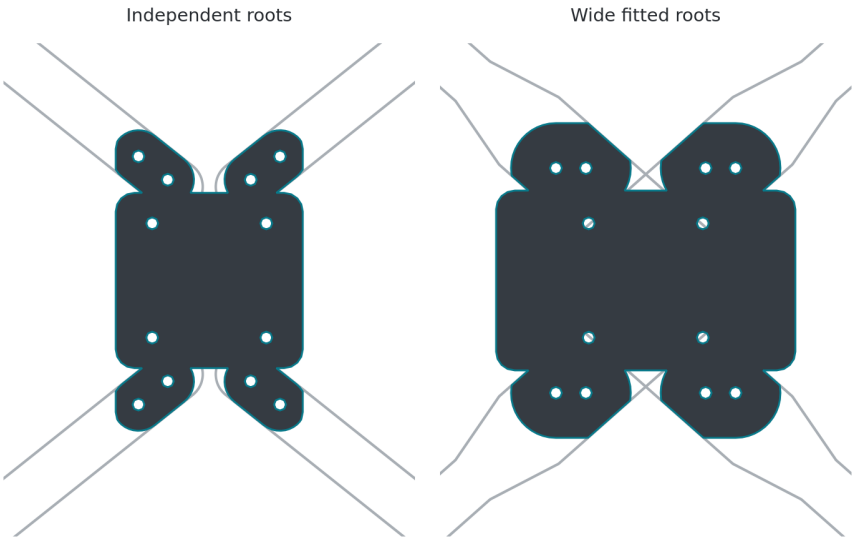


Fitted root joints and the compact key pocket trim the arm tails while retaining the configured bolt and motor centres.

Two-plate clamps and keystone capture

Enable Arm clamp plate (2 plates) to add one continuous clamp across all four roots. Arms above the bottom use one upper clamp, which also carries the FC stack. Arms below the bottom use one clamp beneath the arms, and the FC stack mounts directly on the bottom plate. This choice is independent of the keystone. The eight root bolts secure the bottom/arms/clamp sandwich.

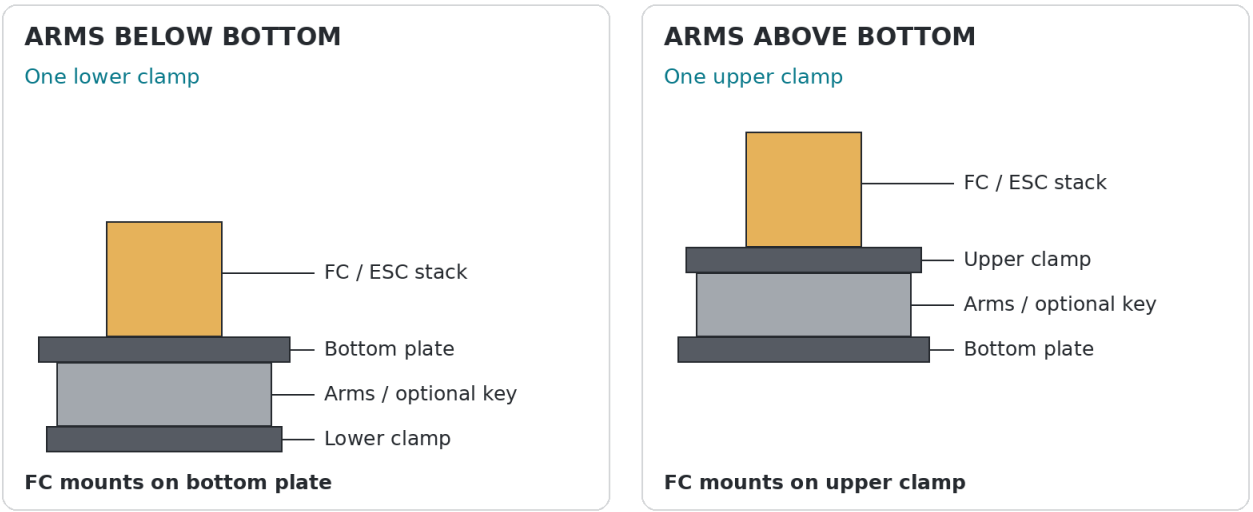
Both clamp ends follow the actual arm roots, using the same trimmed bolt-pad approach as the mid and bottom tabs in three-plate frames. Upper clamps retain stack mounting support and local standoff pads. Lower clamps retain service openings for the screws in the bottom plate. Arm width, taper, fitted joints and bolt locations update the tabs automatically. Upper and Lower clamp thickness retain independent values; only the active value is editable. Update existing frames and re-export after upgrading to remove an old extra upper clamp.



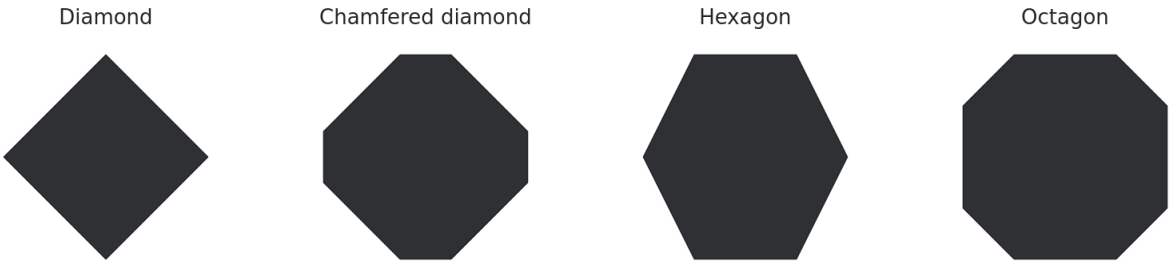
Arm-matched clamp ends: independent roots on the left and wider fitted roots on the right. Grey outlines show the arms; carbon fills show the finished clamp.

In three-plate mode the bottom and mid capture the key. In fitted-miter mode, key width, length and outline define a compact pocket: diamond, chamfered diamond, hexagon or octagon. Each

outline cuts a matching seat in the arms. Independent roots retain the automatic arm-following key. The key has stack-access reliefs; enabled arm-root reliefs also open material that crosses the stack service holes.



Two-plate root sections: one lower clamp for arms below the bottom, or one upper clamp for arms above it. The optional key occupies the arm layer and needs no additional plate.



Four compact fitted-miter key outlines. Matching arm seats use the same shape and the selected clearance.

Control	What it does
Arm position (2 plates)	Choose arms below the bottom plate or above it, inside the frame. Shared standoffs sit above the complete arm sandwich. **Choices:** Below bottom plate; Above bottom plate.
Arm clamp plate (2 plates)	Create one arm clamp opposite the bottom plate: above the arms when they are above the bottom, or beneath the arms when they are below. The lower clamp is independent of the optional keystone. The FC stack mounts on the upper clamp or, with below-bottom arms, on the bottom plate.
Upper clamp thickness	Thickness of the upper arm clamp / FC deck when arms are above the bottom plate. Below-bottom arms use Lower clamp thickness instead; both values remain saved.
Lower clamp thickness	Thickness of the single clamp beneath below-bottom arms, with or without a keystone. Arm bolts clamp it against the bottom plate; stack service holes give access to the stack screws in the bottom plate.

Control	What it does
Arm locking keystone	Add a four-seat key in the arm layer. Bottom and mid capture it in three-plate designs. In two-plate designs, the bottom and the single active upper or lower clamp capture it with the eight arm bolts.
Compact key outline	Choose Diamond, Chamfered diamond, Hexagon or Octagon for a compact key with Fitted miters. The selected outline also defines the matching arm pockets and their clearance. Independent roots retain automatic arm-following seats and disable this selector. Root reach, bolt support, continuous key material and screw-access checks still apply. **Choices:** Diamond; Chamfered diamond; Hexagon; Octagon.
Compact keystone width	Full left/right width of the compact diamond key with Fitted miters. Its matching pocket is cut from the arms. Move roots inward and extend their tails so all four arms reach the key; bolt support is checked.
Compact keystone length	Full front/rear length of the compact key with Fitted miters. It follows the junction of the root joints and uses the same stock thickness as the arms.
Keystone clearance per mating edge	Lateral gap between the key and each arm seat. Zero gives nominal contact; allow a suitable fit for your cutting process.

05 Motor mounting patterns and pad shapes

Motor fixing holes use either a four-hole or three-hole bolt-circle pattern. Enter the measured diameter through the fixing-hole centres. The complete pattern and pad rotate outward with each arm, including Stretched X and Deadcat. Set the actual motor thread spacing and centre opening first, then choose an outline large enough to support them.

Control	What it does
Motor pad outline	Choose a motor bearing outline. The first ten choices through Scalloped shield are retained. Twelve shield variations add shallow edge scallops, varied shoulders and compact crowns.
Motor pad overall width	Nominal motor-pad size. Some shoulders, tails or lobes extend beyond this dimension. Enlarge it when the selected hole pattern lacks edge material.
Motor corner / chamfer / scallop size	Corner radius or chamfer on the listed original shapes; edge scallop size on Scalloped shield. On the twelve shield variations it controls shallow scallop depth. Material checks still apply.
Motor pad rotation pivot	Motor centre keeps each motor and prop centre fixed. Pad beginning pivots at the root-facing extent of the unrotated pad outline, where it meets the beam. The motor, fixing holes, tip features and references move together. A full arm-width elbow joins the beam to the moved pad. Existing designs retain Motor centre. **Choices:** Motor centre; Pad beginning.
Motor pad rotation (mirrored)	Signed rotation about Motor pad rotation pivot. Front-right/rear-left use this angle; front-left/rear-right use the opposite angle. Root bolts and main beam directions remain fixed. Motor centre mode retains motor positions; Pad beginning mode moves motor and prop centres and updates clearance checks. Motor hole rotation remains an independent local offset.
Rotated pad transition radius	Rounds inward junctions near a rotated motor pad using a 2D rolling-radius blend. Adds local supporting material without rounding the carbon thickness edges. Zero keeps the original junction. Reduce if a large radius cannot be built.
Motor tip extension	Extra outward tip reach for Tapered nose and Arrowhead. Replacement platforms use Motor pad neck length for their root-facing necks instead.
Enable motor tip slot	Cut a separate rounded relief slot near the end of a Tapered nose or Arrowhead. It does not replace motor screw holes; airflow settings are independent.
Motor tip slot width	Width of the optional tip relief. Keep the configured material around the outside edge, motor screws and centre bore.

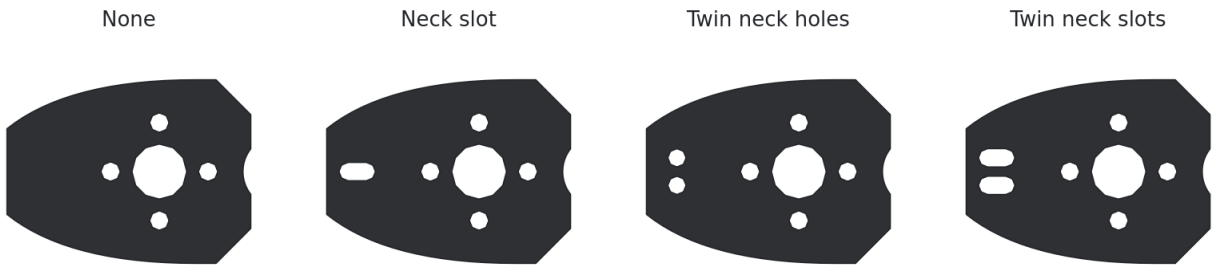
Control	What it does
Motor tip slot total length	Complete end-to-end length, including the semicircular ends. The slot sits one quarter of the extension beyond the normal pad front; increase the extension if the cut leaves too little material.
Motor pad neck length	Length of the root-facing neck on the twelve replacement pad outlines. Optional reliefs are placed here, away from the motor bearing and screws.
Additional pad cutouts	Optional neck relief on the twelve replacement motor-pad outlines. Neck length and the cutout dimensions set the available material. Holes and slots rotate with the pad and are checked against mounting hardware and outline material. **Choices:** None; Neck slot; Twin neck holes; Twin neck slots.
Pad cutout width / diameter	Slot width or circular relief diameter on the new pad outlines. Paired openings keep at least Minimum web between them. The native builder checks remaining edge and screw support.
Pad cutout slot length	Overall slot length along the motor-pad axis. Used by Neck slot and Twin neck slots; ignored for twin circular holes. Must be at least the width.
Motor mounting pattern	Four-hole bolt circle uses 90-degree intervals; three-hole bolt circle uses 120-degree intervals. Enter the measured diameter through the fixing-hole centres. **Choices:** 4-hole bolt circle; 3-hole bolt circle.
Motor bolt circle diameter	Diameter through the centres of the motor fixing holes. Measure your actual motor; this is not square side length.
Motor hole rotation	Clocking angle for either bolt-circle pattern, relative to local pad axes. At zero, one hole points toward the tip; positive turns counterclockwise in top view. Only fixing holes rotate; the pad and centre bore stay fixed.
Motor Hole Diameter	Diameter of the four motor fixing holes. Match the measured motor threads and required bolt clearance.
Motor centre bore	Central opening below the motor shaft or bearing housing. Keep enough material between this bore, the fixing holes and the pad edge.

Shield outlines and optional neck reliefs

The catalogue includes round, faceted and shield families. Shield variations use shallow circular edge scallops. Choose between swept or curved shoulders, a broad or compact deck, paired side recesses and single or split nose crowns. Corner / chamfer / scallop size controls their scallop depth. They are original parametric families rather than replacement arms for named commercial frames.

Motor pad neck length extends the root-facing connection rather than the outward tip. Optional Neck slot, Twin neck holes and Twin neck slots remove material there, away from motor screws

and the shaft bore. Width / diameter and full slot length remain editable. The neck needs enough length for the opening and required edge material. These optional cuts do not move mounting holes. The two retained extended-tip shapes keep their original tip-slot controls.

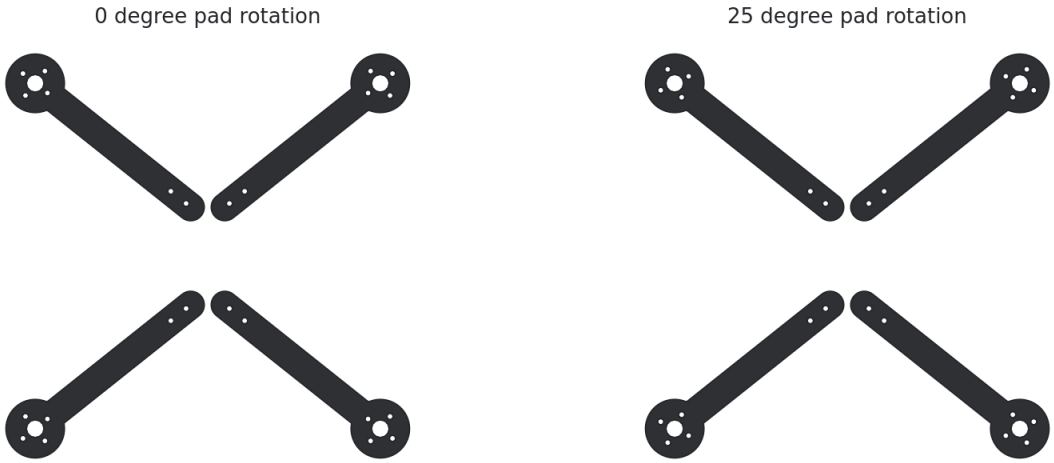


Optional reliefs in the root-facing neck of Round-shoulder shield. The motor bearing and fixing holes are kept clear.

Rotate complete pads with frame symmetry

Motor pad rotation (mirrored), under Arms > Motors, rotates each pad outline, fixing pattern and optional tip slot around its motor centre. Arm root bolts and main beam directions stay fixed. The selected pivot determines whether motor and prop centres move. Zero keeps pads aligned with their own arms. Positive values turn the front-right and rear-left pads counterclockwise in top view, and the front-left and rear-right clockwise. Negative values reverse those directions. The same rule applies to True X, Stretched X and Deadcat, using mirrored rotation signs. Motor centre mode retains the existing motor spans; Pad beginning mode changes the actual motor spans and diagonals.

Motor hole rotation remains a separate adjustment within each pad and uses its own offset. Leave that offset at zero for a fully mirrored three-hole pattern. A manually clocked three-hole offset can intentionally differ from its reflected orientation. With round pads, look at the fixing pattern or tip features to see rotation. The live preview, exact CAD, slots and cutting profiles follow both controls. Check material around holes and the connection to the arm when using large angles.



Plan views: opposite corners share a rotation sign. Motor centres (red) and arm/root positions stay fixed.

Choose where the motor pad rotates

Control	What it does
Motor centre	The default pivot. The outline and its fixing holes turn about the motor shaft centre; motor and prop centres remain fixed.
Pad beginning	The pivot sits at the root-facing extent of the unrotated pad outline, measured along the original arm axis. Shapes with a longer neck therefore pivot farther inward. The motor centre swings around this fixed point, carrying its holes, centre bore, nose slots and motor/prop references.

Rounded connections on rotated pads

Rotated pad transition radius adds a planar rolling-radius blend at inward corners between the beam and a rotated pad. It adds local supporting material, retains square thickness edges, and works at either pivot with mirrored rotation. Zero keeps the original junction. Reduce the radius if the builder cannot construct the blend.

Choose Motor pad rotation pivot under Arms > Motors > Pad outline, then adjust Motor pad rotation (mirrored). The beam retains its root and original axis, ends at the pad beginning and joins the moved pad through a continuous full arm-width elbow. Both the preview and the carbon CAD use this connection, including widened roots and their taper. At zero rotation, both pivot modes produce the same arm. Geometry checks reject disconnected arms and roots which leave too little room for the pad transition.

In Pad beginning mode, the actual wheelbase diagonals, prop clearance and motor spans can change even though the entered base wheelbase stays the same. Review the moved prop discs and minimum gap after turning the pads. Cutting profiles and exports include the same connected shape, transformed fixing holes, pivot coordinates and motor positions. Existing projects keep Motor centre until you select the new mode.

Rotate a bolt-circle pattern

Under Arms > Motors, select a bolt-circle pattern and adjust Motor hole rotation. The angle follows each arm/pad direction in every layout. Use 45 degrees to turn a four-hole cross into square corners, or any signed angle to clock a three-hole pattern. Circle diameter stays constant: a 16 x 16 mm square needs about 22.627 mm bolt-circle diameter. Check hole support in the preview before building or exporting.

Motor and prop sizing actions

Arms > Motors includes Apply motor / prop sizing. Choose a 9, 12, 16 or 19 mm bolt-circle diameter to fill four-hole mounting geometry, pad width, fixing/centre bores, reference motor dimensions and a corresponding nominal prop size. A separate 9 mm bolt-circle / 1103 / 2-inch choice supports small motors. These sizing actions do not change the motor centres or silently resize the whole frame. Adjust wheelbase if the new props overlap. All dimensions remain editable.

Control	What it does
9 mm bolt circle	Nominal 1404 motor body, 20 mm pad, M2 clearance holes and 3-inch props.

Control	What it does
12 mm bolt circle	Nominal 1804 motor body, 25 mm pad, M2 clearance holes and 4-inch props.
16 mm bolt circle	Nominal 2207 motor body, 30 mm pad, M3 clearance holes and 5-inch props.
19 mm bolt circle	Nominal 2806.5 motor body, 36 mm pad, M3 clearance holes and 7-inch props.
9 mm bolt circle	Nominal 1103 motor body, 25 mm pad, M2 clearance holes and 2-inch props. Four holes at 90-degree intervals.

Simple references are approximate clearance cylinders; optional detailed shapes are on Finishing / Detailed references. Neither style is a branded CAD model or a motor/prop performance recommendation. Motor stator size does not uniquely define bolt pattern, bell size, KV, battery voltage or prop compatibility. Measure your actual parts and edit the reference dimensions. The simple prop reference is a full swept disc; the detailed option shows three blades. Guard clearance checks retain the swept propeller envelope. Only 4-hole bolt circle and 3-hole bolt circle modes are available. Circle diameter and square pitch are different measurements.

Control	What it does
Motor reference bell diameter	Outside diameter of the approximate motor body/bell reference. Stator numbers such as 1404 do not specify the outside bell diameter. Does not resize mounting holes or the pad.
Motor reference body height	Motor bell body height. The detailed M5 shaft and self-locking nut extend above the bell.
Prop plane above motor mounting face	Height for simple prop discs only. Detailed props sit directly on the motor bell and ignore this dimension.

Mounting examples reviewed: iFlight XING2 1404 (9 x 9 mm), iFlight XING2 1806 (12 x 12 mm), iFlight XING2 2207 (16 x 16 mm), and T-Motor F90 (larger motor family). The sizing actions use nominal bodies rather than claiming exact fit to these products.

Reading the motor-pad catalogue

Catalogue views use zero pad rotation and share scale and colour: the arm continues left and the tip points right. They use 30 mm pads, 14 mm arms, 22.627 mm four-hole bolt circles at 45 degrees with 3.2 mm holes, a 9 mm centre bore and no tip slots. Pad width is nominal; shoulders and noses can extend beyond it. Each card lists its applicable controls. Corner/chamfer/scallop size affects only listed shapes. The original tip slot applies to Tapered nose and Arrowhead. The twelve shield variations use Motor pad neck length and the separate Additional pad cutouts selector. Outlines keep the entered motor centre and fixing pattern.

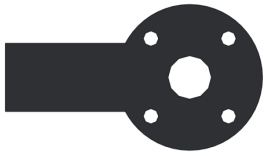
Practical profile references

Commercial arm references reviewed for motor-bearing material, short tips and tapered necks: GEPRC MARK5 frame parts and FlyFishRC Volador spare arms. The shield variations add shallow edge scallops, swept or curved shoulders, and short crowns around a solid mounting deck. They are original outlines rather than dimensioned copies of commercial parts. Measure actual motors and validate the finished design.

Motor-pad catalogue 1 of 4

Arm section to the left; outward motor nose to the right. All views use the same scale and colour.

Round



Circular motor platform with a direct transition into the arm.

Controls: Pad width.

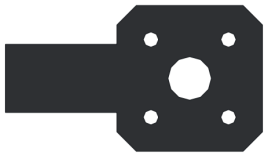
Rounded square



Broad square platform with curved outer corners.

Controls: Pad width and corner radius.

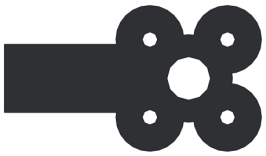
Chamfered square



Square platform with four straight clipped corners.

Controls: Pad width and chamfer size.

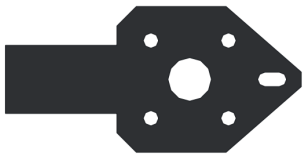
Four-lobe



A central hub with four round ears around the fixing holes.

Controls: Pad width; ear positions follow the motor pattern.

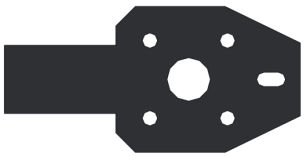
Tapered nose



A long, narrow outward tip with clipped inner corners.

Controls: Width, corner size, tip extension and optional tip slot.

Arrowhead



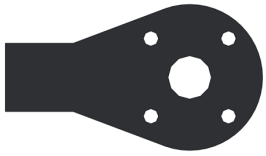
A wider extended tip with sloping sides and clipped inner corners.

Controls: Width, corner size, tip extension and optional tip slot.

Motor-pad catalogue 2 of 4

Arm section to the left; outward motor nose to the right. All views use the same scale and colour.

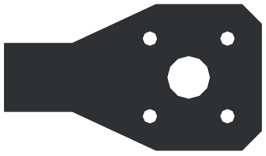
Teardrop



A round bearing platform with a tangent transition into the arm neck.

Controls: Pad width; neck follows arm width.

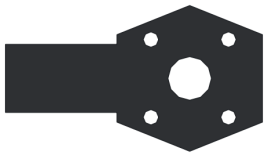
Shouldered



A clipped shield with a tapered neck and broad mounting shoulders.

Controls: Pad width and corner/chamfer size.

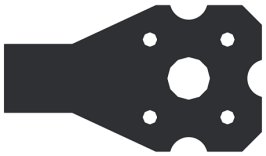
Compact hex



Six straight edges around a compact motor platform.

Controls: Pad width; fixed outline proportions.

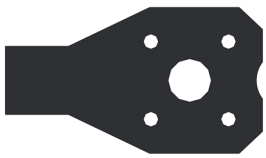
Scalloped shield



A shouldered shield with circular reliefs at both sides and the nose.

Controls: Pad width and scallop/corner size.

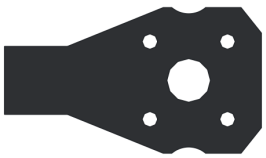
Crown shield



Broad shield shoulders with a single shallow circular recess in the outward nose.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

Twin-scallop shield



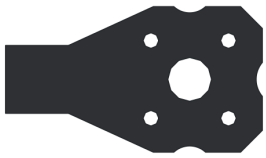
A compact clipped shield with paired recesses in the side edges.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

Motor-pad catalogue 3 of 4

Arm section to the left; outward motor nose to the right. All views use the same scale and colour.

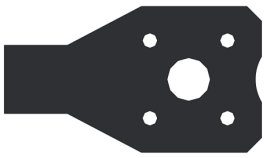
Tri-scallop shield



Three shallow scallops at the sides and nose of a tapered shield.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

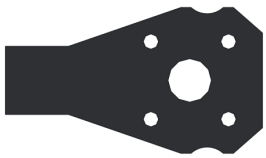
Recessed-nose shield



A wide, shallow nose recess leaves two short protective front corners.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

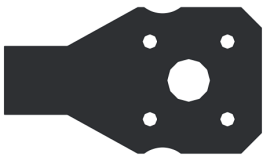
Swept-shoulder shield



Long swept shoulders with paired scallops near the forward side edges.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

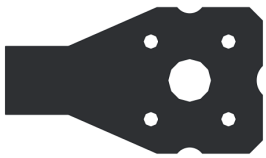
Waist-scallop shield



Rearward side scallops create a waisted transition into the solid mounting deck.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

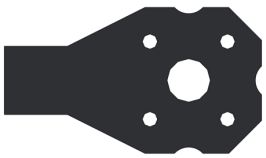
Broad-crown shield



Broad front corners with three small edge scallops and a short flat nose.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

Faceted-scallop shield



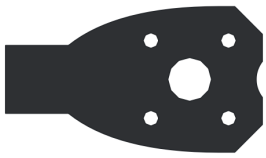
Angular shield facets with three compact circular edge recesses.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

Motor-pad catalogue 4 of 4

Arm section to the left; outward motor nose to the right. All views use the same scale and colour.

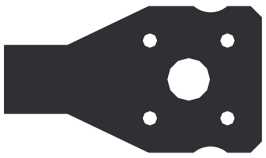
Round-shoulder shield



Curved neck shoulders lead into a shield with a scalloped nose.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

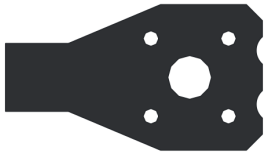
Twin-cheek shield



High front cheeks with paired side scallops and a broad outward edge.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

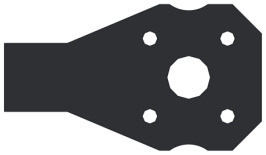
Split-crown shield



Two small outward scallops leave a short central crown between them.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

Compact-scallop shield



A short clipped shield with broad, shallow paired side recesses.

Controls: Pad width, scallop depth (corner size), neck length and optional neck reliefs.

06 FC stack mounting and screw access

Choose one of the two FC stack patterns: 20 x 20 mm or 30.5 x 30.5 mm. Both are centre-to-centre square pitches. Stack spacer height and stack clearance height remain editable.

The stack sits on the bottom in a two-plate design without a clamp or with below-bottom arms. With above-bottom arms and the clamp enabled, the upper clamp carries the stack. In a three-plate design, the mid carries the stack. Stack spacer height is measured above that supporting surface. With FC stack press nuts enabled, the stack starts directly on the exposed flange. The manual spacer height applies only without stack press nuts. This lift is for the inserts; arm clearance still needs to be checked.

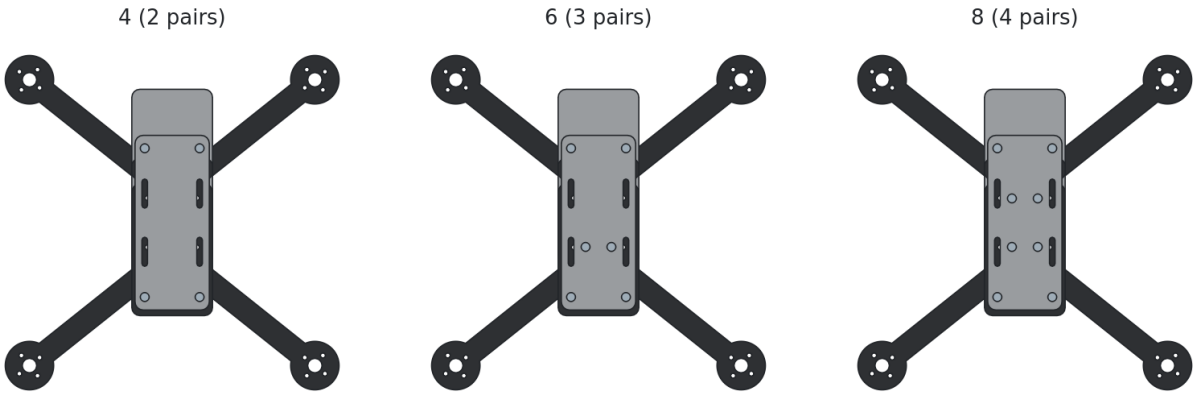
Control	What it does
FC stack mounting pattern	Select 20 x 20 mm or 30.5 x 30.5 mm centre-to-centre FC/ESC screw spacing. Both editor and native project offer these two choices. Stack mount height and stack clearance remain independently adjustable. Older nonstandard numeric settings map to the nearest supported pattern with a warning; review before updating the frame.
Stack hole diameter	Electronics mounting-hole diameter. Match the stack screws and required clearance.
Stack Offset Y	Moves the FC/ESC stack and its mounting holes along frame Y. Positive moves it toward the front.
Stack spacer height	Spacer above the carrier without FC press nuts. With FC press nuts enabled, the stack starts directly on the exposed flange and this value is ignored.
FC / ESC stack envelope height	Height of the FC/ESC electronics envelope above its mounting spacers.
Stack screw access diameter	Default 6 mm service openings beneath the FC carrier: in the lower clamp for below-bottom arms, or in the bottom beneath an upper clamp or mid. The carrier keeps normal mounting holes. Matching arm and keystone reliefs preserve screw access.

Stack screw access diameter makes larger service openings in the plate beneath the carrier. Below-bottom arms use these openings in the lower clamp; an upper clamp or mid uses service openings in the bottom. The actual carrier keeps normal mounting bores, or press-nut bores when enabled. Enable arm-root reliefs if arm material crosses a screw-access position. The key also preserves this access.

Control	What it does
Fit stack height	Raises the linked standoff heights to a common size that clears the stack envelope. It does not change the stack support plate or the explicit spacer height.

07 Standoff counts, positions and heights

Choose four, six or eight standoffs as two, three or four mirrored left/right pairs. Custom positions use pair spacing and Y coordinates. Arm-hole modes snap to the selected inner or outer root bolts and follow later arm adjustments. A nearby hole is not enough: shared holes must be coaxial and have compatible diameters and bearing material.



Extra pairs and shared arm-hole mounts follow the selected support surfaces.

Control	What it does
Number of standoffs	Choose 4, 6, or 8 standoffs in 2, 3, or 4 mirrored left/right pairs. Extra pair positions are adjustable below. **Choices:** 4 (2 pairs); 6 (3 pairs); 8 (4 pairs).
Standoff outside diameter	Outside diameter of the standoff tubes. The mounting surfaces must support their bearing footprints.
Standoff through-hole diameter	Through-hole diameter in the approximate standoff tube and its plate mounting bores. Match the intended screws and shared arm-root holes.
Front pair position	Custom uses the spacing fields. Arm-hole choices snap this pair exactly onto the front arm bolt centres and follow geometry changes. In three plates these standoffs sit on the mid arm tabs. **Choices:** Custom position; Arm inner hole; Arm outer hole.
Rear pair position	Custom uses the spacing fields. Arm-hole choices snap this pair exactly onto rear arm bolt centres. Shared holes require matching diameters and supporting material. **Choices:** Custom position; Arm inner hole; Arm outer hole.
Outer pairs left–right spacing	Left/right spacing of the front and rear custom standoff pairs. Arm-hole position modes instead use the actual mirrored arm bolt coordinates.
Outer pairs front–rear spacing	Front/rear spacing of the two outer custom pairs, measured between pair centres. Arm-hole modes follow the arm bolts instead.

Control	What it does
Pair 3 position mode	Choose a custom pair position or snap it to a front/rear arm inner/outer bolt pair. Near misses and incompatible diameters are rejected. **Choices:** Custom position; Front arm inner hole; Front arm outer hole; Rear arm inner hole; Rear arm outer hole.
Pair 3 position Y	Extra pair centre along Y: negative is rear, positive is front. Keep it between the outer pairs. Its mounting plate determines its length.
Pair 3 left–right spacing	Left/right separation of pair 3 in Custom position mode. Its centre Y is set separately; snapped arm-hole modes override these coordinates.
Pair 4 position mode	Choose a custom pair position or snap it to a front/rear arm inner/outer bolt pair. Near misses and incompatible diameters are rejected. **Choices:** Custom position; Front arm inner hole; Front arm outer hole; Rear arm inner hole; Rear arm outer hole.
Pair 4 position Y	Extra pair centre along Y: negative is rear, positive is front. Keep it between the outer pairs. Its mounting plate determines its length.
Pair 4 left–right spacing	Left/right separation of pair 4 in Custom position mode. It is active only when eight standoffs are selected.
Rear standoff height	Actual rear pair length. Other pairs are linked to keep the top level. Pairs on the mid or combined clamp have different bases from pairs on the bottom. Lengths follow the actual support and keep the top level.

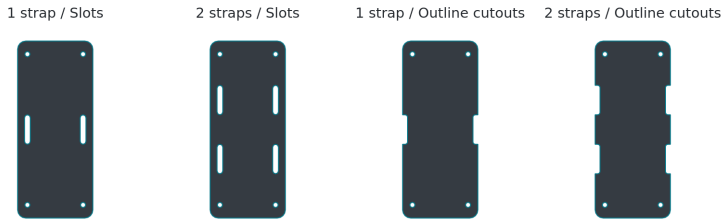
Control	What it does
Front standoff height	Actual front-pair length above its support. Editing it adjusts the rear height to keep the top level.
Common height shortcuts	Choose common hardware lengths such as 22, 25 or 30 mm, or type a custom value. The linked front and rear sizes remain level.
Fit top plate around shared arm mounts	Enlarges top width and length to support the current shared standoff positions. Review the new bounds before building.

A bottom-mounted standoff in a three-plate frame spans the bottom-to-top distance. A mid-mounted standoff is shorter by the intervening arm and mid thickness. Shared arm-hole standoffs sit on the actual clamping surface, including the mid front tabs. Read the per-pair mounting surface and actual length, rather than assuming every front standoff is bottom-mounted.

08 Top-plate strap openings

Hardware > Straps selects Slots or Outline cutouts, and 1 strap or 2 straps. Use the enable checkbox to turn them on or off. Disabled controls retain their dimensions for later use.

Control	What it does
Enable battery strap openings	Enable the selected top-plate opening style; all dimensions remain saved when disabled.
Top plate strap openings	Choose enclosed slots or open left/right outline cutouts while retaining the saved dimensions and positions. Side guards bridge the outline cutouts along the outside plate edge, leaving the strap passages clear. Use the enable switch to omit openings. **Choices:** Slots; Outline cutouts.
Number of top-plate straps	Choose one left/right opening pair or two pairs. Two-strap spacing remains saved when using one strap. **Choices:** 1 strap; 2 straps.
Single strap fore/aft position	Single-strap centre in frame Y: zero is centred, positive is forward, negative is rearward. Used for slots and outline cutouts.



Left to right: one strap with slots, two straps with slots, one strap with edge cutouts, two straps with edge cutouts. Front is upward.

One strap uses its signed position. Two straps use the saved symmetric spacing. Outline cutouts have rounded inner corners. Side guards bridge the unnotched outside edge, leaving the strap passage clear even when separate guard openings are disabled. Mounting, camera-tab and central-web checks remain active.

Control	What it does
Strap Slot Width	Width of each enclosed slot. Used only in Slots mode.
Strap Slot Length	Fore/aft length of each slot or outline cutout. In Slots mode this must exceed slot width.
Slot centre distance from centre	Distance from frame centre to the front/rear slot rows. The four slots form two mirrored left/right pairs.
Slot inset from top plate edge	Inward distance from the top plate side edge to each slot centre. Leave material around the opening and nearby fasteners.
Outline cutout depth	Depth inward from the local top-plate side edge. Outline cutouts use the existing strap length and fore/aft positions. Inner corners are rounded; slot width and inset remain saved for Slots mode.

09 LiPo reference and pad creator

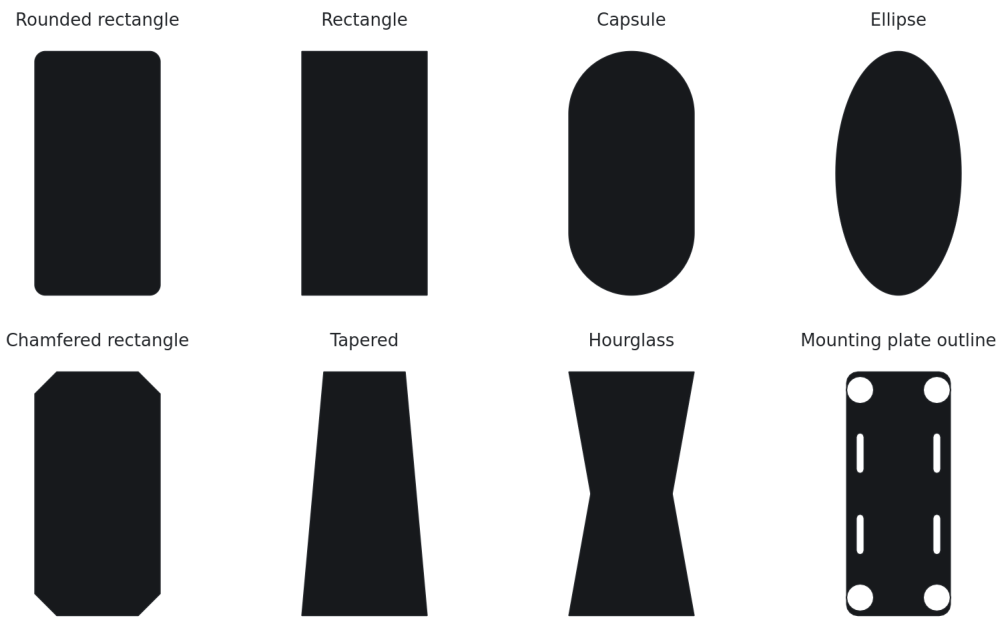
Battery > Placement starts with an editable nominal 1300 mAh 6S reference: 75 x 38 x 38 mm and 208 g. These starting values match the manufacturer specification for the Tattu R-Line 3.0 example, not every 1300 mAh pack. The default reference is a simple clearance box. Optional Detailed LiPo reference adds a shrink-wrapped pack, label, seams, schematic leads and plug. Dimensions refer to the pack body; the displayed leads extend beyond it and do not define routing or swollen-pack clearance. Measure your actual battery and edit its dimensions and mass. Show references controls its visibility together with motors, props and other envelopes.

Choose Top plate or Bottom plate and set X/Y offsets and rotation. Positive X is right, positive Y is forward, and positive rotation turns counterclockwise in top view. The battery and pad share this placement. The battery rests on the pad if enabled, or directly on the selected carbon face. Below-bottom arms or capture plates can restrict the available underside space; resize or reposition the pad if CAD reports an intersection.

Control	What it does
Battery mounting face	Top plate places the pad and battery above the top carbon face. Bottom plate places them below the bottom carbon face, keeping the pad between plate and pack. **Choices:** Top plate; Bottom plate.
Battery / pad X offset	Shared left/right placement of the battery and pad in frame coordinates; positive X is right. The frame carbon remains fixed.
Battery / pad Y offset	Shared front/rear placement of the battery and pad in frame coordinates; positive Y is forward.
Battery / pad rotation	Rotate the battery footprint and pad together around their shared centre in the frame XY plane. Zero places the longer local dimension along frame Y; positive turns counterclockwise in top view.
Show LiPo reference	Display the editable nominal 1300 mAh 6S battery envelope when Show references is on. It is excluded from manufactured parts and bare-frame mass; its mass is used only in the separately labelled frame-plus-battery estimate.
Battery width	Battery envelope width in its local X direction. Default 38 mm; capacity alone does not define pack dimensions.
Battery length	Battery envelope length along local Y before rotation. Default 75 mm.
Battery height	Battery envelope height. Default 38 mm. The pack rests on the optional pad or directly on the chosen plate.
Battery reference mass	Entered reference pack mass, default 208 g. Added to both the bare frame plus battery and equipped plus battery estimates. Weigh the actual pack.

Create a separate pad

Battery > LiPo pad creates a separate material part. Choose a rounded rectangle, rectangle, capsule, ellipse, chamfered rectangle, tapered or hourglass outline; set its width, length, thickness and, where applicable, corner radius. Pad dimensions are independent from the battery dimensions. Follow mounting plate clips the pad to the actual finished carrier, including holes, strap slots and applied cutouts. Turning it off retains the independent outline but still requires contact with the carrier. Empty, disconnected or intersecting pads are rejected.



Eight LiPo pad outlines, including a copy of the finished mounting plate. Existing outline choices remain available.

Mounting plate outline follows the carrier side profile. Plate outline width scale narrows it across frame X. Pad front reach and Pad rear reach are independent millimetre distances from the FC stack centre, at Stack Offset Y. For example, front 25 mm and rear 35 mm give limits at stack Y + 25 and stack Y - 35. Decreasing either reach cuts that end straight across the frame; it does not compress the plate shape along Y. Zero retains the full carrier at that end. Pad width and Pad length are disabled in this outline mode. All grip patterns now cover the complete finished pad automatically, including narrowed sides and independently trimmed ends.

Every pad leaves at least 6 mm radial clearance measured from each standoff-hole centre (a 12 mm diameter exclusion). These access regions stay at their actual positions and can create edge notches. Carrier openings stay aligned. A pad must remain supported and connected. Older length percentages are converted to front/rear reaches when old settings or documents are loaded. Scaled-pad spline edges in DXF/SVG use line segments with 0.01 mm chord deflection; STEP retains CAD curves.

Control	What it does
Create LiPo pad	Create one separate pad solid, not carbon. It is included in the native assembly and assembled STEP, and exported separately as flat STL/STEP and an ungrooved DXF/SVG outline. Every pad automatically leaves at least 6 mm radial space from each standoff-hole centre. This also applies to independent pad shapes; disconnected results are rejected.

Control	What it does
Pad outline	Choose Rounded rectangle, Rectangle, Capsule, Ellipse, Chamfered rectangle, Tapered or Hourglass for an independently sized pad. Mounting plate outline follows the finished carrier sides and openings. Adjust Plate outline width scale for width and independent Pad front reach / Pad rear reach from the stack centre for length. Shortened ends are straight. Zero reach retains that full carrier end. Every pad leaves a 6 mm clear radius around standoff-hole centres. **Choices:** Rounded rectangle; Rectangle; Capsule; Ellipse; Chamfered rectangle; Tapered; Hourglass; Mounting plate outline.
Pad width	Width of an independent pad before carrier clipping. Disabled for Mounting plate outline: set width with the outline scale. Grip coverage always uses the finished pad footprint; this value does not crop the grip region.
Pad length	Length of an independent pad before carrier clipping. Disabled for Mounting plate outline: set its length with independent front/rear reaches. Grip coverage always uses the finished pad footprint.
Pad thickness	Overall pad thickness. Grooves cut into the battery-facing surface and must leave a continuous backing. This height also lifts or lowers the battery reference away from the carbon face.
Pad corner radius	Corner radius for Rounded rectangle. Zero makes square corners; positive values must be smaller than half the shorter pad dimension. Use Capsule for fully rounded ends.
Follow mounting plate	Clip an independently sized pad to the finished carrier footprint. Mounting plate outline always follows the carrier. All pad shapes retain a 6 mm clear radius around standoff-hole centres; empty or disconnected pads are rejected.
Plate outline width scale	Width percentage of the carrier outline, from 1 to 100 percent. 100 keeps full width. Narrowing scales only frame X about the carrier centre and clips back to the carrier. The front/rear profile is not scaled; use separate reaches from the stack to cut straight ends. Actual openings and 6 mm standoff clearances remain at their real positions.
Pad front reach from stack (0 = full)	Independent distance in millimetres from the FC stack centre (Stack Offset Y) toward frame +Y. Zero retains the full carrier at this end. A shorter positive distance clips the pad at a straight transverse edge, without rescaling the plate profile along its length. The opposite end is unchanged. Width still follows the carrier outline. Used only with Mounting plate outline.
Pad rear reach from stack (0 = full)	Independent distance in millimetres from the FC stack centre (Stack Offset Y) toward frame -Y. Zero retains the full carrier at this end. A shorter positive distance clips the pad at a straight transverse edge, without rescaling the plate profile along its length. The opposite end is unchanged. Width still follows the carrier outline. Used only with Mounting plate outline.

Control	What it does
Grip surface	Choose one of 25 grip patterns shown together in the catalogue. Ten additions provide brick joints, herringbone, plus/X grids, square/diamond islands, triangular lattice, ladder grooves, interrupted ribs and fish scales. Each fills the finished pad and is clipped to its outline and clearances. Width, depth and pitch remain editable.
Groove width	Width of each shallow grip channel, measured across the channel.
Groove depth	Depth into the battery-facing surface. Must be smaller than pad thickness so the backing stays continuous.
Groove spacing	Spacing between repeated grip features. Must exceed groove width. Pattern count is based on the finished pad, with a maximum of 512 channel segments; increase spacing if a pattern is too dense. Ring spacing is the inset between successive rectangular rings.
Pad material density	Density used to convert exact pad volume into its mass estimate. Default 1.2 g/cm3 is a nominal solid TPU value. Enter the actual material/effective printed density; infill and adhesive are not modelled.

Export frame includes the pad in FrameAssembly.step and creates a separate LiPoPad folder containing assembled and flat STEP, a flat STL, and an ungrooved DXF/SVG cutting outline. The STL rests at Z = 0 with the grip face upward. The outline represents through cuts only; shallow grooves are carried by STL/STEP. Pad and reference battery profiles are excluded from carbon nesting sheets. The reference battery is excluded from manufacturing exports.

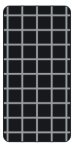
Sizing reference: Tattu R-Line 3.0 1300 mAh 6S manufacturer specification. The reference box is for placement, not a battery model or electrical/performance recommendation.

Grip pattern catalogue - all 25 surfaces

Every pattern fills the finished pad. Width, depth and spacing are shared controls; all grooves leave a continuous backing. The examples use the same pad, scale and colours.


Solid


Transverse grooves


Cross grooves


Longitudinal grooves


Diagonal grooves


Diamond grooves


Chevron grooves


Zigzag grooves


Rectangular rings


Dimple grid

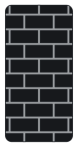

Staggered dimples


Dash grid


Basket weave


Hexagon cells

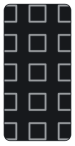

Wavy grooves


Brick joints


Herringbone

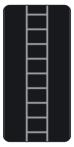

Plus grid


X grid


Square islands


Diamond islands


Triangular lattice


Ladder grooves


Interrupted ribs


Fish scale grooves

10 FPV camera cage, tilt and captured tabs

Enable FPV camera drone on Frame > Layout to add the camera side plates and rear VTX mounting holes. Non-FPV designs omit these parts and holes. Video system selects the reference category; measure the actual camera width, body, side screws and VTX instead of relying on a product name.

The two camera side plates are captured between matching sockets in the bottom and top. Upper and lower contact positions can differ, allowing the side walls to slope when the top is shorter. Choose a camera recess under Frame > Top plate when the lens needs uptilt clearance, while retaining the supporting tab sockets. Straight leaves the front solid.

Body dimensions and side-screw pattern

Control	What it does
Video system	Analog and digital cameras can share widths. Select the actual width, body dimensions and screw pattern below; this label does not force one camera model. **Choices:** Analog; Digital.
Camera body width	Actual camera width between side faces. Common-size shortcuts set body height, centre the camera and choose taller stock standoffs if necessary.
Camera body height	Measured camera body height, excluding the lens. Used to check the tilted body envelope and automatic vertical placement.
Camera body depth (without lens)	Reference body depth excluding the approximate 6 mm front lens. Measure your camera for clearance checks.
Extra width clearance (total)	Total extra width added across the camera cage, shared between the two sides. It is separate from the camera body width and plate thickness.
Camera side screw hole diameter	Through-hole or slot width for the camera side screws. Match the camera threads and intended clearance.
Vertical screw spacing (0 = pivot)	Vertical centre-to-centre spacing on one camera side, measured with the camera level. Use 6.5 mm for that measured pattern, a custom pitch, or zero for a single pivot screw. Camera body width is independent.
Camera side plate thickness	Stock thickness of each camera side plate. Side plates flatten into separate XY cutting profiles during export.

Regular two-screw FPV cameras use a vertical pair of side holes when level. Vertical screw spacing is independent of camera width; for example, a 6.5 mm vertical pitch does not describe the width of a mini camera. Zero spacing selects a single friction pivot. Width shortcuts are starting dimensions, not a guarantee that a digital camera or mounting unit matches every mechanical feature.

Fixed tilt, variable tilt and camera position

Control	What it does
Camera tilt mount	Variable uses a single-screw friction pivot, or a vertical screw pair in upper/lower arc slots. Fixed uses two vertically spaced round holes at the chosen uptilt. **Choices:** Fixed; Variable.
Variable tilt mechanism	Vertically stacked side screws use centred upper/lower arc slots. The alternative fixes the upper screw and moves the lower screw along an exact arc. Enter the measured vertical pitch. **Choices:** Dual arc slots; Upper pivot + lower arc.
Camera uptilt	Displayed camera angle. In Variable mode it must be inside the adjustment range; changing the preview angle does not change the bracket or arc slots.
Minimum adjustable tilt	Lowest intended adjustable angle. Clearance is checked across the configured range. Single-screw pivots rely on friction and have no mechanical end stops.
Maximum adjustable tilt	Highest intended adjustable angle. Two-screw mounts have arc-slot end stops. Fit camera height after changing the range.
Camera fore/aft placement	Centre the camera on the actual side plates at its configured height, or choose a custom camera centre Y. Height can be centred or set independently. **Choices:** Automatic centre; Custom position.
Custom camera centre Y	Custom nominal body centre Y (+Y front). With an upper pivot this is the centre at the middle of the variable range. Automatic centres it on the actual side plates.
Camera forward / backward offset	Additional camera translation in frame Y: positive moves toward the front, negative toward the rear. Added to Automatic centre or Custom position. Moves the camera, fixing holes/slots and flexible interfaces together; captured tabs and tilt settings remain fixed. Automatic travel support can grow the middle side-wall bearing area; disable it to retain a fixed wall outline. Actual component and material clearances still apply.
Support camera travel automatically	Extend only the middle bearing area of each camera side plate when fore/aft travel needs more screw, slot or TPU flange support. Captured top/bottom tabs and their plate sockets keep their current positions. Disable to keep a fixed wall outline. Real body, edge and height clearances remain checked.
Camera height placement	Centre the nominal body and mounting pattern halfway between bottom/top, or set a custom height. Fit camera height raises the standoffs when needed. **Choices:** Automatic centre; Custom height.
Custom camera centre height above bottom	Custom camera centre height above the upper face of the bottom plate. Used when camera height placement is Custom.

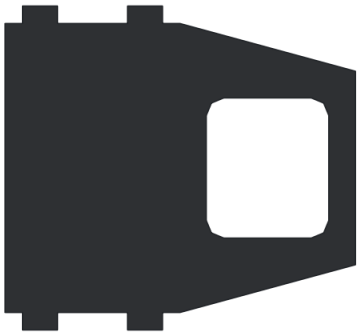
Fixed tilt places the holes at the requested angle. Variable tilt can use dual centred arc slots, which keep the camera body centred, or an upper round pivot with a lower arc, which moves the body around the upper screw. A single-screw friction mount has no mechanical end stops. Check the configured range against the actual camera body and lens.

Camera forward / backward offset adds a relative adjustment to either Automatic centre or Custom position. Positive values move toward the front (+Y); negative values move toward the rear. The reference camera, side holes or tilt arcs, and any flexible pads/inserts move together. Support camera travel automatically grows the middle bearing area of each carbon wall when screws, tilt arcs or TPU flanges need more support. Its top/bottom attachment edges, tongues and plate sockets stay fixed, as do the tilt mechanism, angles and height. Disable it to retain a fixed wall outline. Zero keeps the existing placement wherever support already fits. Use the top-front profile and plate reach to provide clearance over the chosen tilt range. Real body, height, propeller and material clearances still apply, so there is no universal safe travel distance. Native checks cover 10 and 20 mm forward travel with default camera dimensions in all three attachment modes.

Camera offset: 0 mm



Camera offset: 20 mm

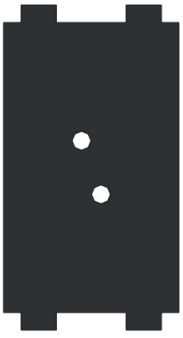


Captured-insert side plates at zero and +20 mm travel: the middle bearing area grows while the top/bottom locating tabs stay at their original positions.

Single side screw

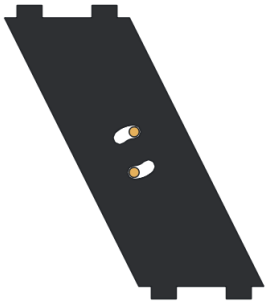


Vertical 6.5 mm side pitch

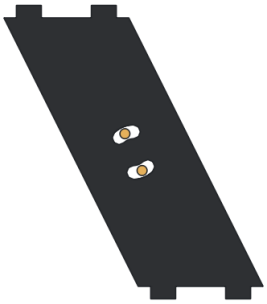


Vertical side-screw spacing and camera width are separate measurements.

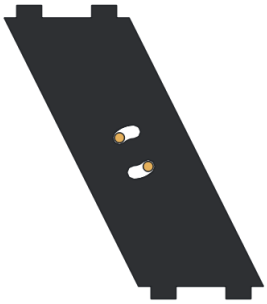
Tilt: 0 degrees



Tilt: 25 degrees



Tilt: 45 degrees



The same dual slots allow different tilt angles. Orange markers show the two vertically spaced camera screws moving through the slots.

Side walls and attachment contacts

Control	What it does
Side plate minimum length	Minimum fore/aft length of each camera side wall. Following the plate ends and fitting the tabs can make the actual wall longer or sloped.
Side plates follow actual plate ends	Fit upper and lower tongues independently inside the actual plate outlines and away from fasteners. A shorter top plate produces sloping side walls. Custom reaches remain exact.
Preferred side-plate centre Y	Preferred cage centre, before fitting the separate upper/lower attachments to the actual plate ends. Camera position within the cage is controlled independently.
Upper attachment rearward offset	Additional distance in millimetres to move only the upper side-plate contact and its matching top-plate sockets toward the rear (frame -Y), after the existing plate fitting. Zero preserves previous behavior. Lower contact, camera centre and camera tilt settings stay unchanged. Works with plate-end following enabled or disabled. The wall slopes to connect the contacts. Excessive offsets are rejected when top support, screw/slot material or hardware clearance is lost.
Lower attachment sizing	Automatic uses the fitted side-plate length at both heights. Custom adds independent front/rear length only to the lower edge resting on the bottom plate, including matching tongue positions and receiving slots. **Choices:** Automatic; Custom.
Lower attachment front extension	Extra contact length toward the nose (+Y), beyond the normal lower edge, in Custom sizing. The top attachment length stays the same. Automatic bottom reach grows when needed; Custom plate reaches stay exact.
Lower attachment rear extension	Extra contact length toward the stack (−Y), beyond the normal lower edge, in Custom sizing. The bottom attachment refits around fasteners; check the live fit. Automatic camera placement follows the resulting cage.
Tab length along frame	Length of each camera mounting tongue along the frame. Matching receiving sockets follow the lower and upper attachment positions.
Tab clearance per side	Per-side clearance between the camera plate tongues and the matching rounded slots. The top plate captures the upper tongues when bolted down.
Relieve camera tab shoulders	Round cutter reliefs beside the upper and lower tabs. Removes material from the side-plate body only; tab width and mating sockets stay unchanged.
Camera tab cutter diameter	Milling cutter diameter for tab-shoulder reliefs. Relief depth into the plate is half this diameter: 1 mm cutter gives 0.5 mm depth, 2 mm gives 1 mm. Use the actual cutter diameter.

Control	What it does
Bottom rear tab offset	Independent tab shift along frame Y in mm: positive is forward. Matching sockets and cutter reliefs follow. Camera, side-wall position and tilt stay fixed. Keep the tab inside its contact edge.
Bottom front tab offset	Independent tab shift along frame Y in mm: positive is forward. Matching sockets and cutter reliefs follow. Camera, side-wall position and tilt stay fixed. Keep the tab inside its contact edge.
Top rear tab offset	Independent tab shift along frame Y in mm: positive is forward. Matching sockets and cutter reliefs follow. Camera, side-wall position and tilt stay fixed. Keep the tab inside its contact edge.
Top front tab offset	Independent tab shift along frame Y in mm: positive is forward. Matching sockets and cutter reliefs follow. Camera, side-wall position and tilt stay fixed. Keep the tab inside its contact edge.

Independent camera tab positions

Bottom rear, Bottom front, Top rear and Top front tab offsets move the four locating tabs along frame Y. Positive moves forward; negative moves rearward. Both side plates use the same offsets. Matching bottom/top sockets and cutter reliefs follow each tab automatically. The camera, tilt, wall outline and lower/upper contact edges keep their positions. Zero restores the original automatic spacing. Keep each tab inside its contact edge, with enough separation for the neighbouring tab and cutter; required plate support is checked. These offsets move the tabs, not the complete camera cage.



Teal highlights the locating tabs. Each upper and lower tab can shift independently inside the same carbon wall outline.

Camera tab shoulder reliefs remove circular cutter clearance beside every upper and lower tongue. The cutter circles lie outside the tab: only the side-plate body is relieved, with a depth equal to half the cutter diameter. A 1 mm cutter produces 0.5 mm depth; a 2 mm cutter produces 1 mm. Enter the actual milling cutter diameter. Tab width, tab height and matching bottom/top sockets remain unchanged. The reliefs appear in the live profile preview, CAD model and DXF/SVG cutting outlines. Disable this option to retain the previous square shoulders.



Lower tab details. Graphite is carbon; teal circles indicate the cutter diameter. The rounded clearance enters the body above the shoulder, preserving the tongue below.

Automatic lower attachment sizing follows the camera wall and captured seats. Custom lower front/rear extensions lengthen only the bottom contact edge; its tongues and matching bottom sockets follow that edge. This supports a sloping wall without forcing the same lower and upper contact lengths.

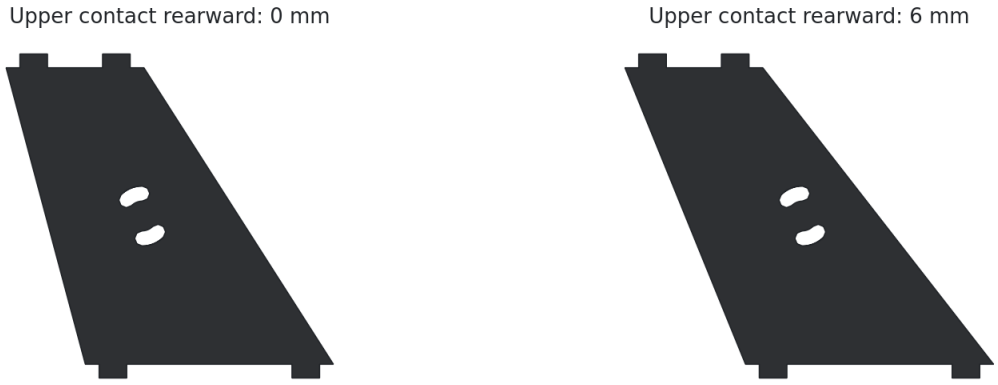


Lower contact extensions let the camera walls reach the bottom while following a shorter or shifted top plate.

Control	What it does
Fit camera height	Centres the camera and raises linked standoffs to a common size when needed to fit its configured tilt envelope. It does not measure the real camera.
Camera width shortcuts	Fill a common body width; use a measured custom width when needed.
Vertical pitch shortcuts	Fill a common two-screw side pitch or choose the single-pivot setting. Width and screw pitch remain independent.

Move only the upper camera attachment rearward

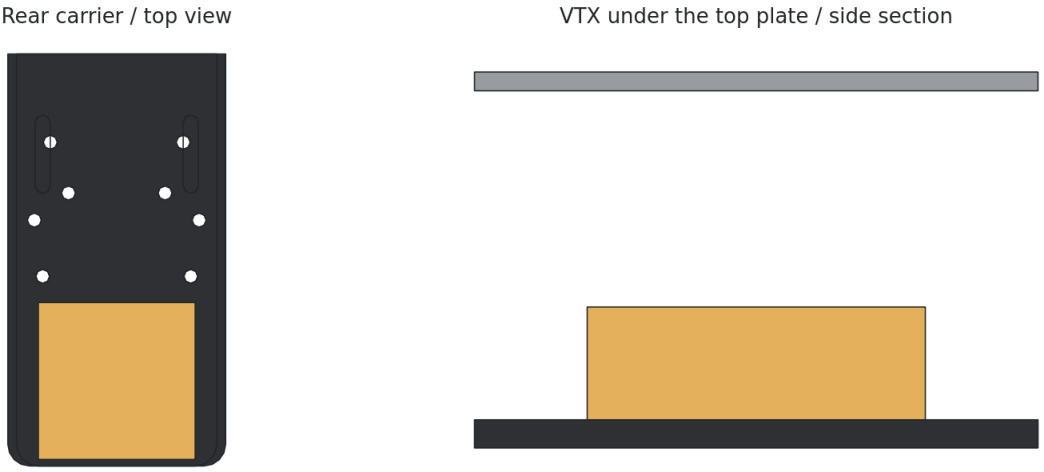
FPV > Side plates > Upper attachment rearward offset moves the top contact and both matching captured sockets back along frame -Y. It is added after the current fitting settings. The lower edge and camera position stay fixed, including automatic camera centring; the side wall changes slope to connect them. Zero keeps the current design. Existing preferred centre, lower extensions, camera position and tilt controls remain available. Reduce the offset if the checks report lost support or insufficient material around a screw or tilt slot.



Left: unchanged upper attachment. Right: upper edge and tongues moved rearward; Rear (-Y) is left; bottom contact and camera screw holes stay fixed.

11 Rear VTX mounting and coverage

The VTX sits directly on the bottom with two plates or on the mid with three. The reference envelope and estimated centre of gravity start at that carrier surface. Select 20 x 20 mm, 25.5 x 25.5 mm or both hole patterns. Both creates eight possible holes; install one matching set of four fasteners. Hole diameter is separate from spacing.



Rear carrier and top-plate coverage in plan, with the VTX envelope in amber. The side section shows the transmitter resting on its carrier below the top plate.

Control	What it does
Rear VTX mounting pattern	Select 20 × 20, 25.5 × 25.5, or both centred patterns. Both makes eight holes; fit one set of four VTX screws. **Choices:** 20 x 20 mm; 25.5 x 25.5 mm; Both (20 + 25.5 mm).
VTX centre position Y	Negative Y is rear. Keep the VTX behind the stack and standoffs; the lower mounting deck extends to support it.
20 × 20 hole diameter	Clearance-hole diameter, independent of spacing. 2.2 mm for M2 or 3.2 mm for M3; match your VTX fasteners.
25.5 × 25.5 hole diameter	Clearance-hole diameter, independent of spacing. Default 2.2 mm for M2. Check material between nested patterns when choosing both.
VTX envelope width	Approximate VTX body width for support and clearance checks. It does not select a mounting-hole pattern.
VTX envelope length	Approximate VTX body length along the frame. Automatic plate reaches provide rear support and top coverage.
VTX envelope height	Approximate VTX body height measured directly above the bottom or mid carrier surface. Check connectors, wiring and heatsinks separately against actual hardware.

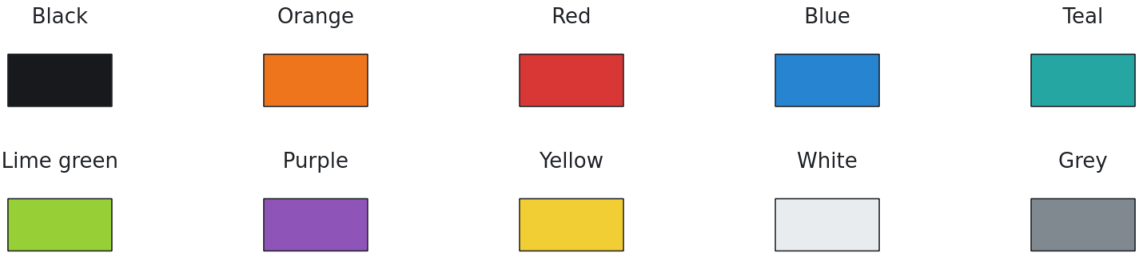
Automatic reaches provide mounting support and top coverage. With Custom reaches, extend the carrier and top if the checks report missing support. For lateral overhang, enable a fully covering VTX clearance opening under TPU > Side guards. Mounting holes still need plate support. Verify the real connectors, heatsink and wiring. The VTX envelope is a clearance reference and is excluded from manufacturing exports.

12 Flexible camera interfaces and protective parts

The TPU category creates separate camera isolation parts, motor guards and electronics side guards. Create / Update builds exact solids; editor previews show lightweight footprints and face views without building solids after every adjustment. All protection is optional.

Colour the printed parts

Open TPU > Appearance and choose Printed part colour. Black, orange, red, blue, teal, lime green, purple, yellow, white and grey apply together to camera compression pads or captured inserts, motor protectors, both electronics side guards and the LiPo pad. Carbon and electronics references keep their own appearance. The selector changes display colour, not geometry, mass, grip texture or material density. STL remains a geometry-only print file; choose the physical filament or silicone colour separately.



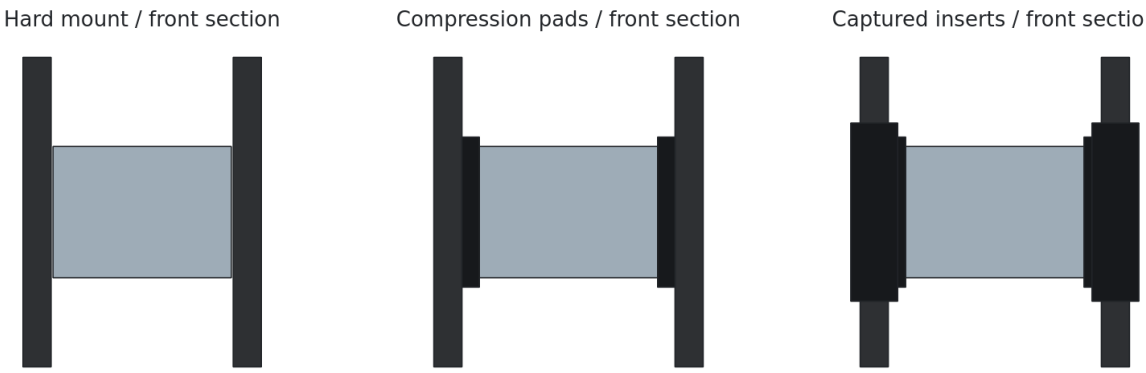
One shared palette for all printed and soft parts.

Control	What it does
Printed part colour	Shared matte display colour for camera isolation pads/inserts, motor protectors, electronics side guards and the LiPo pad. Changes appearance only; dimensions, mass and cutting/export profiles are unchanged.

The overall and individual-part previews use the selected colour. Create or Update applies it to the 3D model and saves it with the frame. Changing colour alone reuses every component and keeps an already calculated weight/CG result. Direct changes in Frame Parameters use the same appearance setting.

Camera attachment without changing the tilt controls

Hard mount keeps the original carbon side plates and fixing paths. Compression pads add one flexible pad between each camera side and its carbon wall. Captured inserts replace the screw openings in each carbon wall with an adjustable rounded window; the printed insert has a neck through that window and retaining flanges on the selected faces. The existing fixed holes or tilt arc slots are formed in the insert. Keep using the existing FPV tilt, angle range, vertical screw pitch and position controls.



Front sections compare hard mounting, compression pads and captured inserts. Graphite denotes carbon, dark grey the soft interface and light grey the camera body.

In either soft mode, each side gap is pad thickness minus the selected compression. The ordinary Camera side clearance applies to Hard mount only. The free printed pad extends slightly into the nominal camera reference by the compression allowance; this is intentional preload. The pad or insert must contain all screw positions over the current tilt range and stay within the supported carbon wall. Increase the pad/window size or cage dimensions if a check reports missing material.

Control	What it does
Camera attachment	Hard mount retains the original side plates. Compression pads widen the cage for two pressed pads. Captured inserts replace the side-plate screw openings with retained flexible inserts; tilt settings and screw paths stay unchanged. **Choices:** Hard mount; Compression pads; Captured inserts.
Camera interface outline	Choose a rounded rectangle or a scaled plain side-wall silhouette, without locating tabs. Captured openings and matching TPU necks use the insert corner radius. Compression-pad outlines retain their existing geometry. **Choices:** Rounded rectangle; Side plate outline.
Camera pad thickness	Free thickness of each pad between the camera and side plate. Soft modes set their own side spacing; the ordinary Camera side clearance applies only to Hard mount.
Camera pad compression	Per-side compression allowance. Printed free material extends this far into the nominal camera side; adjust after a test print.
Camera pad length	Compression-pad size along the frame front/rear direction; captured inserts use the opening size and retaining lip instead.
Camera pad height	Compression-pad vertical size; must contain the complete existing tilt screw paths with supporting material.
Insert opening length	Length of the carbon opening along frame Y, using the selected camera interface outline.
Insert opening height	Vertical opening size in the carbon camera side plates. The original tilt holes and arcs are cut in the insert instead.

Control	What it does
Insert opening corner radius	Round the carbon opening and matching TPU neck, including Side plate outline shapes. Local radii reduce on short edges to keep the rounded profile connected. Tilt screw paths are unchanged.
Insert retaining lip	Flange overlap around the carbon opening on the selected retaining-lip faces.
Retaining lip faces	Inside faces the camera; outside faces away from it. Select a retaining flange on either face or both. The soft camera contact and through-plate neck remain in every mode. **Choices:** Both sides; Outside only; Inside only.
Insert fit clearance	Radial clearance between the printed insert neck and its carbon opening. Positive values loosen the fit.
Flexible material density	Nominal printed TPU/silicone density in g/cm3 for volume-based mass estimates; adjust for your actual print/material.

Choose the flexible interface outline

Rounded rectangle uses a rectangular interface with rounded corners. Side plate outline scales the plain camera wall silhouette to the selected pad length/height or insert opening dimensions, centred around the camera mounting pivot. This preserves the wall shape and slope without copying its full size or the tabs captured between the top and bottom plates. The neck, matching carbon opening and retaining flanges use this outline together. Existing fixed holes and variable tilt slots remain unchanged; enlarge the interface if the scaled shape leaves too little material around them.

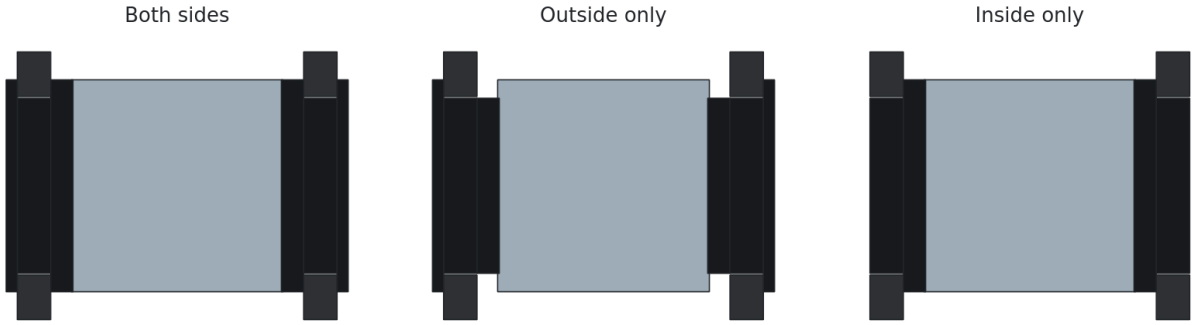


Both captured-insert profiles have rounded corners. Dark grey is the insert; teal traces the matching carbon opening. The side-wall silhouette omits the locating tabs.

Rounded captured openings and retaining lips

Under TPU > Camera isolation > Captured insert, set Insert opening corner radius for the carbon opening and matching TPU neck. The default 1 mm radius suits a 2 mm cutter; 0.5 mm suits a 1 mm cutter. Short outline edges can limit the local radius, so choose a cutter that fits the smallest resulting corner. CAD and DXF/SVG profiles retain circular corner arcs. Compression pads and hard mounts keep their existing attachment behaviour.

Retaining lip faces offers Both sides, Outside only or Inside only. Inside faces the camera; outside faces away from it, mirrored correctly on the left and right plates. Both sides is the default. Outside only keeps a soft contact body against the camera and a neck through the opening, but omits the inner flange. Inside only keeps the inner flange and neck without an outer flange. Insert retaining lip controls overlap on whichever faces are selected. Reference camera screws move to the selected outer contact surface; camera position, height and fixed/variable tilt paths stay unchanged.

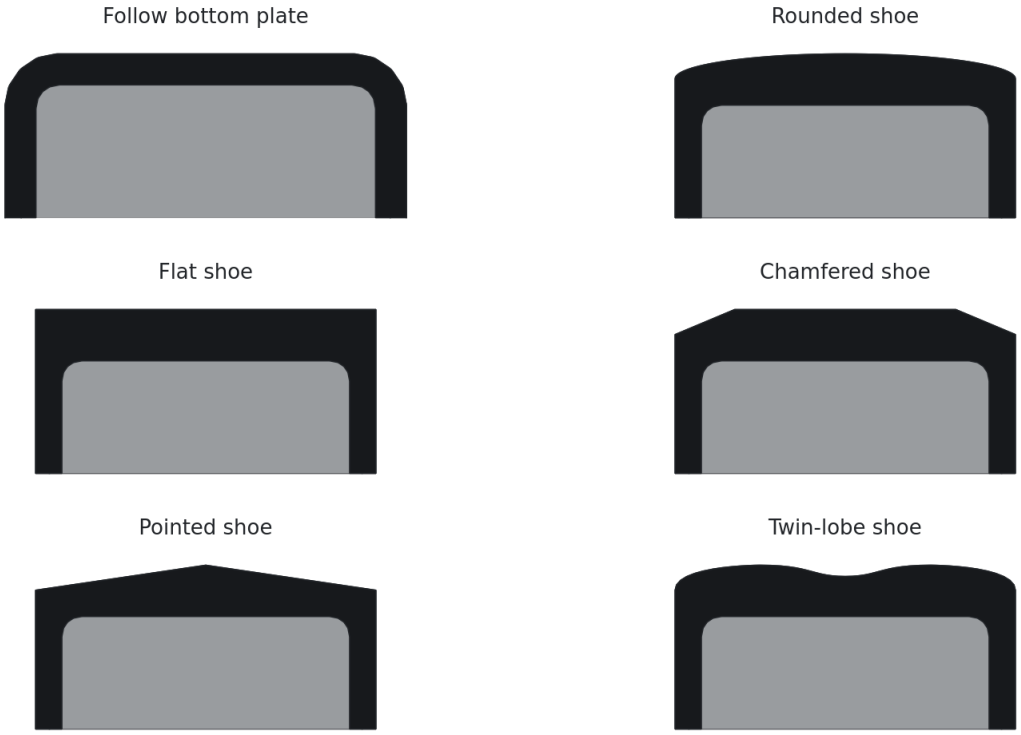


Sections through both side plates: light grey camera, graphite carbon and dark grey TPU. The contact body and neck remain in all three choices.

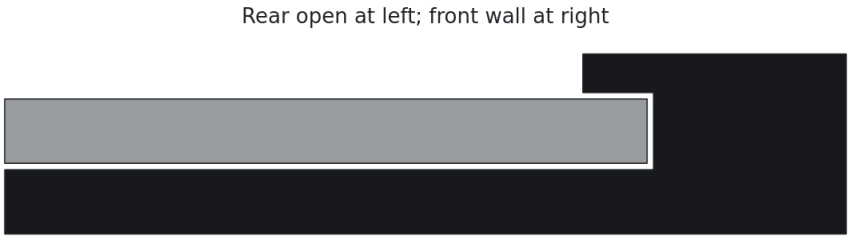
Choose TPU stiffness or silicone to suit the camera and make a fit sample. A flexible interface does not guarantee vibration isolation: over-tight screws, bottomed-out screw threads, or hardware bridging directly to carbon can bypass it. Use suitable screw lengths and compression. Test retention at every working tilt angle.

Bottom-plate front bumper

Enable Create front bumper under TPU > Front bumper to add one slide-on shoe around the finished bottom-plate nose. A sole sits underneath the carbon, a wall protects the edge, and an upper retaining lip captures the plate. The rear is open for fitting. The carbon plate and its existing front-edge profile are not changed. This is a separate printed part in both two- and three-plate frames.



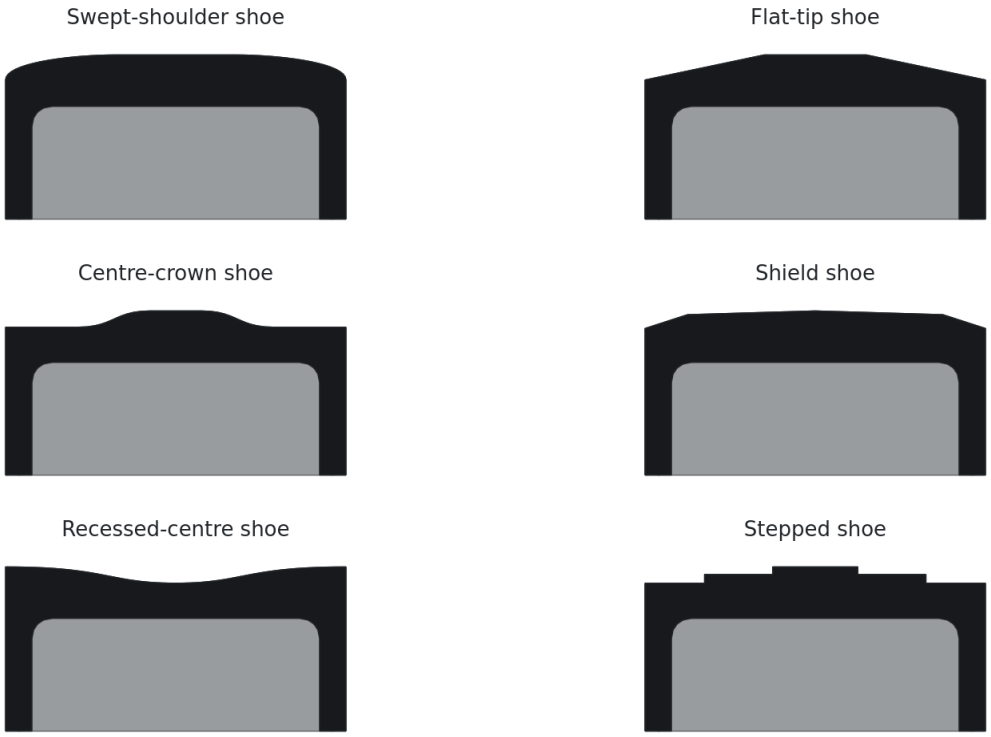
Six core bumper fronts. Light grey is carbon and dark grey is TPU. Follow bottom plate uses the actual outline; the other styles add a protective toe.



Fore/aft section: sole below the plate, fit gap around it, edge wall at the front and retaining lip above. Test the fit with your actual TPU and carbon.

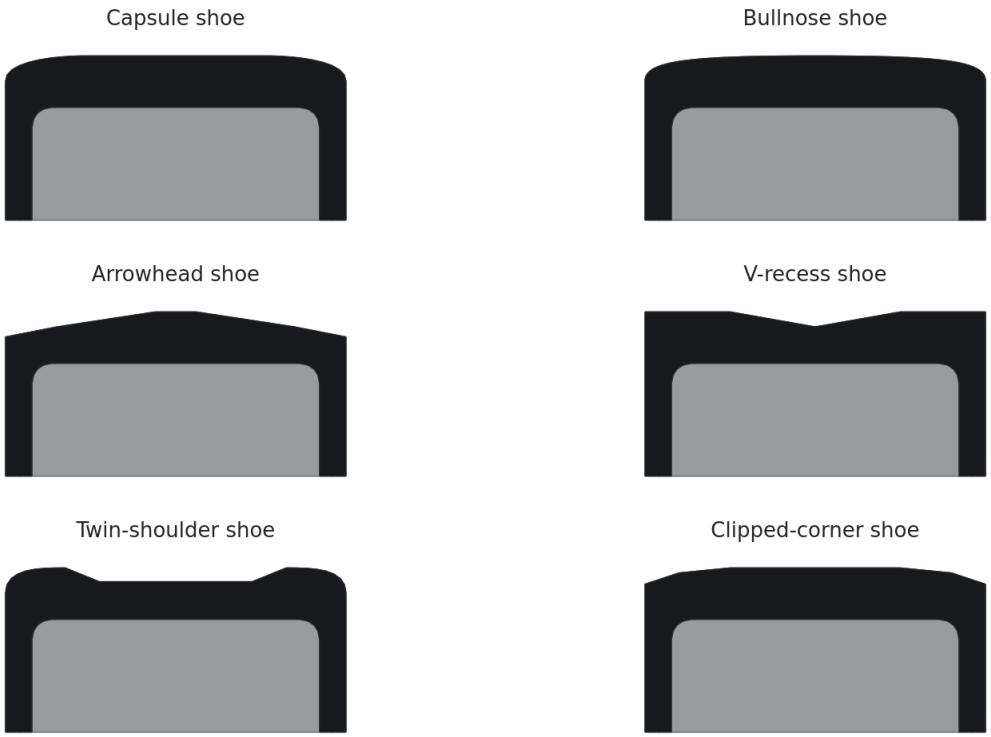
Additional bumper toe profiles

Eighteen bumper shapes include broad swept shoulders, compact flat tips, a centre crown, a faceted shield, a recessed centre and stepped shoulders. Capsule shoe uses a broad flat with rounded ends; Bullnose shoe is a full shallow curved nose. Arrowhead shoe adds a small central flat with swept sides. V-recess shoe leaves an angular centre recess. Twin-shoulder shoe combines rounded outer pads with a recessed flat centre. Clipped-corner shoe uses successive angled shoulders and a flat tip. All toes remain joined to the plate grip through the continuous external shoulder transition.



Six further toe profiles with swept, crowned, shield, recessed and stepped fronts.

Rounded, angular and recessed bumper toes

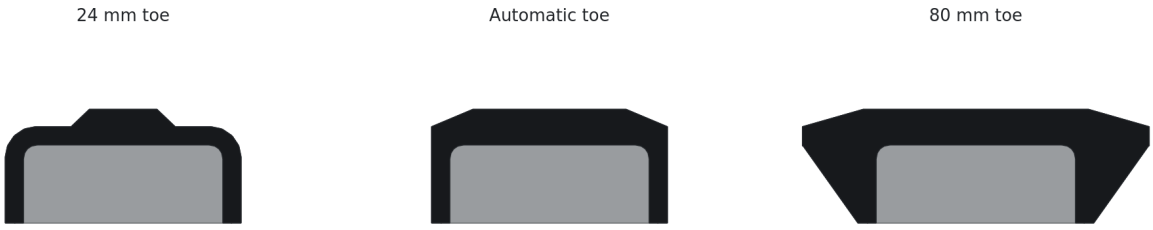


Six further profiles: Capsule, Bullnose, Arrowhead, V-recess, Twin-shoulder and Clipped-corner. Dark grey is TPU; light grey is carbon.

Toe feature width changes the central flat or recess on supported styles, as a percentage of toe width. Toe recess depth changes the cut-back into the added projection on the four recessed styles. Both use the shape proportions at 0. These controls never cut into the plate grip.

Bumper width and continuous shoulders

Bumper toe width sets the full left-to-right width of a shaped protective toe. Set it to 0 for automatic sizing from the plate grip, or enter at least 8 mm for a custom width. Wider toes provide broader front coverage; narrower toes taper into the grip. The external shoulders blend continuously into the toe, preventing side recesses at the join. The toe outline itself can still have deliberate central recesses, scallops or steps.



Narrow, automatic and wide toes on the same carbon plate. The grip retains its original plate fit while the outside connects continuously to the chosen toe width.

Width and extra projection apply to the shaped shoes. Follow bottom plate keeps the actual carbon outline and ignores both controls. Toe width does not scale the plate cavity, wall clearance or retaining lip. A narrow toe can therefore be narrower than the grip behind it. Check camera, wire and side-guard clearances on the finished assembly.

Front shoe controls

Control	What it does
Create front bumper	One printable TPU shoe around the front of the actual bottom plate. Includes an underside sole, edge wall and upper retaining lip; no carbon geometry is changed.
Bumper front shape	Eighteen styles: follow the plate or choose rounded, swept, flat, crowned, shield, twin-shoulder or recessed protective toes.
Bumper toe width (0 = automatic)	Overall width of the added protective toe. Zero uses the plate grip width. The grip keeps its plate fit and blends continuously into wider or narrower toes. Ignored by Follow bottom plate.
Toe feature width (0 = automatic)	Central flat or recess width as a percentage of toe width. Zero retains the style proportions. Applies to chamfered, swept, flat-tip, crown, recessed, stepped and the six additional profiles.
Toe recess depth (0 = automatic)	Depth of central recesses as a percentage of extra projection. Zero uses the style default; 1 to 90 percent customizes Twin-lobe, Recessed-centre, V-recess and Twin-shoulder shoes without cutting the plate grip.
Shoe reach behind nose	Length of bottom plate captured behind its actual foremost edge. The rear remains open for sliding the shoe onto the plate.
Lip capture length (0 = full reach)	Length of upper retaining lip measured backward from the carbon nose. Zero uses the full shoe reach. A shorter lip makes fitting easier while the sole and outside wall keep their original reach.
Extra toe projection	Additional protective material ahead of the wall. Ignored by Follow bottom plate.

Lip capture length limits the upper retaining overlap measured backward from the carbon nose. Zero uses the full shoe reach; a custom length must be at least 2 mm and no longer than the shoe. The sole retains its original reach. Test that the shorter lip still retains the printed shoe securely.

Fit and export the front shoe

Control	What it does
Bumper wall thickness	Protective material outside the plate, in addition to fit clearance.
Shoe sole thickness	TPU below the underside fit gap; printed as the shoe base.
Retaining lip thickness	Vertical thickness of the upper retaining lip.
Retaining lip overlap	Inward reach of the upper lip over the plate edge. Camera tabs are relieved automatically.
Plate fit clearance	Clearance around the carbon edge and above/below its faces. Start with a test fit for your TPU and plate thickness.

Reach is measured backward from the actual foremost carbon edge. The extra toe projection applies only to shaped shoes. Fit clearance surrounds the plate and separates its upper/lower faces from the shoe. Lip overlap controls how far the upper lip reaches inward; lip thickness is its vertical thickness. Use enough wall and sole material for a connected part. A deep notch or short reach can split a follow-outline shoe; change its reach or choose a bridging toe style.

Mounting holes receive at least 6 mm head-access openings through the shoe. Camera tab/side-plate areas and intersecting assembly parts are relieved automatically. Check the finished part around your camera, arm roots and side guards; an unconnected result is rejected. The live view is schematic; created solids and exports contain the detailed clearances. The bumper uses Printed part colour and Flexible material density. Its volume contributes to frame mass and CG. Export writes separate assembled/print STEP files and a bed-aligned FrontBumper.stl in FlexibleParts. Print a fit sample and verify retention before use.

Shape-matched motor protectors and landing feet

Enable Create motor-pad protectors on TPU > Motor protectors to add four separate cups. The outside follows the current motor-pad shape, mirrored rotation and pivot; the cavity clears the actual arm outline and carbon thickness. These are motor/arm-end guards, not propeller guards. Their rim stops below the arm top so it does not form a wall around the motor bell.



Carbon arm profile and matching protector underside. The protector floor, screw access and landing foot are separate from the carbon stock.

Control	What it does
Create motor-pad protectors	Generate four separate TPU cups matching the selected pad shape, rotation and actual arm thickness. They are excluded from carbon cutting profiles.
Protector wall thickness	Material outside the measured motor-pad outline; follows scallops, rotation and neck shape.
Protector underside thickness	Base beneath the carbon arm. Motor-screw retention needs screws longer by this amount; verify motor thread depth.
Protector fit clearance	Radial clearance around the actual carbon outline. Print a fit sample for the selected TPU.
Landing foot height	Extra downward perimeter foot below the cup floor. Zero keeps a flat underside. Screw and shaft access remain open.
Protector screw access	Motor screws keeps small fixing holes. Linked screw access joins larger screw-head clearances to the shaft opening using radial slots; no tie tunnels are created. **Choices:** Motor screws; Linked screw access.

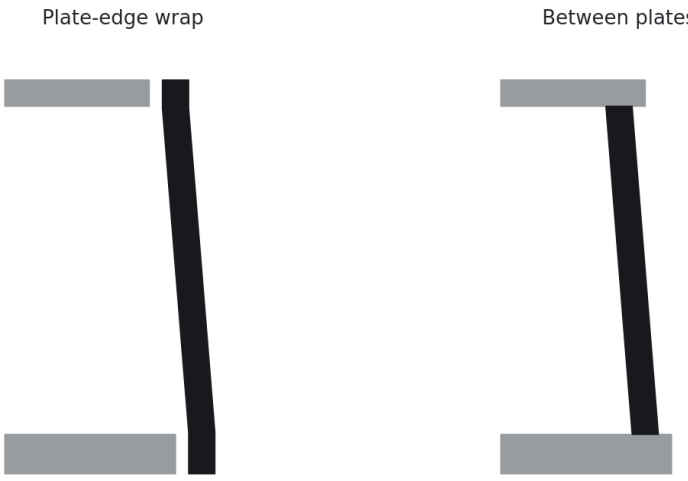
Motor screws uses the active three/four-hole fixing pattern through the floor. Choose screws longer by the underside thickness and verify engagement and winding clearance. Linked screw access makes larger screw-head clearances and connects each radially to the central shaft opening, removing the small bridges between them. Both modes retain the protector perimeter; no tie holes or tunnels are generated. Use suitable hardware and check retention on your actual arm. Landing foot height adds a perimeter foot below the floor; zero gives a flat underside. Print a fit sample before a complete set.



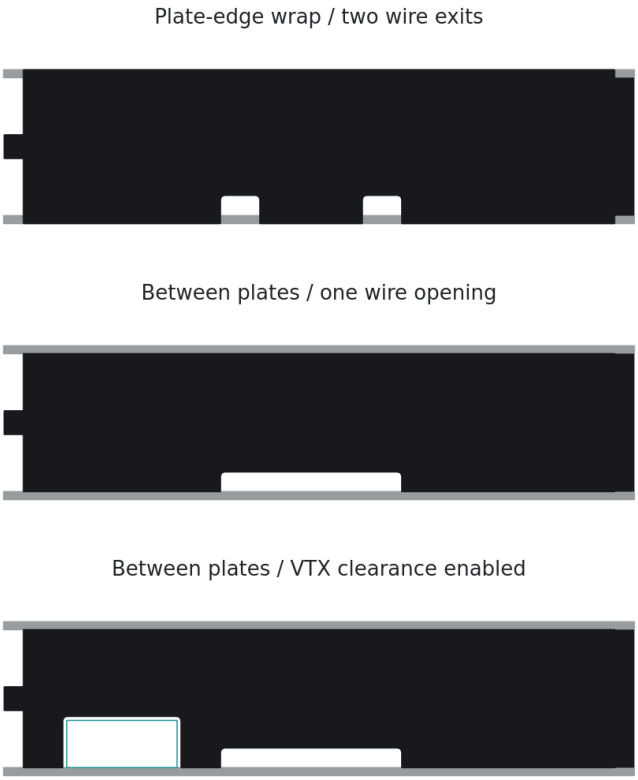
Underside profiles: small motor-screw holes and linked screw-head service openings.

Electronics side protection

Side guard fit offers Plate-edge wrap (the original design) and Between plates. Between plates moves the full wall thickness inward beneath the plate edges. Its ends seat against the upper face of the bottom plate in two-plate frames, or the mid plate in three-plate frames, and the underside of the top plate. There is no vertical assembly gap: the plates capture the TPU when tightened. Forward sections without a mid plate retain the upper strip; an optional camera closure starts above the bottom plate. Existing saved designs default to Plate-edge wrap.



Cross-section through a supported wall: original wrap outside the edges (left); inset wall captured by the top and carrier plate faces (right). Nominal contact, with no vertical gap. The inward setback uses Side guard plate clearance.



Side profiles: original plate-edge wrap with two wire exits; between-plate wall with a joined wire opening; the same wall with additional VTX clearance. Light grey marks the carbon plates. Teal marks the VTX envelope.

Side guards follow the finished top/mid edges in three-plate frames, or top/bottom edges in two-plate frames. In Plate-edge wrap mode, their wall reaches the top surface of the top plate and the underside of the mid plate with three plates, or the underside of the bottom plate with two. Vertical lips cover the plate edges; the wall between them slopes to follow different widths. Carbon, arm and propeller clearances can trim coverage locally. Where no mid plate supports the forward section, the existing upper strip is retained. The foremost row covered by each guard uses the full exposed post height, with 0.15 mm assembly clearance at each end. Intermediate FC-row sleeves use the upper 10 mm and the rear sleeves use the middle 6 mm. The inboard grip surfaces are circular and the outward supports use rounded capsule profiles instead of square front/rear corners. Both walls use the specified thickness independently of motor-wire opening size. Supporting material connects toward the guard wall to leave room around the electronics. Local arm/clamp reliefs keep the sleeves clear of carbon parts while retaining a continuous inner grip. Exact collision checks reject bearing locations inside a prop sweep.

Side guard plate clearance has a mode-specific direction. Plate-edge wrap uses it as an outward gap from the plate edges. Between plates uses it as an inward setback from the edge to the outer face of the wall; 0 mm makes that face flush with the edge while the full wall thickness stays inside. For example, 1 mm clearance and a 2 mm wall place its outer face 1 mm inward and its inner face 3 mm inward. Increasing it does not add a vertical gap. Plate contact is nominal; verify print dimensions and TPU fit on your hardware. Standoff post clearance is a separate fit setting and remains unchanged. After changing clearance on a saved frame, use Update frame to regenerate its guards.

Cover the foremost standoffs

Enable Extend to foremost posts to continue each guard to the actual foremost standoff row, including an added pair ahead of the nominal front pair. Existing FC-row bearings remain, so a six-post frame can have three supporting sleeves per side. With only two physical rows, the guard may already reach that row. The top plate must support the selected front posts. Beyond the end of a three-plate mid plate, the wall rises to an upper strip instead of inventing a new mid-plate extension. Front and rear insets shorten the main panel while the bearing sleeves remain attached. In Plate-edge wrap mode the front sleeve encloses its post over the full exposed height even where the adjacent wall follows an upper strip. If the arm or clamp leaves too little room for a continuous sleeve, reposition the post or adjacent carbon part.

2 plates / carrier underside to top face

3 plates / carrier underside to top face



Sections through the side wall: graphite carrier and top plates, dark TPU wall and grey post. Plate-edge wrap covers the full plate-edge height; grips remain on the exposed standoff.

Top-plate outline strap cutouts stay open beside the guards. Plate-edge wrap bridges the recess along the unnotched outer plate edge. Between plates follows the same smooth reference outline with an inward setback, and clears the actual top-plate strap slots or outline cutouts for 2 mm below the plate so the wall cannot seal their floor. The plate keeps its selected one or two strap cutouts. Widened and tapered plate sides, finishing fillets and unrelated custom outline cuts still determine the surrounding guard profile. The separate Side guard straps controls continue to add optional top- or bottom-edge openings in the guard itself.

Top view / one side of the top plate



Plate-edge wrap top view: the dark guard stays outside the light-grey carbon. The white gap inside the straight guard is the open strap passage.

Side view / full plate-edge coverage

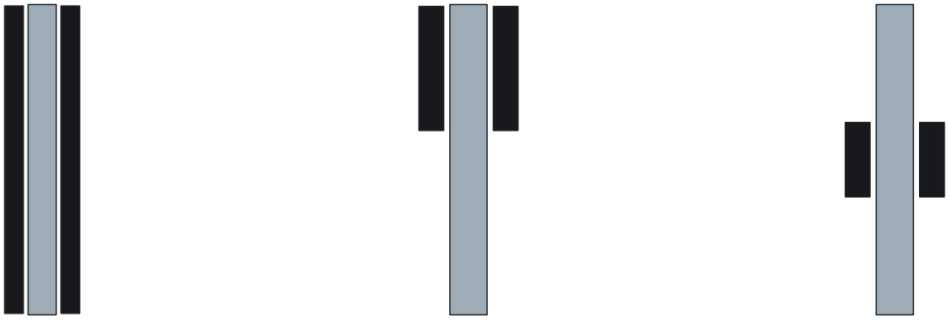


Side profile shows the full-height wall, bearing bands and lower-edge openings.

Front: 32.7 mm grip

Rear: 10 mm grip

Pair 3: 6 mm grip



Front, intermediate and rear bearing bands compared with their metal posts. Assembly clearance remains at the ends.

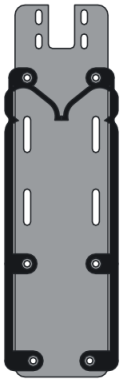
Circular inner bearing

Rounded outward support



Plan sections compare the circular inner sleeve and its rounded outward support.

Both side guards / top view



Plan view of both guards: equal wall thickness and rounded supports at the front and rear posts.

Close the sides to the camera plates

On TPU / Side guards / Forward coverage, enable Close sides to camera plates. This adds a broad solid panel from the foremost standoff row forward to the outside rear edge of each actual camera side plate. In three-plate designs, only this forward extension drops to the bottom plate. Plate-edge wrap reaches the plate underside; Between plates seats on its upper face and moves inward under the plate edge. The FC and front-arm area behind that row keeps its normal carrier-height lower edge and motor-wire openings. The panel tapers inward to the camera cage and follows its sloped rear edge when the top and bottom camera contacts differ. Foremost-post coverage is included automatically. The existing camera position and tilt controls remain unchanged.

Foremost-post forward closure



The forward closure drops to the bottom plate at the foremost standoffs. The FC/arm area behind them keeps its original lower edge.

Camera plate landing length sets the overlap along the rear of the camera side plate, from 1 to 8 mm. The landing touches the outside carbon face without thickening the camera cage. In Between plates mode the panel bears on the frame plate faces and plate clearance sets the inward setback. In Plate-edge wrap mode, set plate clearance to 0 mm for nominal edge contact; positive values leave a lateral gap. The camera side plates must span the bottom-to-top opening, and both frame plates must reach the joining area. Existing arm and plate reliefs remain active. USB and motor-wire openings are still optional; inspect their actual positions and cable clearance. The additional camera-side panel is solid, while the existing FC and VTX regions retain their vent choices.

Clearance for a larger VTX

The optional VTX clearance opening is separate from the vents. On TPU > Side guards > VTX clearance, enable it and choose Left, Right or Both. Automatic dimensions (length or height set to 0) follow the VTX envelope plus the clearance on each edge. Positive dimensions override the corresponding automatic size. Signed fore/aft and vertical offsets move its centre relative to the VTX reference. A low opening continues through the lower guard edge, avoiding a fragile bottom strip. The top web and bearing sleeves remain protected.

A fully covering clearance opening permits a wider VTX to project sideways past the plates. The lower deck must still support all mounting screws, the top and carrier must retain fore/aft coverage, and the generated assembly must clear standoffs, carbon and other parts. A narrow or misplaced custom opening does not waive those checks. If the VTX sits entirely behind the guard, the editor reports that no cut is needed there. This opening requires FPV mode and works with Solid or Vented guards in either fit mode.

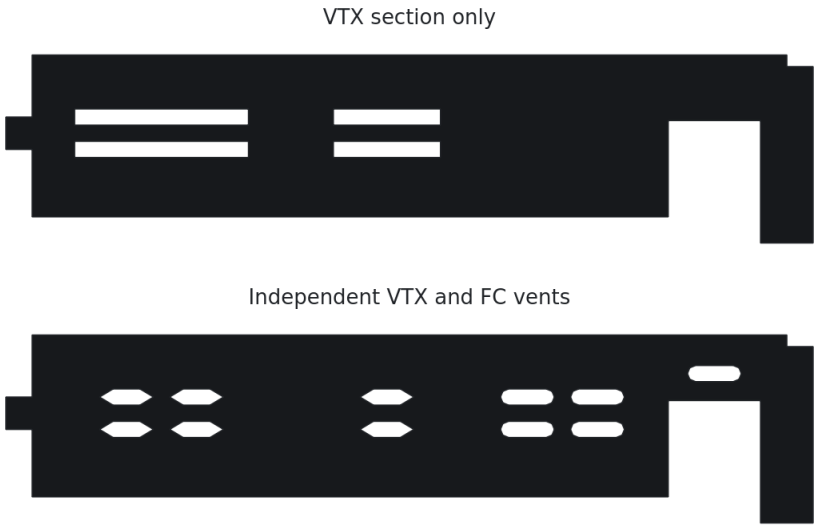
Control	What it does
VTX clearance opening	Leave room for a wider VTX through the selected guard wall. Follows the VTX envelope position and size; independent of ventilation. Requires FPV mode. Openings near the lower edge continue through it; standoff sleeves and a top web must remain intact. A fully covering opening also permits lateral VTX overhang beyond the plates while retaining mounting-hole support and collision checks.
VTX opening side	Choose which guard receives the VTX clearance opening. **Choices:** Left; Right; Both.
VTX opening fore/aft offset	Opening centre offset from VTX centre Y. Positive is forward, negative rearward.
VTX opening vertical offset	Opening centre offset from the VTX envelope centre height. Positive raises the opening; negative lowers it.
VTX opening length	Fore/aft clear length. Zero follows VTX envelope length plus twice the opening clearance; a positive value is the exact custom length.
VTX opening height	Clear vertical height. Zero follows VTX envelope height plus twice the opening clearance; a positive value is the exact custom height.
VTX opening clearance	Extra space on each edge for automatically sized dimensions. Custom length/height already include their desired clearance.

Independent VTX and FC ventilation

Choose Vented style on TPU / Side guards, then open TPU / Side guard vents. Enable VTX section vents for the rear bay and FC section vents for the forward bay. For vents only near the VTX, disable FC vents. Either section can be used alone; disabling both leaves a solid wall. USB and motor-wire openings remain independent of these switches.

The initial section boundary is the rear FC stack mounting-hole row: stack centre Y minus half its spacing. Vent boundary offset moves this line forward (positive) or rearward (negative), limited to

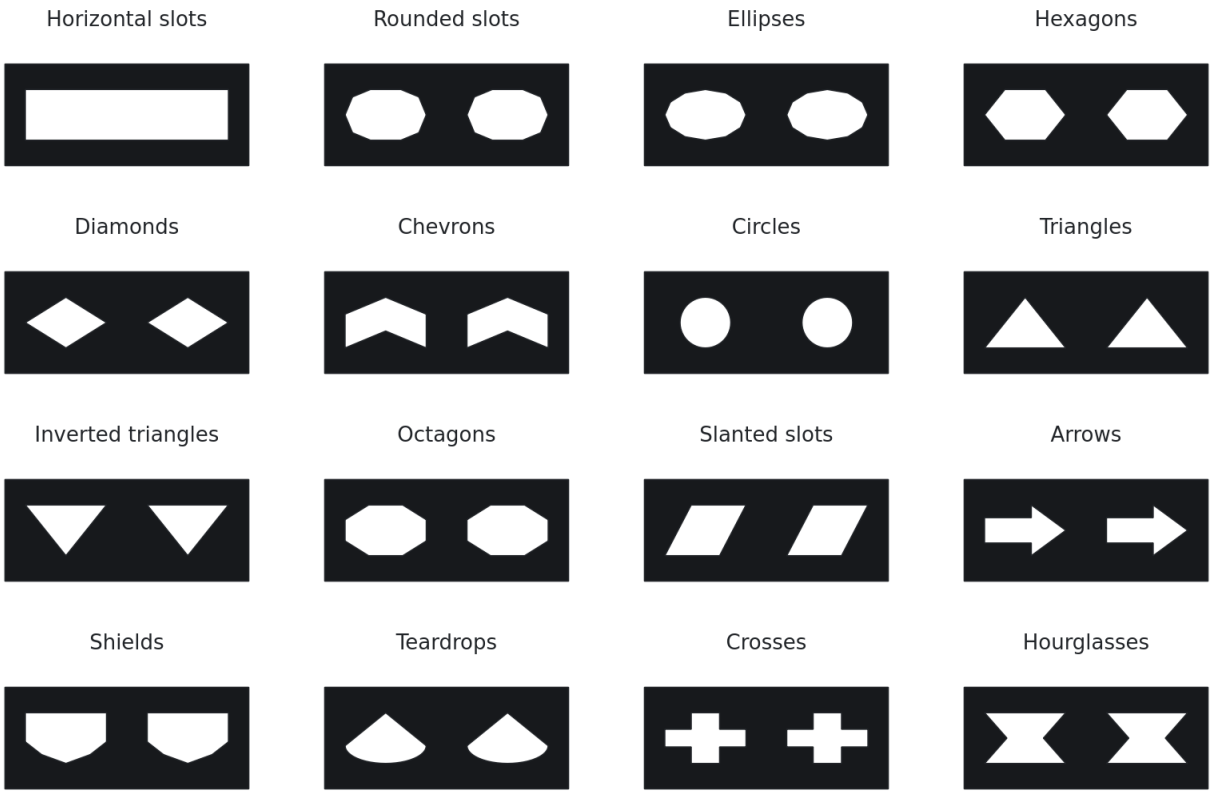
the covered guard length. Adjust it for the actual board edge and VTX location; the mounting row is a reference rather than a universal electronics envelope.



Side profiles compare rear-only ventilation with independently shaped VTX and FC openings.

Each section has its own shape, opening width/height, row count, web, front/rear insets and vertical offset. Vent row counts are unitless whole numbers from 1 to 6; fractional rows cannot be entered. Insets shorten its vented region without shortening the guard. Vertical offsets move rows up or down within the available wall. Travel, row count and height are limited where necessary to retain supporting webs. The shallow strip beyond the mid plate is treated separately. Horizontal slots span each clear section/post bay; repeated shapes use the chosen width. Sleeves and the material around the section boundary remain intact.

The existing six profiles are joined by Circles, Triangles, Inverted triangles, Octagons, Slanted slots, Arrows, Shields, Teardrops, Crosses and Hourglasses. Repeated shapes fit within the width/height envelope and retain the selected web between openings. Narrow details are sensitive to print resolution; check a test print before fitting the guards.



All sixteen profiles are available independently in the FC and VTX sections. Circular openings use the smaller of the requested width and height.

Control	What it does
Vent boundary offset	Shift the section boundary from the rear FC stack mounting-hole row (stack Y minus half its spacing). Positive moves forward; negative moves rearward. Limited to the guard ends.

VTX section controls

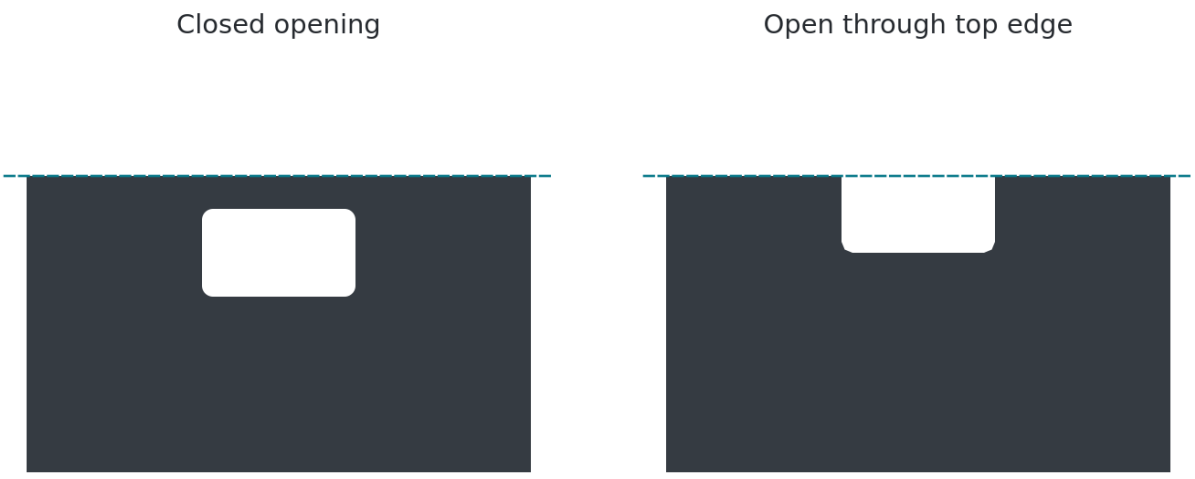
Control	What it does
VTX section vents	Enable vents in the rear section. Requires Vented style.
VTX vent shape	VTX section only. Opening profile used by Vented side guards. Individual repeated shapes use vent width and web; horizontal slots span each clear post bay.
VTX vent width	VTX section only. Fore/aft size of individual repeated vent shapes; horizontal slots use the clear bay length.
VTX vent height	VTX section only. Airflow opening height; limited to retain top and bottom material.
VTX vent rows	VTX section only. Whole-number count of airflow rows (1 to 6); actual openings fit the available panel height.
VTX vent web	VTX section only. Minimum spacing between repeated openings and their bearing regions.
VTX vent rear inset	Extra solid material at the rear edge of the VTX vent region.
VTX vent front inset	Extra solid material at the front edge of the VTX vent region.
VTX vent vertical offset	Move VTX rows up (positive) or down (negative), limited to retain top and bottom webs.

FC section controls

Control	What it does
FC section vents	Enable vents in the forward section. Disable for VTX-only ventilation.
FC vent shape	Opening shape used only in the FC section.
FC vent width	Fore/aft width of repeated openings. Horizontal slots span each clear FC section/post bay.
FC vent height	Requested opening height, reduced to fit the local available wall.
FC vent rows	One to six whole-number rows. Shallow upper strips can use fewer rows.
FC vent web	Minimum material between openings, posts and section edges; never less than guard thickness.
FC vent rear inset	Extra solid material at the rear edge of the FC vent region.
FC vent front inset	Extra solid material at the front edge of the FC vent region.
FC vent vertical offset	Move FC rows up (positive) or down (negative), limited to retain top and bottom webs.

USB cable access

On an extended guard, enable USB access opening and choose Left, Right or Both. Set its fore/aft centre relative to the FC stack, its centre height above the lower carrier, and its width and height. Allow space for the cable plug housing, not just the metal socket. The rounded opening passes through the guard wall. It can now sit close to the top plate. If the remaining strip at the physical top edge would be thinner than the guard wall, the cut automatically continues through that edge. The requested bottom and width stay fixed. Lower-wall support and bearing-sleeve clearance remain required. Keep it disabled when no opening is needed.



USB access near the top: a closed opening with adequate upper material, and an automatically opened top edge. Dashed teal line: physical guard top.

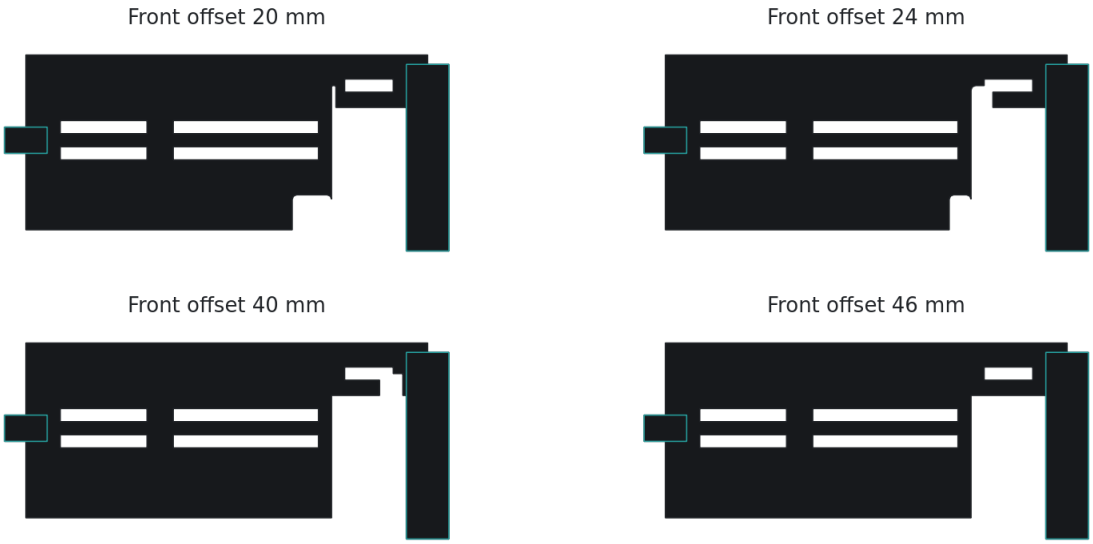
Bottom openings for motor-wire bundles

With Between plates selected, enabling motor-wire openings creates one continuous lower-edge cutout per selected side. It spans both saved front and rear centres, with half the Motor-wire opening width beyond each end. Height remains adjustable. Motor-wire positions is disabled in this mode; its saved Front / Rear / Both choice is restored when switching back to Plate-edge wrap. The combined cut follows stepped carrier edges and must clear the standoff grips, so it does not add a divider between the front and rear wire positions.

Enable Motor-wire bottom openings when the guard covers the FC/ESC bay. Choose Left, Right or Both sides. Plate-edge wrap offers Front, Rear or Both positions. The two fore/aft offsets are independent and use the stack centre as their anchor; positive values move forward. In Plate-edge wrap, width and height apply to each selected opening. Each notch opens through the local bottom edge and has rounded upper corners, leaving the upper wall connected. Without full front closure, the guard beyond the mid-plate end has an upper strip and the notch starts at that strip’s lower edge. With full closure, the portion ahead of the foremost standoff row starts at the bottom plate. Front openings can span these lower-edge transitions: each portion follows its local carrier edge. FC-area openings retain the original carrier height.

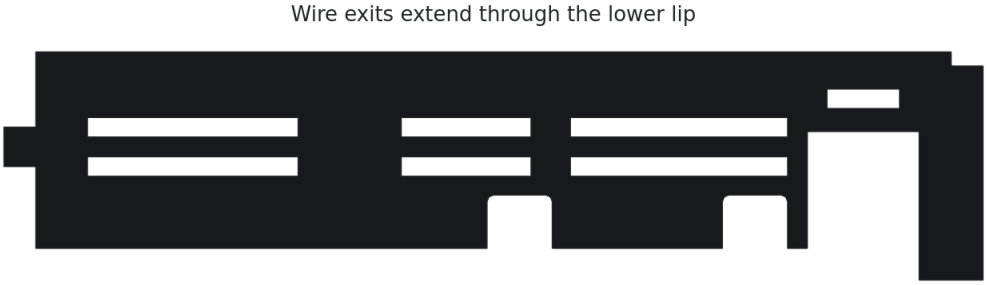
Front openings may approach the front standoff grips. Their cutters are trimmed around the actual bearing sleeves, preserving the sleeve stock and a local 0.8 mm buffer instead of rejecting the complete rectangular opening. The usable opening can narrow beside a grip; increase width or

adjust height/position as needed and inspect the created 3D part for your wire bundle. The side schematic retains the grip silhouette, so an outboard passage can be partly hidden in that view. Rear-opening and USB bearing protection retain their existing checks. Openings outside the guard, excessive heights and disconnected results are still rejected.



Front offsets across the mid-plate edge and toward the foremost post. Teal outlines mark the retained bearing bands; actual pockets curve around the sleeves in 3D.

Motor-wire opening height is measured above the local carrier top. In Plate-edge wrap its lower cut extends automatically through the plate-height lip, so extending the guard does not reduce the existing wire passage. USB and vent locations remain anchored to the internal electronics space.



Side profile: front and rear bottom-edge exits leave room for motor-wire bundles while preserving the circular post grips.

Allow space for insulated wire bundles and their bend radius. Rear pockets may pass under a raised standoff grip and open through the rear guard edge. Rear-pocket clearance checks use each grip’s actual height band, retaining at least one wall thickness between the pocket roof and the grip. A rear pocket that reaches the grip height is rejected. It must still intersect the guard wall; moving it entirely behind the guard has no useful effect. Front openings can approach the forward end while preserving the bearing stock. Fit checks also retain material above each opening. If a position conflicts, move the opening or reduce its size. The editor does not model or route actual wires; check the ESC connector positions and cable movement in the assembled frame. This option is independent of Solid/Vented style and USB access, and is enabled separately when needed.

Control	What it does
Create electronics side guards	Create separate left/right guards. Plate-edge wrap covers the plate edges; Between plates moves the wall inside the plate outlines with its ends bearing against the bottom/mid and top plate faces for clamping. The foremost covered post uses the available height; intermediate FC sleeves use upper 10 mm and rear sleeves middle 6 mm. Inboard grip surfaces are circular to preserve electronics space.
Side guard fit	Plate-edge wrap keeps the original lips covering the plate edges. Between plates moves the wall inward beneath the plate edges and seats its ends against the bottom/mid carrier and top plate for clamping. Plate clearance sets the inward setback. Top-plate strap passages receive 2 mm clearance underneath. Its motor-wire opening joins the front and rear positions into one continuous cutout. **Choices:** Plate-edge wrap; Between plates.
Side guard thickness	Thickness of the protective side panels and post clips.
Side guard post clearance	Diametral fit allowance around the actual metal standoffs. Test the sleeves using your material and printer.
Side guard plate clearance	Plate-edge wrap: outward gap from the finished plate edges. Between plates: inward setback of the outer wall face from those edges; the full wall thickness sits beneath the plates. Between plates has nominal contact with both plate faces, without a vertical assembly gap. Standoff fit clearance remains independent.
Side guard style	Solid panels cover the bay. Vented panels use the selected opening shape, rows and dimensions while preserving bearing regions. **Choices:** Solid; Vented.
Side guard front inset	Shorten the main wall from the foremost covered row while its full-height circular sleeve remains attached by the outboard support.
Side guard rear inset	Shorten the main wall from the rear attachment row; a broad bearing pad retains the complete middle 6 mm sleeve.
Extend to foremost posts	Extend the side guards to the actual foremost standoff row, including an added forward pair, while retaining the FC-row bearings.
Close sides to camera plates	Add a solid full-height panel ahead of the foremost standoffs down to the bottom plate, tapering to the actual camera side plates. Includes foremost post coverage. Requires FPV camera side plates; camera tilt stays unchanged.
Camera plate landing length	Length of the guard landing along the outside rear edge of each camera side plate. Nominal face contact; carbon and camera screw access are relieved automatically. Between plates seats against the frame plate faces; Plate-edge wrap uses plate clearance 0 mm for nominal edge contact.

Control	What it does
Motor-wire bottom openings	Open the lower edge for motor wires. Front and rear positions are independent and referenced to the FC stack. Openings preserve actual grip bands; rear exits may pass underneath raised grips and open through the rear end. Between plates joins the front and rear positions into one wide opening on each selected side.
Motor-wire side	Choose which side guards receive wire openings. **Choices:** Left; Right; Both.
Motor-wire positions	Create front, rear or both bottom-edge openings on each selected side. Used only by Plate-edge wrap; Between plates always spans both saved positions. **Choices:** Front; Rear; Both.
Front wire fore/aft offset	Front opening centre relative to the FC stack. Positive is forward; negative is rearward. Front exits follow stepped lower edges and are trimmed around bearing sleeves with a local 0.8 mm buffer. The usable opening can narrow beside a grip; adjust width/height and inspect the model. In Between plates this sets one end of the combined span.
Rear wire fore/aft offset	Rear opening centre relative to the FC stack. Negative moves rearward, including beneath raised standoff grips or through the rear end if height permits. In Between plates this sets the other end of the combined span.
Motor-wire opening width	Clear fore/aft width for each wire bundle. Leave room for insulation and movement; retain material around the actual grip height. Between plates adds this width to the distance between the front and rear centres.
Motor-wire opening height	Clear height above the local carrier top. The cut extends down through the plate-height skirt automatically, keeping wire space unchanged. Rounded upper corners soften the wire exit.
USB access opening	Cut an adjustable USB cable/connector opening in the extended side guard. Place it clear of the bearing sleeves.
USB access side	Choose the side that carries the FC USB connector. **Choices:** Left; Right; Both.
USB fore/aft offset	Opening centre relative to FC stack Y. Positive moves forward; negative moves rearward.
USB centre above carrier	Opening centre height above the lower carrier plate. Near the guard top, the opening continues through the top edge instead of leaving a fragile strip.
USB opening width	Fore/aft clear opening size, including room for the cable plug housing.
USB opening height	Vertical clear opening size, including room for the cable plug housing.

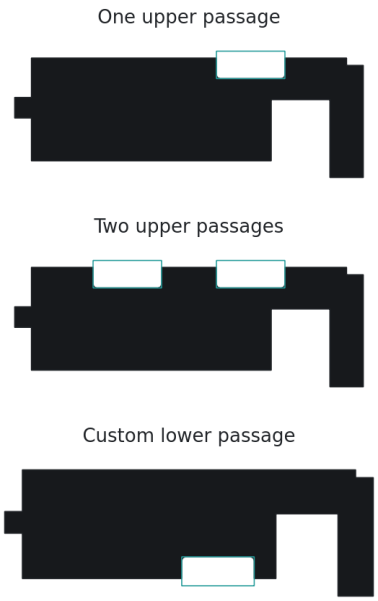
Review real USB position, cable bends, cooling, wiring and print fit. No electronics model specifies a universal USB position. The editor provides dimension and side controls so the opening can match your FC.

Generate, export and estimate flexible parts

Create / Update produces each enabled camera interface, front bumper, four motor guards and/or two electronics guards. Export writes each active part as its own STL and print-positioned STEP in FlexibleParts, with separate assembly-position STEP files and print notes. Camera interfaces and side guards lie on their broad faces; motor cups keep their undersides down. Review supports and orientation in your slicer. These parts appear in assembly STEP and volume-based weight/CG estimates at the flexible-material density, and are excluded from carbon DXF/SVG nesting. Unmodelled antennas, cables and accessories belong in Other equipment / wiring allowance.

Battery strap access through side guards

Under TPU > Side guard straps, enable Battery strap openings to cut one or two passages through both guards. They open through the selected top or bottom edge, so the strap can pass around the carrier plate. Existing guards are unchanged until this option is enabled.



Side-guard-only views: one or two upper passages, and a custom lower passage. Teal outlines identify the strap notches.

Side-guard strap controls

Control	What it does
Battery strap openings	Open one or two strap passages through the selected edge of both side guards. Keep their full width clear of standoff grips.
Strap opening count	Create one front opening or a front/rear pair on each side guard. **Choices:** 1 strap; 2 straps.
Follow plate strap positions	Use +Strap fore/aft offset for one strap and both +/- offsets for two. Positions follow the existing top-plate slots even when those slots are disabled.
Front strap offset	Custom opening centre relative to the FC stack along frame Y. Positive is forward.
Rear strap offset	Independent second opening centre relative to the stack. Used for two straps; negative is rearward.
Strap opening width	Clear fore/aft opening span in mm. Allow room for the actual strap width and assembly movement.
Strap opening depth	Clear depth inward from the chosen guard edge. The cut opens through that edge and retains material at the opposite edge.
Strap opening edge	Top edge lets straps pass beneath/around the top plate. Bottom edge opens the lower carrier skirt (mid plate in three-plate mode, bottom plate in two-plate mode). **Choices:** Top edge; Bottom edge.

Follow plate strap positions uses the top plate single-strap position when one top-plate strap is selected. With two top-plate straps, it uses the front position or both positions according to the side-guard strap count. Disable it to position each opening relative to the stack in frame Y. Opening width is the clear fore/aft span; depth is measured inward from the selected guard edge. Leave material at the opposite edge and keep the full passage clear of standoff grips. Bottom openings in three-plate designs belong in the continuous mid-plate carrier section. Overlapping wire/vent openings may need smaller cuts to keep the panel connected. The preview and printed exports update with the guard; carbon plates and their strap slots are unchanged.

Rear VTX antenna mount

Enable TPU > Rear antenna. This one-piece mount uses the two rearmost upper standoffs. Antenna tilt is adjustable from 0 degrees (horizontal and rearward) to 90 degrees (upright). The default is 60 degrees above horizontal, or 30 degrees from vertical. Dual antennas retain 30 degrees of outward yaw per side. At 90 degrees both axes are vertical, so their outward splay naturally disappears.



Actual CAD: 35 mm sleeves at 60 degrees with a 12 mm rearward holder offset. The grips keep their original height.

The guard rear sleeves resize to form a shared seat directly below the antenna grips. Guard material above that seat is recessed for the full mount grips and connecting bar; the mount grips are not trimmed away. At least 4 mm of guard bearing is retained. Only the standoff grip height is capped to the remaining space below the top plate. Holder length and offset do not change the guard seat or grip height. Without guards, more of the standoff is available. The editor reports the height used. Default grip wall is 3 mm; default bar width is 6 mm. The connecting bar follows the rear plate/guard width. Outward-pointing sleeves can extend beyond that width and above the top plate when clear of the actual frame. Both wrap and between-plates guards accommodate the mount. Actual intersections with plates or guards are rejected; there is no global standoff-height limit on sleeves behind the frame.

SMA and RP-SMA use a bulkhead bore secured with the connector nut. u.FL and MMCX grip the antenna stem or protective tube, not the small VTX connector. Automatic bore diameters are starting values: measure your hardware and adjust the bore and holder length. Route the antenna cable before assembly. Connector type alone does not guarantee mechanical fit.

The mount is included in saved designs, previews, mass calculations and FlexibleParts STEP/STL exports. Full antenna, connector nut and cable envelopes are not modeled: verify their clearance, cable bends and retention on real hardware. The editor shows a plan schematic; the final CAD model includes exact guard relief. Angled holes may need slicer support. Test-print in your chosen TPU before flight.

Standoff hole diameter sets the finished mount hole directly. Zero uses the frame standoff diameter plus Standoff fit allowance; a nonzero diameter overrides that allowance. The hole cannot

be smaller than the modeled standoff diameter. Measure your standoff and tune clearance with a test print. This control is separate from Antenna bore diameter, which holds the antenna itself.

For u.FL/MMCX, Antenna holder length extends the sleeve along its antenna axis from a fixed root. Increasing it does not push the inner end downward. Holder fore/aft offset moves the sleeves horizontally relative to the connecting bar: positive is rearward/out, negative is toward the frame. The connecting supports follow that offset. A long holder behind the plates may rise above the top plate, but cannot pass through the bottom, mid or top plate or a side guard. Adjust offset or tilt if a real intersection is reported.

SMA and RP-SMA now use a broad rounded bulkhead tab rather than a tube. Feed the threaded pigtail connector through Pigtail mounting hole, then secure it with its washer and nut. SMA bulkhead tab thickness is independent of sleeve length (default 2.4 mm, range 2-5 mm). Match the hole and thickness to the actual connector: enough thread must remain for the nut and antenna. Both tab faces provide a 10 mm diameter hardware access region (larger automatically for holes above 9 mm). Check your nut/washer size and test-print the fit; larger hardware may need a custom mount.

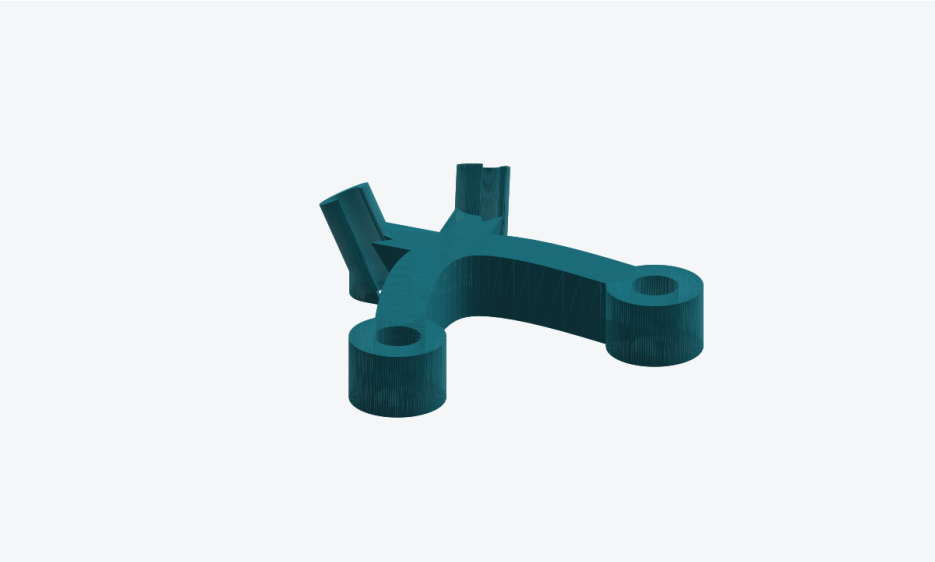


Actual CAD: two outward-angled SMA pigtail tabs with accessible mounting faces. The u.FL/MMCX variants retain their stem sleeves.

Editor changes run numerical validation and a 2D schematic, not exact CAD collision checks. Layout results and unchanged guard geometry are reused. Create/Update, explicit CAD checks and weight calculation still check the solid assembly. The curved bar is extruded from one profile instead of fusing many overlapping solid segments.

For u.FL/MMCX, Stem holder style switches between Closed tube and Side-entry clip. The clip opens toward the front of the frame. The channel runs through the full axial thickness at the clip, including its root. Tube roots are trimmed flush with the underside of the carrier, while the upper sleeve retains its length. Its mouth stops beyond the clip wall rather than cutting into the remote carrier; no additional rear brace is used. At zero tilt, a single horizontal rearward stem uses an upward side opening because the front direction is along its axis. Stem retaining lip sets retention on each side: opening width = bore diameter minus twice the lip. The opening has a flared

entrance. Closed tubes remain the default for existing saved designs. SMA/RP-SMA continue to use pigtail tabs; these clip controls are inactive for them.



Actual CAD: front-facing u.FL/MMCX channels through the supporting stock, without an added rear brace, retaining the entry lips.

Rear grips stack flush by default. Top plate gap lowers the VTX grips from the underside of the top plate; zero gives direct contact. VTX to side-guard gap adds intentional space below the VTX grips. With either rear mount enabled, the side-guard rear grips fill the remaining height between the active mounts or the opposite plate, retaining at least 4 mm of bearing height. Grip-height changes update this stack automatically.

Control	What it does
Antenna connection	SMA/RP-SMA: flat bulkhead tab for a pigtail secured by its nut and washer. u.FL/MMCX: tubular holder for the measured antenna stem, not the PCB connector. **Choices:** SMA; RP-SMA; u.FL; MMCX.
Antenna count	One rear antenna or two with symmetric outward yaw of 30 degrees each. Elevation uses Antenna tilt. **Choices:** 1 antenna; 2 antennas.
Connecting bar shape	Broad rounded connecting bar, triangular brace or filled rectangular platform in top view. **Choices:** Curved; Triangle; Rectangle.
Rearward reach	Distance from the rear standoff row to the connecting bar apex. Holder fore/aft offset moves the sleeves independently.
Standoff grip height	Requested sleeve height. The side-guard rear grip adjusts to support the antenna grip directly. Available height caps the request while retaining at least 4 mm of guard bearing.
Standoff grip wall	Radial sleeve wall thickness. Only grip rings may extend outside plate or guard width.
Standoff fit allowance	Diametral standoff clearance. Tune with a test print.

Control	What it does
Connecting bar width	Fore/aft rib thickness. Minimum 4 mm.
Antenna bore diameter	Zero uses 6.5 mm for SMA/RP-SMA, 4 mm u.FL stems or 3 mm MMCX stems. Override using measured hardware and print allowance.
Antenna holder wall	Radial tube wall for u.FL/MMCX. For SMA/RP-SMA it enlarges the tab when the hole needs more surrounding stock; the tab is at least 12 mm wide.
Antenna holder length	u.FL/MMCX sleeve length along the antenna axis; zero uses 6 mm. Extends outward from a fixed root. Retained but inactive for SMA/RP-SMA, which use bulkhead tab thickness.
Dual antenna spacing	Distance between holder centres; active for two antennas. Must fit the rear width.
Antenna tilt	Elevation above horizontal: 0 degrees rearward and level, 90 degrees upright. Default 60 degrees is 30 degrees from vertical; dual outward yaw remains 30 degrees per side.
Standoff hole diameter	Direct finished hole diameter. Zero uses standoff diameter plus fit allowance. A nonzero value overrides that allowance; must be at least the modeled standoff diameter.
Holder fore/aft offset	Move the antenna sleeves independently of the connecting bar. Positive moves rearward/out; negative moves toward the frame. Actual plate and guard collisions remain blocked.
SMA bulkhead tab thickness	Thickness clamped between the pigtail shoulder and its nut/washer. SMA and RP-SMA only; 2 to 5 mm. Verify enough thread remains for the nut and antenna.
Stem holder style	Closed tube slides over the stem. Side-entry clip opens toward frame front through the full axial thickness at the clip, without an added rear brace. At zero tilt, a horizontal rearward stem uses upward radial entry because front is along its axis. u.FL/MMCX only. **Choices:** Closed tube; Side-entry clip.
Stem retaining lip	Per-side retention for the side-entry opening. Opening width equals stem bore minus twice this lip. Tune to TPU and the measured stem.
Top plate gap	Vertical gap below the top plate. Zero seats the VTX grips flush.
VTX to side-guard gap	Vertical gap between VTX and side-guard grips. Zero gives direct contact.

Control	What it does
Rear VTX antenna mount	One-piece TPU mount on the rearmost upper standoffs.

Design references: DFR generic SMA/RP-SMA pigtail mounts, Walksnail V2 dual mounts and reinforced Matchstick U.FL mounts. RotorForge generates original parametric geometry; it does not

use their STL files. Pigtail retention reference: Team BlackSheep Oblivion SMA U.FL Pigtail (team-blacksheep.com/products/prod:oblivion_ufl_sma); mounting example: DFR Generic SMA/RP-SMA Pigtail Mount (drone-fpv-racer.com/en/generic-smarp-sma-pigtail-mount-tpu-by-dfr-11604.html).

ELRS T-antenna mount

Enable TPU > ELRS antenna independently of the VTX mount. The separate printable mount slides onto the two rear upper standoffs below the side-guard grips, above the bottom plate in two-plate frames or the mid carrier in three-plate frames. Grip height is capped to the available space; at least 4 mm must remain. ELRS height above plate is the intentional gap above the bottom plate (mid plate in three-plate designs); zero seats the grips directly on it. ELRS to side-guard gap sets optional clearance above the ELRS grips; zero gives direct contact. The rear side-guard grip expands or contracts with the mount heights. Without side guards it uses the free space below the VTX grips, or below the top plate when the VTX mount is off.



Actual CAD: extended hooks on two rounded square-section ribs and an adjustable centre pass-through in the connecting bar.

Set ELRS antenna diameter from the insulated horizontal antenna housing, not from the coax or connector. Hook spacing sets their centre-to-centre distance; leave room for the T junction and cable between them. Hook width sets contact length along the antenna. Snap the antenna into the upward openings, or slide it through where the housing shape permits. The small retaining lips and flared mouths are intended for flexible TPU. Confirm retention with a test print; this is not a measured snap-force simulation.

ELRS rearward reach moves the hooks and their connecting bar behind the frame. ELRS hook rearward offset adds movement of the hooks only; two rounded support ribs extend back from the fixed bar. Each connector has equal width and height, sized to the smaller of hook width and available grip height. The extension reaches into the hook body across its full section; the bore and insertion opening remain clear. Choose Curved, Triangle or Rectangle for the bar. Standoff hole diameter and fit allowance work like the VTX grip controls. The bar and hooks stay within the rear plate/guard width; grip rings may extend beyond it. The side guards receive local clearance below their grips. Actual intersections with the plates, guards, VTX mount and other frame parts remain blocked.

Only grip-related changes affect guard relief. Editing hook reach, spacing, diameter or lip keeps unchanged guard geometry cached. The editor uses a 2D schematic; exact connected-solid and collision checks run on Create/Update. The complete ELRS antenna and coax are not modeled: verify the antenna ends, cable routing and propeller clearance on the assembled frame.

ELRS antenna pass-through diameter sets a front-to-back hole through the centre of the bar, aligned with the T-antenna centre leg/coax. It is independent of the hook bore diameter. Size it for the housing or connector that must pass through, not the full horizontal T-bar. Zero disables the hole. At least 1 mm of material must remain above and below it; increase grip height or reduce hole size if the check fails.

Control	What it does
ELRS T-antenna mount	Separate TPU mount on the rear upper standoffs below the side-guard grips.
ELRS bar shape	Connecting bar joining both rear grips. **Choices:** Curved; Triangle; Rectangle.
ELRS rearward reach	Distance rearward from the standoff row to the antenna axis. Moves both hooks and the bar together.
ELRS height above plate	Raise the mount from the lower carrier: bottom plate in two-plate frames, mid plate in three-plate frames. The grip must remain below the guard grip.
ELRS grip height	Requested grip height, capped to the available space below the guard grips. At least 4 mm must remain.
ELRS grip wall	Radial TPU around the standoff hole. Minimum 1.2 mm.
ELRS standoff hole	Finished hole diameter. Zero uses standoff diameter plus fit allowance. Must not be smaller than the standoff.
ELRS standoff fit allowance	Diametral clearance when ELRS standoff hole is zero.
ELRS bar width	Fore/aft connecting rib width, at least 4 mm.
ELRS hook spacing	Centre-to-centre spacing across the two hooks. Leave room between them for the T junction and coax.
ELRS antenna diameter	Finished bore for the insulated horizontal antenna housing. Measure your T-antenna; this is not a connector diameter.
ELRS hook width	Contact length of each hook along the horizontal antenna.
ELRS hook wall	Radial TPU around the antenna. Minimum 1.2 mm.
ELRS retaining lip	Per-side retention. Upward opening equals antenna diameter minus twice this lip. Snap in from above or slide in along the antenna axis; test the fit in your TPU.
ELRS hook rearward offset	Additional rearward movement of the hooks only. The main bar and grips stay in place; two ribs connect the hooks to the bar.
ELRS antenna pass-through diameter	Front-to-back hole through the centre of the connecting bar for the T-antenna centre leg/coax. Measure the part that must pass through. Zero disables it; retain at least 1 mm above and below the hole.
ELRS to side-guard gap	Vertical gap above the ELRS grips. Zero gives direct contact with side-guard grips.

Design references: Brain3D ELRS EP1 Horizontal Antenna Mount (brain3d.com/products/elrs-ep1-horizontal-antenna-mount-1) uses rear standoffs and a horizontal antenna; Rotor Village RVS Dual Direct TPU Antenna Mount (rotorvillage.ca/rvs-dual-direct-tpu-antenna-mount-o4-avatar/) combines VTX and ELRS support. RotorForge uses original parametric geometry, not downloaded STL files.

13 Airflow openings and shaped plate outlines

Open Finishing > Airflow / outlines. This subsection uses the full editor width. The left column contains the saved-pattern list, search, filters, Remove and How to use cutouts. The centre contains pattern settings and Add, Apply, Discard, Copy and CAD-check actions. The right column contains the preview, pattern numbers, Zoom to selected, Fit plate and status. Drag the dividers to resize the columns. The ordinary frame preview returns when you leave this subsection. Long lists and large display scaling may still require scrolling.

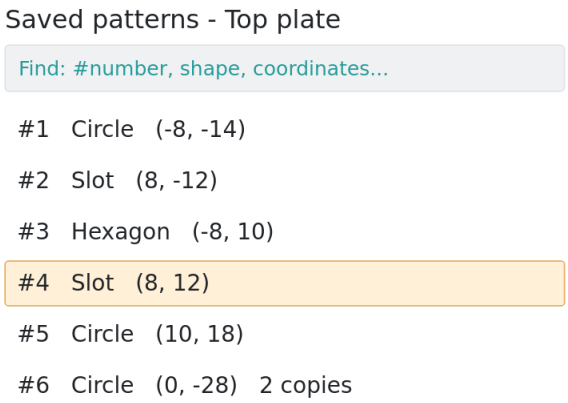
Airflow / outlines stores independently editable cutout patterns by part. Select the part, cutting mode and shape, set its position and dimensions, then press Add pattern. To modify a saved pattern, select it and press Apply to pattern after editing. The preview and generated model use saved enabled patterns.

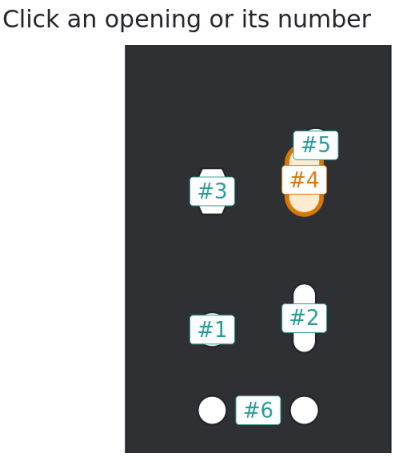
Control	What it does
Part	Chooses the target carbon component. Settings for inactive parts are retained until that component is enabled.
Cutting mode: Interior	Makes an enclosed opening while preserving the required material to existing edges, holes and fasteners.
Cutting mode: Outline notch	Intentionally crosses an outside edge of the bottom, mid, top or a clamp plate. Each instance must overlap material and extend outside. Arms, keys and camera side plates retain Interior patterns.
Cutout type	Chooses one of the twenty profiles described below.
Width / length	Sets the local shape size. Circle uses width as its diameter.
Centre U / V	Horizontal parts use frame X / Y. Camera side plates use U along frame Y from the nominal camera centre, and V as height above the bottom upper face.
Rotation	Rotates the cutout and its centred repetition grid.
Columns / rows	Repeats the shape in a centred grid. Each enabled copy counts toward the frame cutout limit.
Pitch U / V	Spacing between grid centres, before rotation. Pitches smaller than the cut dimensions can create overlapping copies.
Enabled	Keeps the pattern saved while including or excluding it from the model.
Saved patterns / Enabled cuts	Shows the capacity used. Up to 200 patterns can be stored and at most 200 instances enabled, including grid copies.
Add pattern	Appends the current controls as a new saved pattern. Failure shows a warning and preserves the saved list.
Apply to pattern	Replaces the selected saved pattern with the current controls.
Remove	Deletes the selected saved pattern.
Discard unapplied changes	Restores the saved control state without changing existing patterns. Unapplied edits must be added, applied or discarded before checking, saving settings or building.
Copy to other camera side	Adds an independently editable copy of the current camera-side pattern to the opposite camera plate.

Control	What it does
Check plate cuts in CAD	Builds the actual solids and checks complete geometry without creating or updating the frame document objects.

Find an existing cutout

Use Find cutout above the saved list to search by pattern number, shape, part name, cutting mode, dimensions or centre coordinates. Multiple words must all match. Enter #12 to find exactly pattern 12. The part filter narrows the list without changing saved patterns. Selecting a result opens that part and loads its controls. Numbers follow the saved list order and can change after deletion; they are locator labels, not permanent part names.





The list and preview share the same pattern numbers. Orange identifies the selected pattern, even when its opening merges with another cutout. Preview detail shown.

Control	What it does
Click an opening or number	Selects its saved pattern. Clicking any member of a repeated array selects that entire array. Clicking again at the same overlapping opening cycles through the matching visible patterns.
Show pattern numbers	Shows labels for visible enabled patterns on the current preview part. The selected pattern always has its number and highlighted outline.
Zoom to selected / Fit plate	Magnifies the selected pattern and its array copies. Fit plate returns to the complete part.
Filters and hidden selections	The count shows how many patterns match and identifies a selected pattern hidden by the filter. Apply and Remove show the selected number. Clear the search or choose All parts to reveal more results.
Disabled patterns	Remain searchable in the list. Select one there to inspect its dashed outline; it stays excluded from generated cuts until enabled.
Unapplied edits	Apply or discard edits before selecting another pattern. Switching results cannot silently overwrite the current controls. Search, filtering, highlighting and zoom do not build CAD or change saved geometry.

After editing, press Apply to #number to update the selected saved pattern. The existing geometric checks still run when you check, create, update or export the frame. Pattern arrays and overlapping shapes keep their original editing behaviour.

Cutout catalogue - all 20 shapes

These coloured silhouettes show the material removed. They use the exact editor geometry, one scale and an 8 mm width by 12 mm length envelope. Circle uses width as its diameter. Rotation turns the profile and its repetition grid together.



Circle

Width is diameter.



Slot

Capsule, round ends.



Rounded rectangle

Radius = min(W,L)/4.



Hexagon

Six straight sides.



Ellipse

Independent width/length.



Rectangle

Square corners.



Triangle

Point faces local +V.



Diamond

Four pointed corners.



Trapezoid

Narrower toward +V.



Pentagon

Five straight sides.



Octagon

Eight clipped sides.



Chevron

Concave V opening.



Heptagon

Seven straight sides.



Dodecagon

Twelve straight sides.



Parallelogram

Slanted quadrilateral.



Kite

Offset broad shoulders.



Hourglass

Narrow centre waist.



Cross

Four joined branches.



Crescent

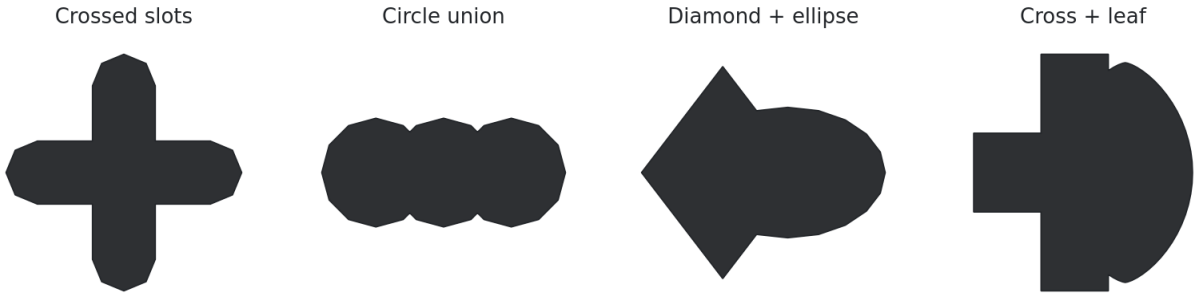
Curved concave opening.



Leaf

Pointed curved oval.

Combine openings and cut plate edges



Combined openings formed by overlapping editable source patterns.

Overlapping patterns on the same part are united before subtraction. Different shapes and grid copies can combine, and an Interior opening can overlap an Outline notch. Merely tangent shapes do not merge: overlap them or leave at least the configured minimum material between separate cuts. Existing holes, sockets and fastener bearing footprints remain protected. A union that disconnects the part or isolates an island is rejected.



The same top plate before and after edge cuts. White cutouts remove material; the coloured plate remains connected around its mounting holes.

A pattern wholly inside the plate belongs in Interior mode. A pattern wholly outside removes no material and is rejected as a notch. In a grid, every notch instance must cross an outside edge. If a saved design already has many patterns, check the capacity readout; use the displayed limits before adding more cuts.

14 Checks, references and dimensions

Calculate frame weight and centre of gravity

Click Calculate frame weight / CG beneath the preview to evaluate the current geometry before creating a document. Apply or discard pending airflow/outline edits first. The calculation uses finished CAD solid volumes, including holes, cutouts and grip grooves. It reports carbon mass separately and adds the modelled hollow standoff tubes and optional pad using their own densities. Active flexible parts, including guards, camera interfaces and the front bumper, contribute at the flexible-material density. Frame CG is the mass-weighted centre of those parts in frame X/Y/Z coordinates. A result can lie inside the structure; the editor marks its top-view projection with a red cross.

The bare Frame and Frame + reference LiPo results remain available. Hardware > Weight estimates adds a separate Equipped result (without LiPo) and Equipped + LiPo result. These include four motors, four props, one FC and one 4-in-1 ESC, plus the camera and VTX when FPV mode is enabled. Standoffs use their actual hollow geometry and density, so longer front standoffs and changed pair counts affect the result without being counted twice. Screw counts follow the mounting geometry; shared arm/standoff bolts replace separate base screws.

Defaults are editable representative planning values, not a market-wide statistical average. FC 10.5 g and ESC 23.5 g follow the SpeedyBee F405 V4/BLS 55A specifications; motor 30.5 g follows the iFlight XING2 2207 specification. Camera/VTX 10 g each, props 4 g each and Other equipment 15 g are nominal allowances. Change masses for your actual motor size and analog/digital equipment. Zero omits an item. Visibility never changes mass inclusion.

CG uses mass-weighted component centres. Motor positions follow the actual rotated pad centres; FC/ESC centres follow the stack deck; camera follows its body centre; VTX follows its rear envelope. Other equipment is concentrated at the stack centre. Screw positions and lengths are approximations. The red cross uses Equipped + LiPo CG when available, Equipped CG without the reference battery, or bare Frame CG when equipment estimates are off. Verify final aircraft mass and balance with the actual hardware.

Control	What it does
Include equipment estimates	Include nominal motors, propellers, FC, ESC, enabled FPV camera/VTX, automatically counted screws and an extra equipment allowance. Bare-frame results stay separate. Reference visibility does not affect masses. The CG marker uses loaded CG when the battery is enabled, equipped CG otherwise; with this option off it shows bare-frame CG.
Motor mass (each)	Mass of each motor including wires, multiplied by four and located at its actual motor centre (including pad rotation). Default 30.5 g represents a 2207 motor. Change it for smaller or larger motors; changing mounting pitch does not guess a new mass.
Propeller mass (each)	Mass of each of the four propellers, placed at its motor XY and prop reference height. Default 4 g is a 5-inch planning allowance; enter the actual prop mass.

Control	What it does
Flight controller mass	Mass of the flight controller. Default 10.5 g is a representative 30.5 mm FC value. Its assumed centre is at 75 percent of the stack envelope height above the selected stack support.
4-in-1 ESC mass	Mass of one 4-in-1 ESC, default 23.5 g. Its assumed centre is at 25 percent of the stack envelope height. It follows the bottom, clamp or mid stack deck automatically.
VTX mass	Mass of the VTX, included only in FPV mode. Default 10 g is a nominal allowance, not a digital-unit specification. Enter the mass of your actual analog or digital VTX. Located at the rear VTX envelope centre.
Camera mass	Camera mass, included only in FPV mode and placed at the tilted camera body centre. Default 10 g is a nominal small-camera allowance. Enter your actual camera mass.
Other equipment / wiring allowance	Combined allowance for wiring, receiver, antenna, nuts, spacers, straps and other equipment. Default 15 g. Placed at the centre of the FC/ESC stack envelope; this simplified placement limits CG accuracy. Set zero to exclude or replace with a weighed total.
Screw mass multiplier	Multiplier for automatically counted arm, shared-arm/standoff, top/base, stack, motor and FPV screws. Default 1. Steel mass uses 7.85 g/cm3, 80 percent of the nominal threaded-cylinder volume and a cylindrical head of diameter 1.7 times the screw diameter and height 0.65 times its diameter. Assumed engagement is 2-4 mm depending on location. Lengths follow grip dimensions. Set zero to omit, or scale to your hardware. This is a mass approximation, not a fastener length specification.

Where nominal values come from

Manufacturer references: SpeedyBee F405 V4 / BLS 55A specifications; iFlight XING2 2207 specifications. Other defaults are declared allowances. Actual builds vary, especially digital video hardware, heatsinks, wiring and motor size.

Calculations do not create a document. Editing parameters marks the result out of date and removes the old preview marker. Calculate again to refresh it. Create / Update also stores the current results in Frame Parameters. Show estimated CG marker and Show references control the 3D cross; they do not include the marker or reference envelopes in frame mass. Export writes the values, material densities and scope to MassProperties.json and ExportInfo.json.

Control	What it does
Minimum material around holes	Geometric edge / hole spacing check. This does not predict structural strength.
Carbon density estimate	Used only to estimate carbon mass. Hardware, electronics, tolerances, and coatings are excluded.

Control	What it does
Standoff material density	Density used for the modelled hollow standoff tubes. Default 2.7 g/cm3 is nominal aluminium. Screw/nut and strap masses are not included.
Show estimated CG marker	Show the estimated CG cross when references are visible. Uses equipped plus battery CG when component estimates and the battery are enabled; equipped CG without a battery; bare-frame CG when component estimates are disabled.
Show prop / motor / stack references	Reference volumes are approximate and are excluded from manufacturing exports.

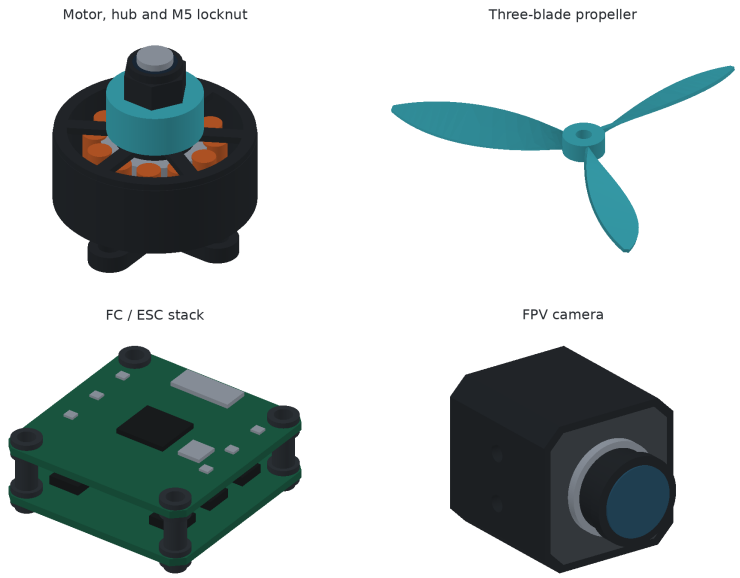
Control	What it does
Status	Reports whether the current generated geometry is valid or which error prevents building it.
Warnings	Lists layout conditions to review. A valid model can still have warnings.
CarbonMassGrams	Finished carbon-only mass from exact solid volume and carbon density.
FrameMassGrams / FrameCG	Carbon, modelled standoff tubes and optional pad; CG is the mass-weighted X/Y/Z position in mm. Unspecified hardware and electronics are excluded.
FramePlusBatteryMassGrams / FramePlusBatteryCG	Frame estimate plus the reference battery only, using the entered battery mass at its envelope centre. Not full aircraft weight/CG.
EquippedMassGrams / EquippedCG	Bare frame plus enabled nominal equipment, screws and extra allowance, without battery.
LoadedMassGrams / LoadedCG	Equipped estimate plus reference LiPo, when both options are enabled.
MinimumPropGap	Calculated closest propeller edge-to-edge gap.
GeneratorVersion	Plugin version that generated or recomputed the frame.

Errors block creation when the fast checks detect invalid dimensions or missing support. Detailed CAD checks run on Create / Update, explicit cut checking and Export. Warnings describe conditions to review, such as a smaller-than-desired prop gap. Neither a warning-free model nor the carbon mass estimate calculates crash strength, stiffness or vibration behaviour.

Reference envelopes show approximate props, motors, FC/ESC stack, camera and VTX. They are excluded from STEP and cutting profiles. Carbon mass uses the configured density and generated carbon volume; equipment weights are assigned separately through the nominal estimation controls. Check real mounting dimensions, connectors and screw engagement against the intended parts.

Optional detailed presentation references

Finishing > Detailed references contains independent switches for motors, three-blade props, FC/ESC stack, FPV camera, VTX and LiPo. Leave them off to keep the original simple references. Enable Show references first; camera detail also requires FPV. Create or Update applies the selected style to the 3D document. The editor keeps its fast schematic plan view.



Generic presentation models built from the current dimensions: outrunner with a seated 6 mm hub and M5 locknut, 3.7-inch-pitch three-blade prop, two populated PCBs with silicone grommets, and FPV camera.

Control	What it does
Detailed motor references	Replace the cylinders with radial mounting feet, a tall stator and copper windings, bell, M5 shaft and self-locking nut. Diameter/height describe the bell body; the shaft and nut extend above it. Display only.
Three-blade prop references	Curved three-blade references with 3.7-inch geometric pitch, 6 mm hubs and 5.1 mm bores, seated directly on the motor bells. Adjacent motors have mirrored props. The simple-disc height is ignored; clearance uses the full swept blade-height range.
Detailed FC / ESC stack	Display two component-populated PCBs and silicone grommets at the existing stack mounting pattern and height. A generic visual model, not an exact electronics footprint.
Detailed FPV camera	Display a shaped camera housing, side fixing recesses and stepped lens with glass. Keeps the current body size, camera position and tilt. The original envelope remains the clearance check.

Detailed hubs sit at bell height; simple-disc height is ignored. The grey stator starts at the lower bell edge, with a dark support down to the feet and 0.5 mm radial mounting-screw clearance. Windings stay inside the bell. Stack/camera settings, mass estimates and manufacturing exports are unchanged. Models stay cached; detailed-prop clearance covers the swept blade-height range.

FC USB reference and guard alignment

The detailed upper FC board now includes a hollow USB-C shell and internal tongue, using a generic side-centred commercial-style location. The SpeedyBee F405 V4 manufacturer layout is a reference for a side-facing connector; this is not a dimensionally exact model of that FC. The shell is 9 mm wide, 3.2 mm high and 7.5 mm deep, extending 0.8 mm past the reference PCB. Its bottom sits on the upper PCB face.



Detailed FC/ESC stack with the USB-C receptacle on the left edge of the upper board. Left/right placement and fore/aft offset are adjustable.

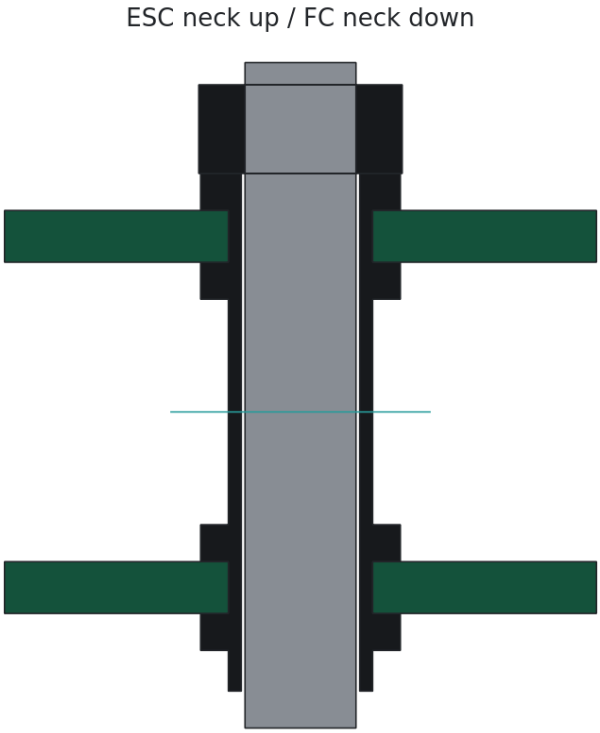
Control	What it does
FC USB side	Side-facing USB-C socket on the upper PCB. Left is negative frame X; right is positive X. Generic side-centred commercial-style placement, not an exact board footprint. **Choices:** Left; Right.
FC USB fore/aft offset	USB socket centre relative to the stack centre. Positive moves forward; negative moves rearward. Adjust to match the actual FC before locating a side-guard USB opening.

The editor summary reports the matching side-guard USB centre Y offset and Z height, including stack spacers and carrier height. Enter those values under TPU > Side guards > USB access, select the matching side, and allow extra width/height for your actual cable plug. Guard USB access requires forward guard coverage. Changing the reference does not automatically move or resize an existing guard opening. The connector is visual only; check the actual FC and cable, since the simple stack collision envelope does not include this protrusion.

Source: SpeedyBee F405 V4 60A Stack manufacturer manual, Layout and Dimensions: https://www.speedybee.com/f405-v4-60a-stack-download/

FC and ESC grommet contact

The detailed stack uses two PCBs. ESC grommet necks point upward and FC necks point downward, meeting between the boards. Only a short retaining flange sits above the FC PCB. The upper nuts remain seated at the selected total stack height, directly above that flange; no extra stack offset is added. Lower grommets start at the existing stack mounting level, including any press-nut adjustment.

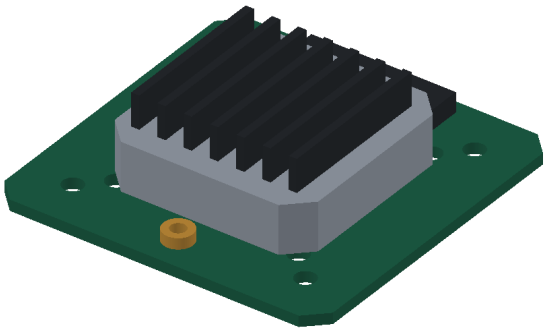


Side section of one screw position: green PCBs, dark grommets and nut, grey screw shank. The teal line marks the contact between the opposing grommet necks.

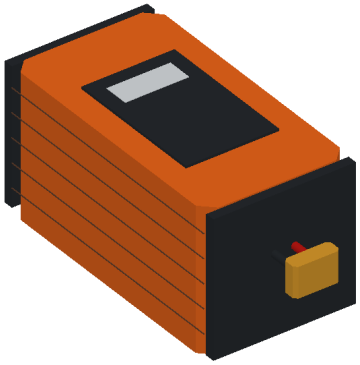
Optional detailed VTX and LiPo references

The additional power references use the same independent switches and Show references control. VTX detail requires FPV; LiPo detail requires Battery reference enabled. Their appearance is independent of Printed part colour.

VTX reference



LiPo reference



Generic VTX board with shield and cooling fins, and shrink-wrapped LiPo with label, seams, leads and plug. These are presentation models, not exact commercial products.

Power reference controls and placement

Control	What it does
Detailed VTX reference	Generic populated VTX board with RF shield, heatsink fins, cable header and RF sockets. Follows the existing VTX size, mounting pattern and plate level; FPV required.
Detailed LiPo reference	Generic shrink-wrapped LiPo with end protection, cell seams, label, leads and an XT-style plug. Dimensions describe the pack body; schematic leads extend beyond it. Follows its position, rotation and top/bottom mounting; requires the LiPo reference.

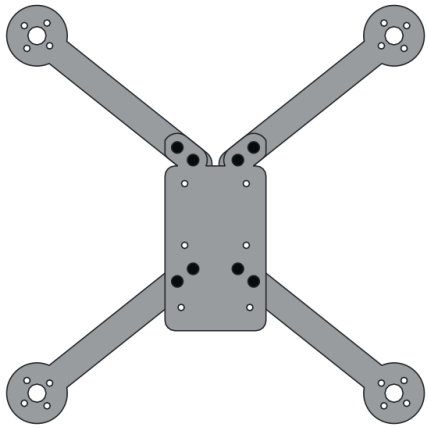
The VTX stays on its existing carrier plate, follows the current mounting-hole pattern and uses its current width, length and height. Analog and digital modes show one or two generic RF sockets. The LiPo pack body follows Battery width, length and height, its X/Y position, rotation and top/bottom mounting; the label faces away from the frame. Schematic leads and plug extend beyond the body and require their own routing clearance. Body-envelope checks, weight/CG estimates and manufacturing exports stay unchanged. Keep simple references for fast editing and enable detail when preparing a design image.

15 Hardware references and nut retention

Low-profile round screw references

Select a generated frame and choose Hardware references from the RotorForge menu or toolbar. Select Arm bolts, Motor screws, Standoff screws, Stack hardware, Camera screws or VTX hardware. Create / refresh references builds simplified round button-head screws at those mounting positions. Nominal thread engagement adds 0.5–10 mm into threaded motor, camera and standoff holes. References have no modelled threads and are not rounded to stock screw lengths; check the purchased hardware and real thread depth. To preserve choices in a reusable design, create or refresh the desired groups, then use Save settings in the frame editor. The JSON stores the selected groups, engagement, retaining-hardware option and visibility. After loading it, Create / Update restores the references automatically; a separate refresh is not needed.

Round button heads at arm mounting positions



Black round button heads at the arm mounting positions. Hardware is a display reference, excluded from manufacturing solids and any second weight estimate.

Reference screws, conventional nuts and press nuts have a black finish. Refreshing existing references applies the finish while retaining unchanged fastener geometry and visibility. This is display appearance only; it does not change hardware dimensions or mass. Show arm retaining hardware controls the arm nut and press-nut reference shapes. Stack hardware includes conventional upper nuts and the selected optional lower press nuts. VTX hardware uses screws only. Hide references and Show references change visibility only. Save the FCStd to retain reference settings. Once created, hardware references follow valid frame changes automatically, including upper nuts and optional press nuts. Your selected hardware groups and hidden states are retained. Unchanged fasteners reuse their existing geometry. Invalid edits pause the references until the frame is valid again. Existing equipment and screw mass allowances remain independent.

Receiving-plate nut seats

Open Hardware > Nut retention in the frame editor. Arm nut retention selects External hex nut, Recessed hex nut or M3 press nut. FC stack press nuts independently enables M3 inserts at the four stack mounting holes. The choices change the real receiving-plate geometry after Update frame; they are not just display settings. Arm retention applies to non-shared arm bolts. Shared arm/standoff bolts continue to enter the threaded standoff. Motors and camera bodies retain their own threaded holes. The FC stack keeps conventional upper nuts even when lower press nuts are selected. Motors, cameras, standoffs and VTX joints receive no retention inserts from these options.

With three plates, the mid plate receives arm nuts. In a two-plate design with above-bottom arms, the upper clamp receives nuts when enabled; otherwise the underside of the bottom plate receives them. With below-bottom arms, the upper face of the bottom plate receives nuts, including when a lower clamp is enabled. VTX screws run up from beneath the carrier into the transmitter mounting points; no VTX nuts or retention cutouts are generated. Only one four-hole VTX pattern is populated when both patterns are drilled.



Section examples in a 3 mm plate. A hex pocket partly sinks a conventional nut. A press nut uses a mounting bore through the receiving plate; its shank depth is an insert dimension, not a blind-hole depth.

Control	What it does
Arm nut retention	Retention for non-shared arm bolts only. Shared arm/standoff bolts keep their threaded standoffs. **Choices:** External hex nut; Recessed hex nut; M3 press nut.
FC stack press nuts	Use M3 press nuts in the stack carrier. Places the stack directly on the exposed flange, overriding manual spacer height; upper conventional nuts follow the stack.
Hex nut recess depth	Partial hexagonal pocket depth into the receiving face. At least 1 mm of plate thickness must remain.
Nut pocket fit allowance	Added to hex nut across-flats width and to circular flange-pocket diameter. Not a radial allowance.

Nut pocket fit allowance adds to the nominal hex nut width and to a circular flange-pocket diameter. The recess may be partial: do not remove all the plate thickness merely to make a nut flush. The builder requires at least 1 mm of carbon below a hex pocket and checks the bearing rim against the finished plate outline and other cuts. A pocket with insufficient stock is rejected; widen

the plate or change the bolt position. Account for the cutter radius and nut corners in CAM; a sharp mathematical pocket may require corner relief or a small cutter to seat a real nut.

M3 press nuts and the correct mounting bore

The defaults follow the documented KF2/KFS2 M3 broaching-nut family: 4.22 mm mounting hole (+0.08 mm), 1.53 mm maximum shank and minimum sheet thickness, 5.56 mm flange diameter and 1.50 mm flange height. These are a specific insert family, not universal dimensions for every M3 press nut. FPV insert suppliers also specify different fits, including 4.50–4.55 mm holes. Match the purchased insert drawing before manufacture; M3 describes the thread, not the required carbon mounting bore.

Control	What it does
M3 press-nut mounting bore	Through-hole diameter in the receiving plate. Default KF2/KFS2 M3 mounting hole: 4.22 mm (+0.08 mm). Other FPV inserts may need 4.55 mm; use the purchased insert drawing.
Press-nut flange diameter	Default KF2/KFS2 M3 E dimension, used for the reference and optional flange pocket.
Press-nut flange height	Default KF2/KFS2 M3 T dimension. The flange remains proud unless recessed.
Press-nut shank envelope diameter	Default KF2/KFS2 M3 C dimension. A smooth reference envelope for the broaching teeth; interference with the smaller mounting bore is intentional.
Press-nut shank depth	Maximum inserted shank length used in the simplified hardware reference; this is not a blind-hole depth.
Press-nut flange recess depth	Optional circular flange counterbore. Zero leaves the flange outside. Remaining plate must meet the insert minimum thickness.
Press-nut minimum plate thickness	Default KF2/KFS2 minimum supporting sheet thickness after flange recess. Change only to match a specific purchased insert.

The mounting bore passes through only the receiving plate. Set flange recess to zero to leave the flange outside, or machine a circular flange counterbore to lower it. The remaining plate must satisfy both the minimum sheet thickness and shank length. M3 press nuts require a compatible M3 joint and FC board holes when used at the stack. The receiving plate must retain the configured minimum-material margin, with a minimum of 1 mm, around the full press-nut flange or flange counterbore, including near other openings. With FC stack press nuts enabled, the stack underside rests directly above the exposed flange, without an added vertical gap. Round button-head screws start beneath the supporting plate, pass through the press nuts and stack, and retain conventional nuts above the stack. Fit stack height includes those upper nuts when selecting taller standoffs. Broaching/press-in inserts are different from long blind-rivet nuts used for sheet metal; installation and pull-out strength in carbon must be verified with your actual insert and laminate.

Dimensional references: PennEngineering KFS2-M3; Jinjiuyi KF2/KFS2 M3; FPV press-nut fit guidance.

Manufacturing files and preview scope

STEP and the native solids contain the real blind pockets and enlarged press-nut mounting bores. DXF/SVG profiles contain through cuts only: they do not turn a shallow hex or flange pocket into a through hole. MachiningOps.csv lists each secondary pocket, receiving face, frame and individual DXF coordinates, depth and either across-flats width or counterbore diameter. Add these pocket operations in CAM before cutting stock. The editor profiles show through bores; inspect the generated 3D parts for shallow seats. Carbon mass uses the machined volume.

16 Save, reload and update a design

Continue editing a saved frame

To save reference hardware in JSON, create or refresh the desired hardware groups, then use Save settings in the editor. The file retains group selection, engagement, retaining-hardware choice, whether references were created, and visibility. Load settings followed by Create / Update restores these references. Hidden references stay hidden. Saving before rebuilding preserves the loaded preferences. Exported FrameParameters.json contains the same information.

Older JSON files lack hardware preferences. Create / refresh the desired references and save again with version 0.89.6 or later to add them. Loading an older file does not automatically enable reference hardware. Loading settings and cancelling the editor do not modify the current frame.

Opening an FCStd keeps its saved solids. Loading JSON refreshes settings and the 2D preview without building a new assembly. Apply Create / Update when ready. Direct parameter edits in FreeCAD regenerate when the document recomputes. Calculate weight / CG, Check plate cuts in CAD and Export also request exact geometry. Correct reported errors before retrying a failed build.

Reuse unchanged components

RotorForge rebuilds affected parts and their dependencies. A plate change can affect its camera contacts, pad or guards; a layout change can affect most of the frame. Vent, USB and wire edits can reuse the solid guard shell and grips. Weight allowances update estimates without remaking carbon. Identical updates leave component Shapes in place. Validation and assembly checks remain active.

Control	What it does
LastBuildSeconds	Geometry build time for the last update; document display and export time are additional.
RebuiltComponents	Components rebuilt for the last update.
ReusedComponents	Components served from the dependency cache.

These read-only values appear under Frame Parameters / 09 Results. Times depend on the design and computer. Preview responsiveness, exact CAD construction and FreeCAD 3D display are separate costs; a faster preview does not remove the time needed for a first complex solid build.

For faster repeat edits, RotorForge keeps a bounded in-memory component cache and a local cache of successful builds. Unchanged components are reused when their actual parameter and upstream-shape dependencies match. The cache is disposable: missing, stale or invalid data falls back to normal generation, and full geometry checks still run before the frame is accepted.

Reuse geometry after restarting

Save the FCStd after updating to retain verified component geometry. RotorForge also keeps a bounded local cache of successful builds, so creating the same JSON settings again can reuse native parts after a restart. The cache holds at most eight designs and 128 MiB. Full settings and generator, FreeCAD and OpenCASCADE versions must match; stored data and shapes are verified before use. Missing, incompatible, damaged or unwritable cache storage falls back to normal generation. The first build of a new design, on another computer or after an upgrade can take longer. JSON files themselves contain settings, not solids.

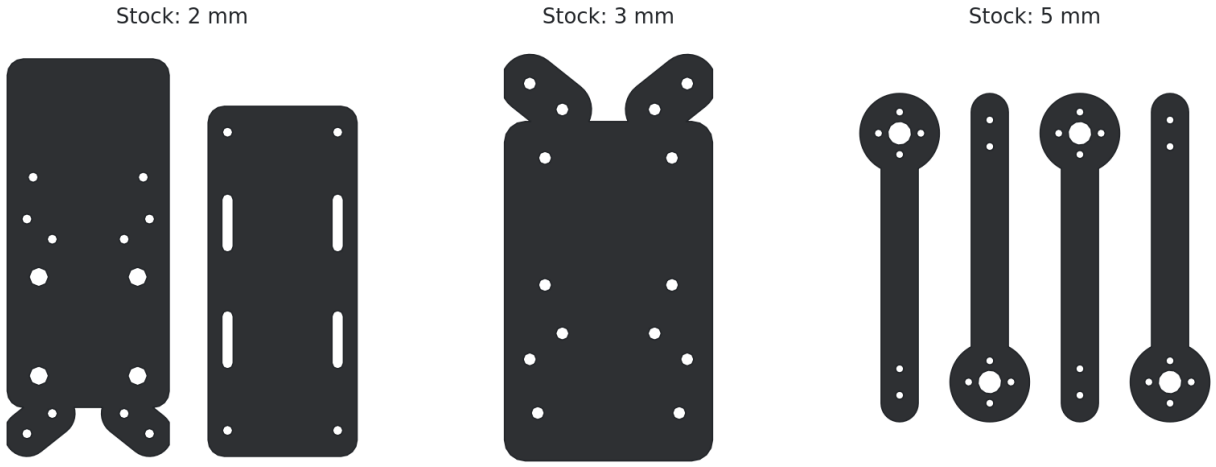
17 Export and prepare a CNC cutting layout

Select a frame, one of its parts, or its cutting-layout objects and choose Export frame. Choose an output directory. A new timestamped subfolder is created for every export, so previous files are not overwritten. Export also refreshes this frame's **Cutting profiles (DXF / SVG)** group in the same FreeCAD document. Save the project to retain that group.

Native cutting profiles in the open project

The cutting group is placed beside the assembly and contains one stock subgroup per thickness. Each physical carbon part appears once as an exact planar face on XY, including its outer perimeter, holes, slots, airflow openings and notches. Camera side plates are flattened from their assembled YZ orientation. There is one contour set shared by the DXF and SVG representations, rather than duplicate overlapping imports.

The root group has a movable placement. Stock subgroup origins correspond to the lower-left origins of the matching stock files. Each child links to its source part and records thickness. Re-export refreshes the snapshot instead of adding duplicate groups; a manually moved root origin is retained. Undo can restore the previous snapshot. Editing the frame does not automatically update a snapshot: export again when ready.



The export planner separates stock thicknesses and alternates arm ends. Sheets use independent display scales.

Create the CAM job from a stock subgroup using its local origin. Verify millimetre units, stock/material and cutter access to holes and slots. Cut internal contours before the outside perimeter.

Spacing, stock thickness and scale

All files are 1:1 in millimetres. The four arms are rotated upright and placed side by side in one row. Motor ends alternate top, bottom, top, bottom, with 4 mm bounding-box clearance between arms. Rotation preserves each arm's handedness and hole geometry; no part is mirrored. Other parts keep their flattened axes and use 12 mm gaps. All sheets have a 10 mm margin. Remaining parts use shelf rows targeting 420 mm; a complete arm row or oversized part can exceed that width without scaling. Different thickness sheets are separated by 30 mm in the combined overview.

This is an alternating arm arrangement with conservative bounding-box clearance, rather than contour-interlocking nesting or G-code. Choose actual stock dimensions, fibre direction, holding tabs and cutter compensation in CAM. The profiles have no cutter-radius or kerf compensation, dogbones or automatic machining allowances. Stock thickness metadata describes the parts; it does not create a CAM stock model.

Control	What it does
FrameAssembly.step	Assembled carbon parts, standoff tubes, optional LiPo pad and active flexible parts, including the front bumper and side guards. Reference envelopes are excluded.
CuttingProfiles / part DXF and SVG	An individual local-origin profile for each physical carbon part.
CuttingProfiles / Stock_<thickness>.dxf and .svg	A separate sheet for each stock thickness, with the same local part positions as its native stock subgroup. These sheets contain no text that could be mistaken for a cutting contour.
CuttingProfiles / AllCarbonParts.dxf and .svg	An overview of all thickness sheets, separated into bands. Keep different thicknesses separate when machining.
CuttingProfiles / Layout.json	Exact units, margins, gaps, stock-sheet sizes and part placements.
FrameParameters.json	Editable settings, including battery/pad parameters and all saved cutout patterns.
MassProperties.json	Bare-frame, nominal equipment and loaded masses and CG, per-item equipment assumptions, material densities and explicit scope.
LiPoPad / STEP, STL, DXF and SVG	If enabled, a separate pad assembly/flat model, printable STL and through-cut outline. Shallow grip grooves appear in 3D files; pad profiles do not enter carbon nesting.
EquipmentMass.csv	When enabled, equipment and screw estimates with mass and X/Y/Z centre for each item. Together with the bare frame these reproduce the Equipped result; the battery is added only to Loaded.
PartsList.csv	Carbon/pad and modelled standoff unit masses, stock thicknesses and hardware summary. Multiply mass by Quantity; blank masses are unassigned. Fastener grip is not total screw length.
ExportInfo.json	Generator version, warnings, mounting geometry, hole patterns, clearances and cutting-layout data.
READ_ME_FIRST.txt	Manufacturing notes for the exported design.
CameraPlates / STEP and STL	In FPV mode, separate assembled STEP and flat printable STL files for the camera sides. Their carbon cutting profiles are also included in the normal sheets.
MachiningOps.csv, when present	Secondary operations such as countersinks, separate from the through-cut profiles. Follow the per-export notes.

18 Common tasks and troubleshooting

If Help shows an older version, close FreeCAD and replace the complete RotorForge folder. The Help button opens freecad/rotorforge/resources/docs/RotorForge_Help.pdf; it must match the PDF at the release root. A small nonzero CG X can result from asymmetric features such as a one-sided USB opening. Compare the design with and without that part.

Control	What it does
A loaded design will not accept another cut	Check the capacity readout. The editor accepts up to 200 patterns, with at most 200 enabled instances. Reduce grid copies or disable an existing pattern if the enabled budget is full.
The preview or model ignores my latest pattern controls	Press Add pattern for a new cut or Apply to pattern for an existing one. Discard unapplied changes restores the saved state. Then check and update the frame.
An outline notch is rejected	Place it across an actual outer edge of a supported plate. Check centre widening, existing holes, captured sockets and standoff bearing material. Every grid copy must cross an edge.
A combined cut is rejected	Overlap shapes instead of merely touching them, or increase the material between separate openings. Avoid cuts that separate the plate or isolate a carbon island.
Arm roots do not meet	Use Fitted miters, bring root references inward and increase tail extension. Root width enlarges material, but joint trimming does not automatically move or extend the roots.
The key is too large	Use the compact fitted-miter key controls. Arrange roots and tails so they meet the intended key pocket, then verify bolt support and screw access.
A widened arm creates plate overhangs	Review the main-body width and reach separately from the arm-matched tabs. Mid front and bottom rear tabs follow their respective arms; an intentionally extended main body is not trimmed away.
The camera hits a plate or cannot tilt through the range	Check actual dimensions, measured vertical screw pitch, cage clearance and top camera cutout. Use Fit camera height if more height is needed.
The lower camera wall does not reach far enough	Enable plate-end following and use Custom lower attachment sizing to extend the front/rear bottom contact independently.
The stack or VTX has no clearance	Check the supporting plate and electronics envelope. Increase standoffs for headroom; Fit stack height adjusts for the FC/ESC. Extend Custom reaches to support the VTX and its top coverage.
Shared standoff holes are rejected	Use an arm-hole snap mode and compatible hole diameters. Check the actual carrier and its full bearing footprint; approximate manual alignment is insufficient.
The cutting layout looks outdated	Re-export the frame. The document group is an export snapshot, not an automatically recomputed dependent model.
I need a different CNC origin or sheet position	Move the cutting root or stock subgroup as needed, and use the matching sheet origin in CAM. Re-export rebuilds the planned child placements and retains the root placement.

Control	What it does
A saved project cannot find the workbench	Install the complete RotorForge folder for that FreeCAD user and restart FreeCAD. See the separate Installation PDF for platform paths.
A help image or guide is missing	Reinstall the complete plugin package. The Help command opens the PDF stored beside the plugin code.